The Price of Delay: Oil Field Lock-In and Delayed Carbon Pricing

Léo Jean*

November 10, 2026

Click here for the latest version

Abstract

Oil projects involve large development costs and comparatively low operating costs to produce. Once development costs are sunk, a carbon tax high enough to have deterred a project can be too low to stop its production, so that delaying climate policy locks in emissions that the delayed policy can no longer remove. Existing models of oil supply give each field a single cost per barrel, which rules this lock-in out. I separate the decision to develop a field from the decision to keep it producing for 25,336 oil fields covering the world's reserves. Delaying a 1.5°C-aligned carbon tax from 2016 to 2030 overshoots the oil carbon budget by a fifth, mostly through overproduction during the delay. A 2030 tax calibrated with existing models misses the emissions locked in during the delay: it overshoots the remaining budget by 13 $GtCO_2eq$, 10% of that budget, although it is only 3.5% too low. Almost half of this overshoot would remain even if investors had correctly expected the tax in 2030. Because locked-in fields barely respond to a higher tax, meeting the budget strands other producers instead, thus redistributing the cost of the delay. Lock-in is concentrated in a few hundred capital-intensive projects, mostly offshore, so that targeted bans can remove nearly half of it.

KEYWORDS: Climate change, fossil fuels, oil, carbon tax, climate policy delay, carbon lock-in, stranded assets, infrastructure lock-in, supply-side climate policy.

JEL CODES: Q35, Q52, Q54, Q58, L71.

*CNRS–CIRED, PSE. Email: leo.jean@psemail.eu. I am grateful to Fanny Henriet for her guidance throughout this project. I thank the members of my thesis jury, Lassi Ahlvik, Julien Daubanes, Gilles Lafforgue, Mar Reguant and Anna Créti, and François Libois, member of my thesis committee, for their helpful comments. I also thank Francis Bloch, Adrien Fabre, François Fontaine, Renaud Coulomb, Pierre-Cyrille Hautcoeur, Sylvie Lambert and Hassan Benchekroun for their suggestions. This paper benefited from comments by participants at the IOEA 2023 summer school, IAEE 2023, EAERE 2024, NAERE 2025, WCERE 2025, the PSI-PSE seminar and the DTU internal seminar, and particularly from my discussants Davin Raiha, Juni Singh, Emmanuel Asane-Otoo, Charlotte De Cannière and Joseph Stemmler. This work benefited from a French government subsidy managed by the Agence Nationale de la Recherche under the framework of the "Investissements d'avenir" programme reference ANR-17-EURE-001.

1 Introduction

In 2015, the Paris Agreement set the objective of holding global warming well below 2°C and pursuing efforts to limit it to 1.5°C. Meeting this objective requires leaving a large share of known oil reserves in the ground (McGlade and Ekins, 2015; Welsby et al., 2021), and carbon pricing is widely regarded as the most effective instrument to achieve it. In the decade since, however, carbon prices have remained limited in coverage and far below the levels these objectives require (Metcalf, 2019), and investors do not expect them to rise. Firms invest according to the carbon price they find credible, which under policy uncertainty is well below the announced one (Dolls et al., 2026). Oil companies, for instance, screened new projects with internal carbon prices that covered only the emissions of their own operations (CDP, 2016). The years since Paris are therefore best described as a delay in carbon pricing that investors do not expect to end. During this delay, the oil industry keeps committing capital to new fields that will produce for decades.

Delaying climate policy, when investors do not expect it to end, affects emissions in the oil sector through two channels. The first is overproduction: fields already in operation keep producing barrels that a timely tax would have retired. The second is investment, as firms develop new fields that a timely tax would have deterred. The two differ in when their emissions occur. Overproduction stops when the tax arrives, since costlier fields are then phased out. Investment has long-lasting effects, because development costs are sunk once a field is built. From then on, the field only needs to cover its operating costs, and a field that is cheap to operate can keep producing for decades. A tax high enough to have prevented its development is therefore not necessarily high enough to stop its production once it exists. The delay thus leaves behind a stock of committed emissions that the delayed tax cannot remove. To meet the same carbon budget, the tax must then rise beyond what overproduction alone would justify.

This paper measures the cost of the second channel, the lock-in of emissions by the investments made during the delay. Existing models of oil supply rule this lock-in out by construction. For the delay since Paris, I find that the carbon tax these models would set in 2030, which ignores lock-in, leaves emissions 13 $GtCO_2eq$ above the remaining carbon budget. This gap is 10% of the budget remaining in 2030, even though the tax is only 3.5% too low. Almost half of it would remain even if investors had correctly expected the tax in 2030. Lock-in also redistributes the cost of the delay across countries and firms: locked-in fields barely respond to a higher tax, so meeting the budget strands production elsewhere. It is concentrated in a few hundred capital-intensive projects, mostly offshore, which restrictions on new development can reach.

The size of lock-in depends on the economics of oil fields, and is therefore an empirical question. Typical oil projects require large investments long before they produce. A deepwater project takes three to five years of investment in facilities and wells before its first barrel, and once built it produces at low operating costs for decades. Shale is the opposite case. Little capital is sunk before production. Drilling continues for as long as the field produces, and each well yields most of its oil within a year or two. Both kinds of projects account for large reserves among the fields developed during the delay: 2,756 shale and tight oil assets hold 57 Gbbl, and 219 deepwater projects hold 46 Gbbl. Whether a field locks in emissions, and how much, follows from how its capital intensity, operating costs and carbon intensity combine.

Existing models of oil supply cannot capture lock-in, because they collapse the decision to develop a field and the decision to keep it producing into a single cost per barrel. Integrated assessment models represent fossil fuel supply as a cost curve in cumulative extraction, with no memory of past investment (Grubb et al., 2021). Asset-level models of oil supply have brought field-level detail into the analysis of climate policy (Heal and Schlenker, 2019). They have measured two inefficiencies in the way oil is supplied: extracting costly barrels before cheaper ones (Asker et al., 2019), and carbon-intensive barrels before cleaner ones (Coulomb et al., 2026). Yet these models give each field a single cost per barrel that averages its development capital over its production, so that the cost of a barrel does not depend on whether the capital behind it has been spent. A field then occupies the same position on the supply curve before and after it is built. Any tax high enough to have deterred a project is also high enough to shut it down, and lock-in cannot arise. Applied to the cost of inaction, these models count the overproduction that occurs during the delay, but miss a third, long-lasting inefficiency. Fields developed while policy was absent keep producing once it arrives, and only a supply curve with memory can show it.

I first develop a simple model of a field facing a carbon tax that is implemented late. Before development, a field must expect to cover both its development cost and its operating costs, which rise with age. Once the development cost is sunk, only operating costs remain, and the highest tax the field can bear jumps upward. Tax levels in between, high enough to have prevented the field but too low to stop it once built, form its *lock-in zone*. A field developed during the delay locks in emissions if the tax that is eventually implemented falls inside its zone. To measure the cost of lock-in, I compare the tax needed after the delay to meet the carbon budget with the tax set by a planner who gives each field a single cost per barrel, as existing models do. I call it the *naive planner*: it treats locked-in fields as if they could still be deterred, and sets the tax too low. The cost of lock-in can then be measured in tonnes, as the emissions the naive planner leaves above the budget, which I call the *residual*

overshoot. It can also be measured in dollars, as the additional tax needed to close the gap, the *carbon tax lock-in premium*.

The model describes a single field. Applying its logic field by field, I build a model of world oil supply from Rystad Energy's UCube database, covering all 25,336 oil fields with positive oil reserves in 2016. For each field, development costs are separated from operating costs, which increase stage by stage as the field depletes. Once a field is built, its production profile is fixed, in line with evidence on the geological constraints on production (Anderson et al., 2018). Its lifecycle carbon intensity, covering extraction, refining and final use, is estimated from engineering models and satellite observations of flaring. Its oil is sold at a discount to Brent that reflects its quality. These dimensions vary widely across fields. Whether a field locks in emissions depends on how they combine within it, which a supply curve built on averages would hide. The path of the delay follows the investments observed up to 2025 and Rystad's forecast thereafter, and no tax is levied before 2030. The two worlds compared in the paper share the same fields, costs, carbon intensities and production paths, and differ only in the date at which the tax arrives. Every difference between them therefore comes from the delay. The reference Brent price is \$80/bbl and the discount rate 10%.

A carbon tax aligned with the oil carbon budget of Welsby et al. (2021) and implemented in 2016 would reach \$80.4/tCO_2eq in 2030.[1] Delaying it to 2030 overshoots that budget by 20%, mostly through overproduction released before the tax arrives. Meeting the budget after the delay requires a 2030 tax of \$98.4. Of this \$18.0 increase, \$14.6 compensates for overproduction, which a supply curve without memory captures. The other \$3.3 compensates for lock-in, which such a curve misses. Each dollar of tax removes about 4 GtCO_2eq of emissions after 2030, so missing this small premium leaves the 13 GtCO_2eq of residual overshoot.

Two features of locked-in fields shape the policy response. They are heavily concentrated: half of the residual overshoot comes from 32 projects, and four fifths of it from offshore fields. They are also resilient, as raising the tax to the level the budget requires removes less than a tenth of their emissions. The premium therefore strands production elsewhere, mostly in the long-lived, low-cost fields of the Middle East and Russia, which already lose through overproduction. North America, by contrast, gains through both channels. For Brazil and Guyana, lock-in reverses the sign of the delay's incidence: they lose through overproduction but gain more through their locked-in deepwater projects. The same features make targeted restrictions effective: a ban on new deepwater approvals from 2016 would have cut the residual overshoot by 44%. This ban would not need to be global. Applied to the projects

[1] All tax levels in the paper are the levels a tax path rising at 2% per year reaches in 2030; a level of x in 2030 corresponds to $x/1.32$ in 2016.

located in OECD countries, or operated and majority-owned by firms headquartered there, it would reach more than half of the deepwater reserves not yet approved in 2016.

The model is one of oil supply: in the main specification, the oil price does not respond to the tax, which is entirely borne by producers. One could expect a price response to weaken lock-in, since part of the tax would then fall on consumers. It does not remove it, because a field whose development cost is sunk keeps producing as long as its operating margin covers the tax. Higher producer prices let locked-in fields produce for longer, and the extra supply they bring lowers the price again. With an isoelastic demand for oil and elasticities between -0.4 and -0.8, the residual overshoot ranges from 7% to 16% of the remaining budget, against 10% at a constant price. The design of the tax matters far more for the total cost of the delay than for lock-in. A constant tax nearly doubles the overshoot of the budget, mostly through overproduction, while the residual overshoot only rises from 13.2 to 14.9 $GtCO_2eq$.

Related literature. This paper relates to three literatures. The first, as mentioned above, models the supply of fossil fuels. A theoretical strand departs from the Hotelling framework by treating supply as a sequence of fixed investments, each followed by a production path that cannot be freely adjusted (Venables, 2014; Okullo and Reynès, 2016; Kellogg, 2026). Anderson et al. (2018) provide the empirical support for this view, showing that producers respond to prices by drilling more or fewer wells rather than by adjusting the flow from wells already drilled. Wachtmeister and Höök (2020) illustrate its consequences with representative profiles of conventional fields and tight oil wells. The empirical strand of asset-level models discussed above has also been used to study the role of market power (De Cannière, 2025) and the excess emissions caused by extracting carbon-intensive barrels first (Asker et al., 2024). Its single cost per barrel is a reasonable simplification when the question is how a given quantity of oil is allocated across fields. It becomes restrictive when the question is the timing of policy. This paper bridges the two strands. It takes the theoretical view to the data, with each field's own costs, carbon intensity and production path. Bringing fixed investments and fixed production paths into an asset-level model makes lock-in emerge when climate policy arrives late.

The second literature studies the cost of delaying climate policy, mostly with integrated assessment models. These studies find that delaying ambitious policy until 2030 raises its cost substantially (Riahi et al., 2015; Kotlikoff et al., 2021). In the models with the richest representation of inertia, the delay can put the objectives out of reach (Jakob et al., 2012; Luderer et al., 2013; Bertram et al., 2015). Inertia in these models comes from fixed investments on the demand side, mostly in power generation, while fossil fuel supply, as noted above, has no memory of past investment. This paper adds inertia on the supply side and

traces it to the specific investments that create it.

The third turns flows of emissions into stocks, either as emissions committed by existing infrastructure or as stranded assets. The literature on carbon lock-in (Unruh, 2000; Seto et al., 2016) measures the emissions that existing infrastructure will release whatever the policy in place. It assumes that this infrastructure operates over its historical lifetime and at its historical utilization rate. Following the accounting of Davis and Socolow (2014), Tong et al. (2019) show that existing energy infrastructure already commits most of the 1.5°C budget. Erickson et al. (2015) and Trout et al. (2022) apply the same accounting to fossil fuel supply, and the latter find that fields already producing or under construction commit more emissions than a 1.5°C budget allows. These studies measure what existing infrastructure would emit if it kept operating as in the past, which tells how much of a carbon budget is already committed. They are not designed to measure what a delay in policy costs. That requires a counterfactual in which policy arrives on time, and an account of the commitments that a tax can still remove. Comparing a timely tax with a delayed one, and counting only the production that remains profitable under the tax, provides both. It measures the part of these commitments that policy can no longer remove after a delay, together with the year each commitment became irreversible and the tax needed to retire it.

The literature on stranded assets has established the size of unburnable reserves (Meinshausen et al., 2009; McCollum et al., 2014) and their uneven distribution across countries (McGlade and Ekins, 2015; Welsby et al., 2021), which depends on extraction costs and carbon intensities (Coulomb et al., 2026). It has also traced the corresponding losses to their ultimate owners (Mercure et al., 2018; Semieniuk et al., 2022), and examined how owners respond when they expect the policy (Coulomb and Henriet, 2018; van der Ploeg and Rezai, 2020). In these studies, policy arrives at a given date, and stranding is computed on a supply curve with no memory of investment. This paper shows how a delay reallocates stranded assets through two channels, overproduction and lock-in. It also shows how the stranding the delay requires builds up with each year of inaction.

The remainder of the paper is organized as follows. Section 2 presents the model, Section 3 the data, Section 4 the results and their robustness, and Section 5 the incidence of the delay and the effect of targeted policies. Section 6 concludes.

2 A simplified model of investment and lock-in under delayed carbon taxation

This section presents a simple model of investment in oil fields when a carbon tax is implemented late. It serves two purposes: defining the objects measured in the empirical part, and showing which field characteristics determine their size. The *lock-in zone* of a field is the range of carbon taxes that would have prevented its development but cannot stop its production once the development cost is sunk. *Committed emissions* at a given tax are the future emissions of existing fields that survive this tax, and *overcommitted emissions* are the committed emissions of fields that a timely tax would have deterred. The *lock-in premium* is the additional tax required after the delay to offset overcommitted emissions, on top of the increase needed to compensate for overproduction during the delay. I use a constant tax to keep the model analytically tractable, and show in Section 2.6 that the main results carry over to an increasing tax. Closed-form expressions, which require no discounting and a parametric cohort of fields, are in Appendix A.1.

Production profiles are fixed by geology and by the initial design of the field, so that the tax affects when a field starts and stops producing, but not how much it produces in a given year. Okullo and Reynès (2016) model supply in the same way: production is bounded by installed capacity and by a geological limit on the share of reserves extracted each year, and capacity only expands through investment. Anderson et al. (2018) show that production from existing wells hardly responds to prices, while drilling does. With fixed profiles and development dates taken from the data, producers who expect the tax cannot accelerate extraction before it is implemented, which rules out the green paradox and makes the total cost of the delay a lower bound. Expecting the tax can also deter investment during the delay, which reduces lock-in, as shown in Section 2.5.

2.1 Setup

A carbon tax τ is levied on producers. Under the timely policy it is in place from 2016; under the delayed policy it is implemented at date Δ. From Δ onwards, both policies apply the same tax.

A field is characterised by a development cost C, paid when the field is developed; a production flow $\bar{q}$; a carbon intensity θ, so that it emits $\theta\bar{q}$ per period; and a selling price p, constant over time.[2] Its operating cost per barrel increases with its age a, counted from the

[2]The selling price is the common reference price minus a discount that depends on the characteristics of

start of production,

$$c(a) = c_0 + \gamma a, \qquad \gamma > 0. \tag{1}$$

Operating costs rise as a field gets depleted because reservoir pressure declines and more water is produced with the oil, which requires more energy to lift each barrel (Chermak and Patrick, 1995; Leighty and Lin, 2012), and the production breakevens computed from the data increase accordingly across production stages (Section 3). Capital intensity $\kappa \equiv C/\bar{q}$ is the development cost per barrel of annual capacity. Firms discount profits at rate δ, while emissions are not discounted.

2.2 Two breakevens and the lock-in zone

Once developed, a field of age a earns $p - c_0 - \gamma a - \theta\tau$ per barrel. It keeps producing as long as this margin is positive, that is as long as the tax does not exceed its *production breakeven* $\bar{\tau}_P(a)$, and it is phased out at age $a^*(\tau)$:

$$\bar{\tau}_P(a) \equiv \frac{p - c_0 - \gamma a}{\theta}, \qquad a^*(\tau) = \frac{p - c_0 - \theta\tau}{\gamma}. \tag{2}$$

The production breakeven only depends on operating costs, since the development cost is sunk once the field produces. It is highest at start-up, $\bar{\tau}_P(0) = (p - c_0)/\theta$, and falls by γ/θ each year.

Before development, investors decide to invest in a field if its net present value is positive:

$$-C + \int_0^{a^*(\tau)} e^{-\delta a}\, \bar{q} \left(p - c_0 - \theta\tau - \gamma a\right) da \;\geq\; 0. \tag{3}$$

Equivalently, the discounted profit per barrel of capacity must cover the capital intensity of the field,

$$\bar{v}_\delta(\tau) \;\equiv\; \int_0^{a^*(\tau)} e^{-\delta a} \left(p - c_0 - \theta\tau - \gamma a\right) da \;\geq\; \kappa. \tag{4}$$

The discounted profit $\bar{v}_\delta(\tau)$ decreases with the tax, so the condition holds if and only if the tax does not exceed the *development breakeven* $\bar{\tau}_D$, defined by

$$\bar{v}_\delta\left(\bar{\tau}_D\right) \;=\; \kappa. \tag{5}$$

$\bar{\tau}_D$ is the model counterpart of the maximum viable tax computed field by field in Section 3. Since $\kappa > 0$, $\bar{\tau}_D < \bar{\tau}_P(0)$: the taxes between the two breakevens would have prevented the

the oil (Section 3.3).

development of the field but do not stop its production once it exists. This interval is the field's lock-in zone. I describe it by its depth, its width and its height, which Figure 1 illustrates. Panel (a) plots the two breakevens against capital intensity for a field of a given age at implementation, and panel (b) plots them against the age at implementation for a field of a given capital intensity. In both panels, region II is the set of taxes at which the field is locked in.

Depth. The depth of the zone is the distance between the two breakevens at start-up, $D \equiv \bar{\tau}_P(0) - \bar{\tau}_D$. Expressed per barrel, $\theta D = p - c_0 - \theta\bar{\tau}_D$ is the margin at start-up at which the field just recovers its development cost, and from (4) it solves

$$\int_0^{\theta D/\gamma} e^{-\delta a} \left(\theta D - \gamma a\right) da \;=\; \kappa. \tag{6}$$

The depth reflects how a field's costs split between fixed and variable costs. The production breakeven only has to cover variable costs, while the development breakeven also has to cover the fixed development cost, so the larger the fixed cost per barrel of capacity, the further apart the two breakevens. In panel (a), the development breakeven falls as capital intensity rises and region II opens up; in panel (b), a more capital-intensive field has a lower horizontal line $\bar{\tau}_D$ and a taller region II. Expressed per barrel, the depth depends on capital intensity, on the slope of the operating cost path and on the discount rate, but not on carbon intensity or on the level of operating costs. Without discounting, $\theta D = \sqrt{2\gamma\kappa}$. Discounting lowers the value of developing a field without changing the decision to keep it producing, so it lowers $\bar{\tau}_D$ and deepens the zone: the undiscounted depth is a lower bound (Appendix A.1).

Width. A field only locks in emissions if it still produces when the tax arrives. In panel (b), its production breakeven falls with age with slope γ/θ, and the zone closes where it crosses the development breakeven, at age $a^*(\bar{\tau}_D) = \theta D/\gamma$. A field developed during the delay can therefore carry overcommitted emissions only if it is young enough at implementation, $a_\Delta < a^*(\bar{\tau}_D)$. The zone is wider for fields whose operating costs rise slowly, since a flatter production breakeven crosses the development breakeven later, and for capital-intensive fields, since their development breakeven is lower.

Height. The tax that meets the carbon budget must also fall inside the zone. The height of the zone is set by its top, $\bar{\tau}_P(0) = (p - c_0)/\theta$. A lower operating cost at start-up raises both breakevens by the same amount and moves the whole zone up without changing its depth; a lower carbon intensity raises both breakevens and also deepens the zone in tax

units. In panel (b), this shifts both lines upwards. Oil sands illustrate why height matters: their capital intensity is close to that of deepwater fields, but their high operating costs and carbon intensity place a deep zone below the taxes the budget requires. Deepwater fields have a deep zone that sits high, while shale wells have a shallow zone that sits low.

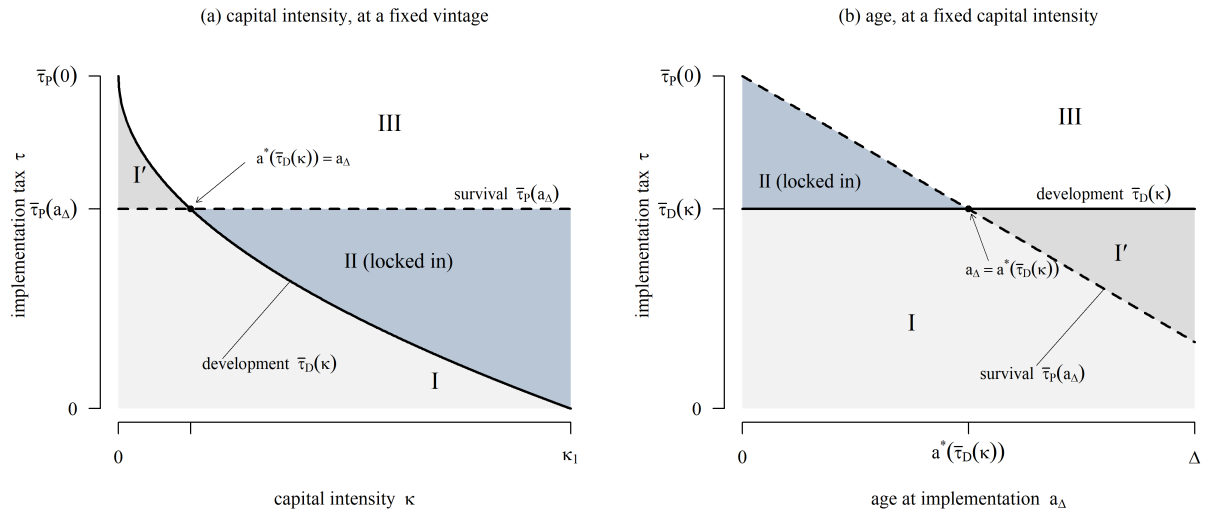


Figure 1: Which fields lock in.

Notes: The implementation tax is on the vertical axis. The development boundary is the development breakeven $\bar{\tau}_D$, the survival boundary the production breakeven at implementation $\bar{\tau}_P(a_\Delta)$, and only region II carries overcommitted emissions after implementation. (a) At a given age at implementation, the lock-in zone deepens with capital intensity, and the two boundaries cross where $a_\Delta = a^*(\bar{\tau}_D)$. (b) At a given capital intensity, the survival boundary falls with age and the zone closes at $a_\Delta = a^*(\bar{\tau}_D)$. Drawn without discounting, with $p - c_0 = 10$, $\theta = \gamma = 1$, $\Delta = 8$, $a_\Delta = 4$ in (a) and $\kappa = 8$ in (b).

2.3 Which fields lock in, and the overshoot they cause

Fields developed during the delay differ in their costs and in their age at implementation: a field developed at date $s \in [0, \Delta)$ is $a_\Delta = \Delta - s$ years old when the tax arrives. Its contribution to the overshoot, compared with the timely policy, depends on two questions. Would a timely tax have allowed its development? From (4), it would not if $\kappa > \bar{v}_\delta(\tau)$, that is if its discounted profit per barrel of capacity under the tax does not cover its capital intensity, or equivalently if $\tau > \bar{\tau}_D$. Fields developed under both policies are called *intensive*, and fields developed only because of the delay are called *delay-induced*. Is the field still producing at implementation? It is if $a_\Delta < a^*(\tau)$. The two questions give four types of fields, which correspond to the four regions of Figure 1. Table 1 reports their excess emissions compared with the timely policy, per field and in units of $\theta\bar{q}$.

Type	Condition	released during the delay	overcommitted after Δ
I	$\kappa \leq \bar{v}_\delta(\tau)$, $a_\Delta < a^*(\tau)$	0	0
I′	$\kappa \leq \bar{v}_\delta(\tau)$, $a_\Delta \geq a^*(\tau)$	$a_\Delta - a^*$	0
II	$\kappa > \bar{v}_\delta(\tau)$, $a_\Delta < a^*(\tau)$	a_Δ	$a^* - a_\Delta$
III	$\kappa > \bar{v}_\delta(\tau)$, $a_\Delta \geq a^*(\tau)$	a_Δ	0

Table 1: Excess emissions of each type of field relative to the timely policy.

Notes: Per field, in units of $\theta\bar{q}$. Types I and I′ are developed under both policies (intensive), Types II and III only because of the delay (delay-induced).

Only Type II fields emit after implementation what a timely policy would have avoided. They are capital-intensive and cheap to operate, like deepwater projects. Type III fields are also delay-induced, but their operating costs are too high for them to survive the tax at implementation. Some are cheap to develop, like shale wells, and others are not, like oil sands. Type I′ fields would have been developed under both policies, but the timely tax would have stopped them before Δ, because they are cheap to develop but costly to operate. The table also applies to fields already producing in 2016, whose development cost is sunk and which can be treated as having $\kappa = 0$. They are of Type I′ if the timely tax would have stopped them before Δ, and of Type I otherwise. Type I fields, typically the onshore conventional fields of the Middle East, are cheap to operate and cheap to develop or already developed, and they produce under both policies.

Summing over all fields splits the overshoot into three parts: the overshoot at the intensive margin, released before Δ by Type I′ fields, the overshoot at the extensive margin released before Δ by Type II and III fields, and the *lock-in wedge*

$$W(\tau) \equiv \sum_{\text{Type II at } \tau} \theta\bar{q}\left(a^*(\tau) - a_\Delta\right), \tag{7}$$

the overcommitted emissions of the cohort at tax τ. The wedge is defined for any tax, independently of the budget. It is zero when there is no tax, since no field is then delay-induced, and it is zero again at taxes high enough to stop every field at implementation. In between, a higher tax brings more fields into their lock-in zone, but shortens the remaining life of those already in it, so the wedge first increases and then decreases with the tax. Appendix A.1 gives its closed form for a parametric cohort.

2.4 The residual overshoot and the lock-in premium

Let B_Δ denote the carbon budget left at implementation, that is the budget minus the emissions released during the delay. The tax that meets this budget depends on the supply curve used to compute it.

The naive planner. The naive planner described in the introduction assigns each field a single cost per barrel, which spreads its development cost over its lifetime production, so that the cost of a barrel does not depend on whether the capital behind it has been spent. For this planner, a field developed during the delay stops producing as soon as the tax exceeds its development breakeven. Its lock-in zone disappears: in Table 1, Type II fields become Type III fields, and in Figure 1, region II merges with region III. The naive planner therefore expects no overcommitted emissions after Δ.[3] Let $E(\tau)$ be the emissions of the cohort after implementation, that is the sum of $\theta\bar{q}(a^*(\tau) - a_\Delta)$ over all fields still producing at Δ, and $E^N(\tau)$ the same emissions as computed by the naive planner, which leaves out Type II fields. Then

$$E(\tau) - E^N(\tau) = W(\tau). \tag{8}$$

E depends on neither the discount rate nor the distribution of capital intensity, since emissions after implementation only depend on the fields that exist. The tax τ^* that meets the budget, defined by $E(\tau^*) = B_\Delta$, is therefore independent of both.

The residual overshoot and the premium. Let τ^N be the tax that meets the budget according to the naive planner, $E^N(\tau^N) = B_\Delta$, which I call the naive tax. Implementing τ^N leaves a *residual overshoot* of the budget

$$\Omega^N \;\equiv\; E(\tau^N) - B_\Delta \;=\; W(\tau^N). \tag{9}$$

Meeting the budget requires increasing the tax by the *lock-in premium*

$$\pi \;\equiv\; \tau^* - \tau^N. \tag{10}$$

The wedge is a function of the tax, and the residual overshoot is its value at τ^N. The difference matters in the data, because the wedge takes different values at the timely tax, at τ^N and at τ^*, and only its value at τ^N measures what the naive planner leaves unabated.

[3] I only apply the single cost per barrel to the decision of delay-induced fields to produce after Δ. Applying it to the producing life of every field would shorten these lives in the planner's calculation, lower its estimate of emissions further and widen the gap with actual emissions, so this choice is conservative.

The residual overshoot and the premium measure the same error, in tonnes and in dollars. To a first approximation,

$$\pi \approx \frac{\Omega^N}{|E'(\tau^N)|}, \tag{11}$$

where $|E'|$ is the reduction in emissions after implementation obtained with one more dollar of tax. For a given residual overshoot, the premium is small when emissions respond strongly to the tax and large when they respond weakly. Section 4.2 reports both numbers and the slope that relates them.

Two properties of the premium hold in general. Appendix A.1 gives a closed form, without discounting and with capital intensity uniformly distributed, in which they are explicit. First, $\pi > 0$ if the delay led to the development of any field the tax τ^* would have deterred. Second, the premium increases with the discount rate. A higher δ lowers $\bar{v}_\delta(\tau)$, so that more fields are delay-induced and E^N falls while E is unchanged: τ^N decreases, τ^* does not change, and both W and π increase. The undiscounted case is therefore a lower bound, and the sensitivity analysis of Section 4.5 shows the same ordering with discount rates of 5% and 15%.

Who pays for the delay. Locked-in fields hardly respond to the premium. When their zones are deep, their production breakevens lie well above τ^N, so an additional dollar of tax does not stop them, and the budget is met by stranding production on fields that a timely policy would have left in place. Overproduction shifts stranding in the same way. Emissions released during the delay reduce the budget left at Δ, so that even with the right supply curve, the tax after the delay must be higher than the timely one, and this increase strands production that a timely policy would not have stranded. The delay thus shifts stranding through two channels: overproduction, which raises the tax to τ^N, and lock-in, which raises it further to τ^*. Section 5 measures both.

2.5 Anticipation

So far, investors have been assumed not to anticipate the tax during the delay. Suppose instead that they screen projects with an anticipated tax $\tau^A \in [0, \tau]$, which can combine a probability of implementation, an expected implementation date and an expected level. The baseline corresponds to $\tau^A = 0$ and full anticipation to $\tau^A = \tau$. The only change concerns which fields are developed during the delay: a project is developed if $\kappa \leq \bar{v}_\delta(\tau^A)$, that is if $\tau^A \leq \bar{\tau}_D$. The delay-induced fields are then those whose development breakeven lies between

the anticipated and the actual tax, $\tau^A \leq \bar{\tau}_D < \tau$. The four types, the decomposition of the overshoot, the wedge and the premium are defined as before.

With anticipation, a field is no longer developed if its development breakeven, $\bar{\tau}_D = \bar{\tau}_P(0) - D$, which is the bottom of its lock-in zone, lies below the anticipated tax. Both the depth and the height of the zone matter here. A field with a deep zone can still be developed under anticipation if its zone sits high enough, as for a deepwater project that is cheap to operate and relatively clean, while a field with a shallower zone can be deterred if its zone sits low. The residual overshoot and the premium remain positive as long as some fields developed during the delay have a development breakeven between τ^A and τ^*, and disappear only when investors screen their projects with the tax the budget turns out to require, applied from the date they invest. Expectations are rarely of this form. Investors may expect the tax at a later date, or at the level that a shorter delay would require, and each of these expectations amounts to screening projects with a tax below τ^*. It reduces lock-in without removing it, as the anticipation scenarios of Section 4.4 show.

2.6 Increasing tax and field heterogeneity

The empirical analysis departs from the simple model in four ways. Each of them affects the results in a direction the model predicts.

An increasing tax. In the empirical analysis, the tax increases at rate r. Under a constant tax, the tax only enters the development decision through the margin at start-up, which is why the depth of the zone does not depend on the tax. With an increasing tax, it also enters through the path of the tax after start-up, and the development breakeven depends on the date of development, $\bar{\tau}_D(a_\Delta) = \bar{\tau}_D^0 e^{r a_\Delta}$, where $\bar{\tau}_D^0$ is the development breakeven of a field developed at Δ. Fields developed early in the delay would have faced a lower tax under the timely policy, and are therefore more likely to have been developed anyway. Figure 2 shows the resulting partition. The production breakeven, the four types and the identity $W = E - E^N$ are unchanged, but the closed forms are lost. Appendix A.2 gives the details.

An increasing tax affects lock-in in two opposite ways. It raises the development breakeven of young fields, so more of them become delay-induced, and it shortens the life of every locked-in field after implementation. After Δ, the zone now closes from both sides: the production breakeven falls by γ/θ each year while the tax rises by about $r\tau e^{r\Delta}$. At the increase rate used here, the shortening effect dominates, but only slightly, so that lock-in and the premium are of similar size under a constant and an increasing tax. The total cost of the delay differs

much more. A tax that meets the same budget while rising over time starts at a lower level, so a timely policy retires fewer fields before Δ, and the overproduction added by the delay before implementation is smaller. Section 4.5 quantifies both.

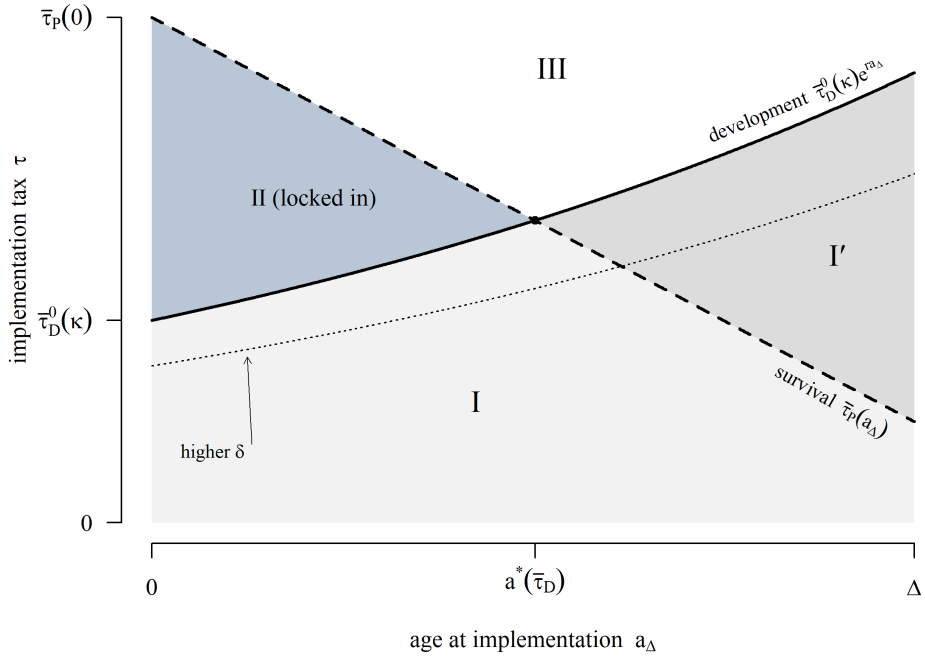


Figure 2: The taxonomy under an increasing tax, at a given capital intensity.

Notes: The survival boundary $\bar{\tau}_P(a_\Delta) = (p - c_0 - \gamma a_\Delta)/\theta$ is the same as in Figure 1. The development boundary is $\bar{\tau}_D(a_\Delta) = \bar{\tau}_D^0\, e^{r a_\Delta}$, increasing with age because fields developed early in the delay would have faced a lower tax under a timely policy. Region II, the set of locked-in fields, lies between the two boundaries and closes where they cross. The dotted line shows the development boundary at a higher discount rate: δ shifts it down and enlarges region II, r tilts it, and neither moves the survival boundary. Drawn with $p - c_0 = 10$, $\theta = \gamma = 1$, $\Delta = 8$, $r = 0.10$ (exaggerated for legibility; the empirical rate is 2%), $\bar{\tau}_D^0 = 4$.

Heterogeneous production, cost paths and development periods. Three other features of the data affect the results in one direction each. First, the overcommitted emissions of a locked-in field, $\theta\bar{q}(a^* - a_\Delta)$, increase with $\bar{q}$, whereas whether a field is locked in does not depend on $\bar{q}$. The size of fields therefore changes the amount of lock-in but not which fields carry it, which is why the concentration documented in Section 5.1 reflects the size of a few projects. Second, capital-intensive offshore projects also have lower operating costs and flatter cost paths than shale. A lower c_0 raises the zone and a lower γ widens it, so heterogeneity in operating costs reinforces the concentration of lock-in in deepwater fields. Third, developing a field takes several years, and some fields are only partly built when the tax arrives. Appendix A.3 adds a development period: a partly built field is completed if its production value covers the development cost it still has to pay, and abandoned otherwise.

Since a development period can only add fields to the locked-in set, the model without it gives a lower bound.

The empirical analysis includes the increasing tax and these three features.

3 Data and methods

3.1 Oil field data and sample

The analysis relies on proprietary asset-level data from Rystad Energy. Rystad's UCube database covers the vast majority of oil and gas production in existing fields worldwide since 1900, together with fields expected to be discovered and developed up to 2100, over 100,000 assets in total across oil, gas, LNG and biofuels. The analysis is conducted at the asset level, the smallest geographic scale available in the database, which typically corresponds to a field or to a portion of a field. For each asset, the data contain annual production by oil type and costs by category (operating expenses, split into production, transportation and SG&A opex; capital expenses for facilities and wells; exploration expenses; and government take), together with administrative, geographic and geological characteristics. Production and costs are observed up to 2025 and forecast by Rystad thereafter, and all monetary values are in constant 2025 dollars. Appendix B.1 reproduces Rystad's definition of each cost item and details the cost assumptions.

The estimation sample keeps oil-producing assets discovered by 2025 with positive remaining oil reserves in 2016.[4] This leaves 25,336 assets, of which 16,746 were already producing in 2016. Reserves in this paper are oil resources as defined by Rystad: all economically recoverable oil over the lifetime of the asset, which is typically larger than 2P reserves because it includes reserve growth. Because they are Rystad's current estimates of what each asset will produce over its life, assets discovered after 2016 have positive reserves in 2016, and the filter excludes none of the assets developed before 2030. Oil is defined as any type of crude (bitumen, extra heavy, heavy, regular, light or sour), together with condensate and natural gas liquids, which are counted as oil because (i) they are liquids and sometimes mixed with crude oil and (ii) their price is set mainly by the reference oil price.

Table 2 describes the sample. Panel A splits it by vintage: fields already producing in 2016, on which the delay acts only through overproduction; the delay cohort, the 5,438 fields

[4]Gas assets are excluded. Assets that Rystad models as not yet discovered are excluded from the main sample because their production, costs and reserves are entirely modelled; the sensitivity analysis of Section 4.5 adds them back, with their exploration expenses treated as development costs.

on which some development takes place between 2016 and 2029, and which are the only fields a delay can lock in; fields approved in 2030 or later; and fields never developed on the baseline path, which fail their development test even without a carbon tax, emit nothing and are left out of the totals. Panel B splits the delay cohort by supply segment, and Panel C splits all developed fields by region. For each group, the table reports the medians of capital intensity κ (development capex per barrel of peak annual capacity), of the development and production breakevens $\bar{\tau}_D$ and $\bar{\tau}_P(0)$, of the production breakeven in 2030, $\bar{\tau}_P(a_\Delta)$, when the delayed tax arrives, and of the depth of the lock-in zone, at start-up, $\bar{\tau}_P(0) - \bar{\tau}_D$, and in 2030, $\bar{\tau}_P(a_\Delta) - \bar{\tau}_D$, the part of the zone still open when the tax arrives. Medians are weighted by 2016 reserves, so that they describe the typical barrel rather than the typical field.

	Fields	Reserves 2016 (Gbbl)	Emissions post-2016 ($GtCO_2eq$)	Emissions post-2030 ($GtCO_2eq$)	κ (\$/bbl)	$\bar{\tau}_D$ (\$/t)	$\bar{\tau}_P(0)$ (\$/t)	$\bar{\tau}_P(a_\Delta)$ (\$/t)	$\bar{\tau}_P(0) - \bar{\tau}_D$ (\$/t)	$\bar{\tau}_P(a_\Delta) - \bar{\tau}_D$ (\$/t)
Panel A. All fields, by vintage										
Producing by 2016	16,746	1,072.9	463.1	278.0	26.7	–	283.9	110.6	–	–
Delay cohort	5,438	260.7	120.6	87.9	50.8	94.0	149.5	123.9	41.3	16.9
Approved 2030 or later	2,429	211.8	90.6	90.6	39.7	82.0	104.1	–	24.3	–
Never developed	723	10.1	–	–	350.8	−18.8	76.6	–	93.3	–
Total, excluding never developed	24,613	1,545.4	674.3	456.6	34.0	85.9	196.9	112.3	31.2	21.3
Panel B. Delay cohort, by supply segment										
Onshore conventional	1,567	79.2	36.9	27.3	42.4	118.5	160.1	144.2	44.1	16.5
Offshore shelf	616	58.7	25.4	18.9	66.7	114.8	163.8	154.4	49.0	29.9
Offshore midwater	230	13.3	6.2	4.4	77.6	87.7	153.3	130.8	47.8	33.7
Offshore deepwater	219	45.5	21.1	14.1	83.6	88.6	146.8	123.9	51.6	37.7
Shale and tight oil	2,756	56.6	26.8	20.0	3.5	61.9	81.3	64.6	13.1	1.3
Oil sands	50	7.3	4.2	3.2	59.4	57.2	95.5	70.9	31.9	11.6
Panel C. All developed fields, by region										
Middle East	1,507	711.3	295.1	224.6	24.9	93.4	304.9	137.5	26.7	21.3
North America	9,356	360.5	162.1	106.1	4.5	58.8	88.1	52.6	18.6	6.7
Russia	2,891	146.7	66.1	40.7	61.7	100.6	208.1	107.6	45.6	25.4
Asia	3,663	115.2	51.1	28.8	63.0	82.3	220.7	93.4	49.3	39.1
South America	2,340	102.3	47.9	30.3	73.1	81.1	139.8	108.0	47.7	36.7
Africa	2,035	78.0	38.3	20.2	61.7	82.3	217.3	116.5	44.4	33.1
Europe	2,161	26.8	11.8	4.9	66.7	97.4	211.3	113.9	53.2	29.7
Oceania	660	4.7	1.9	1.1	47.5	126.5	188.9	140.2	32.6	22.5

Table 2: Summary statistics.

Notes: Medians are weighted by 2016 reserves and are not additive: the two depth columns are computed field by field. Reserve-weighted means give the same ordering of supply segments in Panel B. Emissions are the lifecycle emissions of Rystad's baseline production, without a carbon tax, after 2016 and after 2030. $\bar{\tau}_P(a_\Delta)$ is the production breakeven in 2030, when the delayed tax is implemented. Taxes are 2030 levels of the increasing path, in $\$/tCO_2eq$. The delay cohort is the set of fields on which some development takes place during the delay (start-up after 2016 and approval by 2029). Never-developed fields fail their development test even without a tax, so they emit nothing on the baseline path and are left out of the total of Panel A.

Three features of the table matter for what follows. First, the delay cohort is almost twice as capital-intensive as the fields already producing in 2016, with a median κ of \$51 per barrel of capacity against \$27, and its lock-in zone is deeper, \$41 per tonne at start-up against \$24 for the fields approved from 2030 onwards. Second, within the cohort, capital intensity separates shale from the rest. At \$3.5 per barrel of capacity, shale has a zone only \$13 deep at start-up, and by 2030 the zone of the median shale barrel has almost closed. The capital-intensive segments, at \$42–84 per barrel of capacity, have zones of \$32–52 at start-up

and still \$12–38 in 2030, the deepest being those of deepwater projects. The development breakevens of the median deepwater and midwater projects, about \$88, lie between the timely tax, \$80.4 at its 2030 level, and the lock-in-adjusted tax of 2030, \$98.4, which is why offshore lock-in is only partly activated at the timely tax (Section 4.1). Third, the regional panel reads the same way. North America, where shale and oil sands are concentrated, has a median production breakeven of \$88 at start-up and \$53 in 2030, so most of its output is retired by the tax in force after 2030, while South America and the Middle East have medians of \$140 and \$305. The regional medians describe where a region's production sits relative to the tax; where its lock-in sits is discussed in Section 5.1.

3.2 Field-level carbon intensities

Rystad's UCube data contain no carbon intensity. I estimate a lifecycle carbon intensity for each asset by combining upstream, midstream and downstream emissions. Appendix C gives the full procedure. For upstream emissions, I use the input data of Masnadi et al. (2018) for 737 oil fields, matched to 1,136 Rystad assets, and run the Oil Production Greenhouse Gas Emissions Estimator (OPGEE). The main source of upstream heterogeneity is flaring, which is not reported at the asset level. I use flared volumes derived from the satellite observations of NOAA's VIIRS instruments over 2012–2024 (Elvidge et al., 2016), matched to Rystad assets following Ghosh et al. (2025), and predict flaring-to-oil ratios for assets whose production had not started by 2025. A regression on flaring, steam-to-oil and gas-to-oil ratios, offshore status and oil type then assigns an upstream intensity to every remaining asset, including those not yet developed. For midstream and downstream emissions, I use the Petroleum Refinery Life-Cycle Inventory Model (PRELIM) and the Oil Products Emissions Module (OPEM), matching 518 publicly available crude assays to Rystad crude streams and assets, which cover 64.4% of 2024 production, and predicting intensities for unmatched assets from API gravity, sulphur content and predicted refinery configuration. Summing the three stages gives one lifecycle intensity per asset. In 2024, the production-weighted average is 466 $kgCO_2eq/bbl$, with a weighted standard deviation of 112. Weighted by 2016 reserves, upstream emissions account for 13% of the lifecycle total, refining for 7% and the transport and combustion of oil products for 80%. Carbon intensity does not enter the per-barrel depth of the lock-in zone: it sets the height of the zone, and converts barrels into emissions. The sensitivity analysis replaces the asset-level estimates with a uniform value to show how much of the result depends on them.

3.3 Costs, prices and discounting

Development and production costs. Only the expenses attributed to oil operations are kept. The development cost of an asset, the empirical counterpart of C, is the facility and well capital spent up to two years after the start of production, following Rystad's definition; for shale and tight oil, where drilling recurs throughout the life of an asset, only facility capital counts as development. Production costs include all operating expenses and the capital spent after the first three years of production. The capital spent in these three years, from the start-up year to two years after it, is counted as committed at start-up, so that the decision to develop a field and the decision to keep it producing never apply to the same year (Appendix B.1). Development expenses are discounted from the year in which they are incurred, so the years between the first expenses and the first barrel weigh on the value of a project. Appendix B.1 details each cost category, the timing of development expenses and the construction of production costs.

The two-year rule matters for lock-in, because it determines how much of a field's capital counts as sunk once production starts. Section 4.5 tests a stricter rule, which counts only capital spent before start-up as development. It makes every lock-in zone smaller and therefore gives a lower bound on the residual overshoot. Under this more restrictive rule, results remain close to the main specification, with a residual overshoot of 12.6 $GtCO_2eq$ and a premium of \$3.5.

Prices. The reference Brent price is constant at \$80/bbl. What this paper measures is driven by investment decisions taken during the delay, so the relevant price is the long-run price investors use to screen those projects, not the price of any particular year.[5] Two arguments support this level. First, the stability of the oil market rests mainly on the three core countries of OPEC, Saudi Arabia, Kuwait and the United Arab Emirates (Okullo and Reynès, 2016; Pierru et al., 2018), and the IMF puts the fiscal breakeven price of Saudi Arabia, the price at which it balances its budget, at \$80/bbl in the recent past and \$84/bbl in projection, and those of the other two lower (International Monetary Fund, 2024). These countries are also those a carbon tax affects least, since their fields combine low costs and low carbon intensities, so their fiscal breakeven is a reasonable anchor for the long-run price. Second, \$80/bbl keeps the investment margin studied here active: lower prices reduce the number of economically viable projects, even without a carbon tax. Although the price level

[5]Price volatility is not modelled for the same reason. OPEC's use of its spare capacity reduces volatility (Pierru et al., 2018; Fattouh, 2007), Berntsen et al. (2018) find that volatility has no significant effect on oil and gas investment in Norway once the expected price level is accounted for, and Tang et al. (2017) argue that volatility changes the timing of investment rather than the decision to invest.

sets the pool of viable assets, changing it moves the tax that meets the budget more than the overshoot, because both policies are calibrated on the same budget. Finally, since the price does not respond to the tax, the tax levels reported in the paper are those borne by producers. Section 4.5 adds a demand response and uses Brent prices of \$60 and \$100/bbl.

Asset-specific price discounts. Reference prices such as Brent describe one particular crude stream, and each asset sells its oil at a discount or a premium relative to that reference, which depends on crude quality. Every oil type sells at a discount on average, which reflects the high quality of the crudes constituting the Brent stream. Discounts are also very heterogeneous: heavier oils, which are more difficult to refine into the same products, carry larger discounts, from about 3% for light crudes to 29% for bitumen, and NGL sells at roughly half the price of oil. The reserve-weighted average discount is 12.9%, so that with a reference price of \$80/bbl the average selling price in the model is closer to \$70. Appendix Table 5 reports the discounts by oil type, and the sensitivity analysis replaces them with a common selling price across assets.

Discounting. Net present values are computed at a 10% discount rate. This is the most common rate used in the oil and gas industry to screen projects, the rate at which the industry reports its breakeven prices, and the one Rystad uses (Bailey et al., 2000; Jahn et al., 2008; Wood Mackenzie, 2013). It is also the rate at which listed producers must discount the future cash flows of their proven reserves in their financial statements, the so-called standardized measure or PV-10 (Financial Accounting Standards Board, 2010). The model of Section 2.2 says in which direction this choice matters. A higher rate lowers the value of developing a field without changing the value of keeping it in production, so it deepens and widens every lock-in zone and raises both the premium and the residual overshoot. The sensitivity analysis reports results at 5% and 15%.

3.4 From the model to the data

The empirical model is a sequence of yearly decisions taken for each asset independently of the others: whether to invest in the development of an undeveloped asset, and whether to keep an active asset in production.

Investment decision. An asset is developed if the expected net present value of the project is non-negative, given what investors anticipate about the carbon tax τ:

$$\sum_{t=T_1}^{T} \frac{P(1-d_i)\,q_{it} - c_{it}}{(1+\delta)^{t-T_D}} - \sum_{t=T_D}^{T_1-1} \frac{C_{it}}{(1+\delta)^{t-T_D}} - \sum_{t=T_\tau}^{T} \frac{\tau\,\theta_i\,q_{it}\,(1+r)^{t-2030}}{(1+\delta)^{t-T_D}} \geq 0, \tag{12}$$

where $T_D = T_1 - \min(T_1 - T_A, 6)$ is the first year of development, T_A the approval year, T_1 and T the first and last years of production, C_{it} the development expenses of year t, T_τ the anticipated year of implementation, $P(1-d_i)$ the reference price net of the asset's discount, c_{it} total production costs in year t, θ_i carbon intensity, τ the 2030 level of the anticipated tax and r its increase rate. Appendix B.1 explains how expenses incurred before T_D and in the first years of production are assigned to T_D and $T_1 - 1$. Development is assumed to start in the approval year, but no more than six years before first production, the 90th percentile of the time between approval and first oil in the delay cohort, weighted by 2016 reserves. The cap binds for a tenth of the cohort's reserves, mostly projects whose development stopped and resumed years later; their earlier expenses are counted in the first year of the window, so that a project is not valued as if its revenues came a decade or more after the decision to develop it. Investors choose T to maximise the value of the project. Rearranging (12) gives the maximum anticipated tax consistent with a non-negative net present value, the empirical counterpart of the development breakeven $\bar{\tau}_D(\kappa)$ of equation (5).

Development often takes several years, and in some cases can stop and resume later. The maximum viable tax is therefore computed at each development stage, on the expenses that remain to be spent, and the asset is developed only if the anticipated tax is below it at every stage. When the tax is implemented, an asset under development that no longer clears its remaining-cost threshold is abandoned, and the capital already spent is lost. For a given tax, the first year in which the remaining-cost threshold exceeds the tax is the year of investment responsible for the asset's lock-in (Appendix B.1).

Production decision. Once developed, an asset produces in a given year if the price net of the tax covers the breakeven of its current stage of production, and is phased out for good in the first year in which it does not. Breakevens rise from one stage to the next, because both operating costs and capital expenses per barrel rise as production declines. Appendix B.1 describes how the stages are built. The breakeven of the current stage is the empirical counterpart of $\bar{\tau}_P(a)$ in equation (2).

Investment and production paths. The path of investment and production of each asset is the one observed up to 2025 and forecast by Rystad thereafter. Production rates depend mainly on the initial investment and cannot be changed at will afterwards, because of geological and technological constraints; changing the path would mean changing development expenses and all subsequent costs. During the delay, the tax is not, or only partially, anticipated, which makes the path forecast by Rystad the relevant path for the delay. A field's production profile is therefore taken as given from its start-up year.

3.5 Scenarios, budget and parameters

The analysis compares a scenario in which the tax is implemented in 2016, the timely policy, with one in which it is delayed to 2030 without being anticipated by investors, which is equivalent to business as usual until 2030. These are the dates of the delay since Paris, but nothing in the model is specific to them: any delay in carbon pricing locks in emissions in the same way, and Section 4.3 repeats the exercise for every year of implementation between 2016 and 2030 and for a delay that starts in 2025. From its implementation, the tax rises at 2% per year, the social discount rate. Under a cumulative carbon budget, the cost-effective tax rises at the discount rate (Dietz and Venmans, 2019), while Lemoine and Rudik (2017) obtain a slower-rising path when warming lags emissions, which a cumulative budget abstracts from. A social discount rate below market rates can be justified on ethical grounds (Stern, 2007) or by uncertainty about growth and interest rates, which lowers the certainty-equivalent rate at long horizons (Weitzman, 1998; Gollier, 2012). This literature points to rates between 1% and 2% in the distant future, while Nordhaus (2007) argues for a market-based rate of about 3%. I adopt 2%, the upper end of the first range, and the sensitivity analysis uses rates from 1% to 6%. All tax levels in the paper are reported at the value the tax reaches in 2030, whatever the implementation date, so that timely and delayed policies can be compared directly; a level of x corresponds to $x/1.32$ in 2016.

Anticipation scenarios. I explore alternative anticipation scenarios in which investors either expect the tax to be put in place at any point in time, thus screening their project assuming implementation from the next year, or correctly anticipate the delay until 2030. I also explore scenarios in which the right tax level is underestimated. Appendix B.3 defines them, Section 2.5 maps them onto a single anticipated tax τ^A, and Section 4.4 reports the results.

The 1.5°C-aligned budget corresponds to emissions that Welsby et al. (2021) allocate to oil, 747 Gbbl over 2018–2100, extended to 2016–2100 with the observed production of 2016 and 2017 and converted with the downstream carbon intensity of their model, the IPCC

default of 430 $kgCO_2$/bbl (IPCC, 2006).[6] The resulting budget is 350.3 $GtCO_2eq$ from 2016. Under a timely policy, the tax that meets it starts at \$60.95/tCO_2eq in 2016 and reaches \$80.4 in 2030. Under the delay, the budget remaining at implementation, B_Δ, is the initial budget net of the 217.7 $GtCO_2eq$ released between 2016 and 2029 on the business-as-usual path, that is 132.6 $GtCO_2eq$.[7] Leakage towards other fossil fuels is not modelled, and the results should be read as if similar policies applied to other fuels at the same time. The sensitivity analysis varies the budget between 75% and 125% of its central value.

An asset's excess emissions under the delay are counted on the intensive margin if the asset produces under the timely policy or was already producing in 2016, and on the extensive margin, as a delay-induced field, if it does not produce under the timely policy. Since expectations affect only investment decisions, the anticipation scenarios differ only at the extensive margin.

3.6 Fiscal regimes

Government take is excluded from costs, except for taxes reported as opex by companies, which producers treat as operating costs unrelated to profitability. Profit taxes change the size of an asset's net profitability but not its sign, and bonuses are paid before development decisions are made. The key assumption concerns royalties, which are levied on gross revenue and can therefore make a project unprofitable. I assume that, in the presence of a worldwide carbon tax, governments renegotiate royalty rates so as to keep viable assets in production and keep collecting a share of the rent, without subsidising assets that are no longer viable. Two cases bound this assumption. On the one hand, governments could absorb the whole carbon tax by lowering royalties.[8] On the other hand, contractual rigidities in existing production-sharing agreements and concession contracts may prevent renegotiation in the short term, causing the carbon tax to pile on top of the existing fiscal regime and further constrain investment and production beyond what the model predicts. The assumption made in this paper implicitly lies between these two extremes: governments are assumed to

[6] This factor covers combustion only, while the emissions of each field here also include extraction and refining. It is, however, higher than the downstream intensity of the sample, 375 $kgCO_2eq$/bbl on average weighted by 2016 reserves, and close to its lifecycle intensity, 466 $kgCO_2eq$/bbl (Section 3.2). Converted at the first, the same barrels would give a budget 13% smaller, and at the second a budget 8% larger. The residual overshoot defined in Section 2.4 is 13.2 $GtCO_2eq$ with a budget 10% smaller and 12.6 $GtCO_2eq$ with a budget 10% larger (Section 4.5), so the choice of conversion factor matters little.

[7] B_Δ changes in the anticipation scenarios and under early restrictions, which alter emissions before 2030.

[8] A government at its Ramsey optimum could in principle respond to a worldwide carbon tax by adjusting fiscal regimes so as to leave the equilibrium unchanged, substituting carbon tax revenue for royalty revenue. Fully restoring the pre-tax equilibrium would, however, in some cases require subsidising development or production.

absorb the carbon tax to the extent necessary to keep viable assets in production (potentially capturing a higher rent in the process) without subsidising uneconomical ones, so that assets not resilient to the carbon tax are either phased out earlier than in a no-tax world or simply remain undeveloped.[9] The same assumption makes Rystad's forecast paths, which embed current fiscal terms, the relevant counterfactual without a carbon tax.

4 Results

All results in this section rely on a tax rising at 2% per year, under the no-anticipation scenario, with a Brent price of \$80/bbl. The timely tax, \$80.4 in 2030, is the anchor used for every tax below.

4.1 The cost of the delay

Figure 3 reports, for any level of tax, the overshoot of the carbon budget: the excess of cumulative emissions under the delay over cumulative emissions under the timely policy. The overshoot is split into the cells of the decomposition of Section 2.3: (i) emissions released before 2030 by fields that produce under both policies but that a timely tax would have retired earlier (the intensive margin), (ii) emissions released before 2030 by delay-induced fields and (iii) emissions released after 2030 by delay-induced fields that the tax arriving in 2030 cannot stop. Appendix Figure 22 draws the same bands on top of cumulative emissions under the timely policy.

The overshoot rises with the tax, slowly below about \$20 and steadily above. At low levels, a tax removes little production, during the delay or after it, because oil fields earn large rents. At the tax that meets the budget, \$80.4/tCO_2eq, the delay overshoots the oil carbon budget by 71.1 GtCO_2eq, 20.3% of the budget and about four years of global oil emissions at their current rate. Of this overshoot, 49.8 GtCO_2eq are released before 2030 by fields on the intensive margin, while the extensive margin represents 11.0 GtCO_2eq before 2030 and 10.3 GtCO_2eq after. Delay-induced fields, those that exist because of the delay, therefore account for 30% of the overshoot, and their emissions after 2030 alone for 14%. Six sevenths of the overshoot is already in the atmosphere when the tax arrives, in 2030. The

[9] On the interaction between extraction taxes and rent capture, see Daubanes and Lasserre (2023), on the response of investment to oil taxation, Ahlvik and Harding (2025), Ahlvik et al. (2026) and Stefanski et al. (2025), and on compensation for halted upstream projects Paletta et al. (2025).

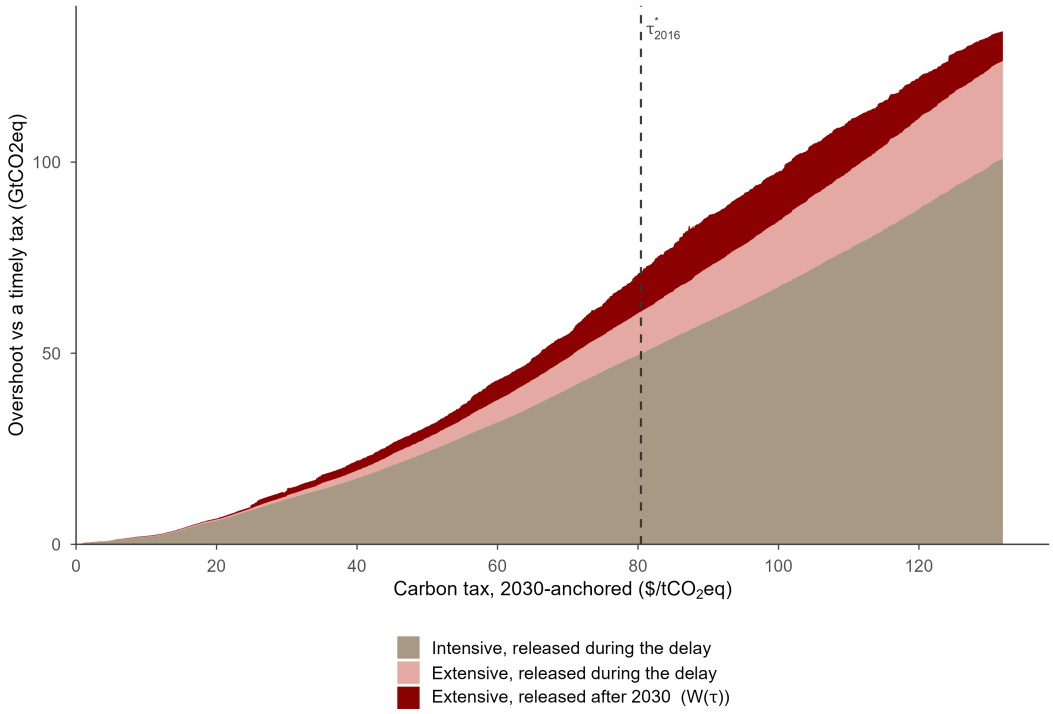


Figure 3: Carbon budget overshoot caused by the delay, by margin and date of emission.

Notes: The horizontal axis is the 2030 level of the tax rising at 2% per year; the dashed line is the tax that meets the budget under a timely policy, $\tau^*_{2016} = \$80.4$. Intensive emissions after 2030, the fourth cell of the decomposition, are zero and have no band. The top band read at \$80.4, 10.3 GtCO$_2$eq, counts the post-2030 emissions of delay-induced fields.

rest of the paper focuses on emissions after 2030, which a policy implemented in 2030 can still act on.

The 10.3 GtCO$_2$eq of the top band only count the fields that the timely tax, \$80.4, would have deterred. The tax set after the delay is higher, and falls inside the lock-in zone of projects that a timely policy would have built, mostly offshore projects whose development breakeven lies between the two taxes. Lock-in measured at the taxes the budget requires after the delay is therefore larger, about 13 GtCO$_2$eq, and Section 4.2 uses this measure.

Supply segments and regions. The overshoot originates from different kinds of fields (Figure 4), whose participation differs widely. Shale and oil sands are shut down by the tax as soon as it arrives, whether in 2016 or in 2030. Within the delay cohort, their production breakevens are the lowest of any segment, with medians of \$65 and \$71 in 2030, below the timely tax (Table 2). Under a timely tax, the two segments would have emitted 7.7 and 0.8 GtCO$_2$eq from 2016 onwards. The delay lets them produce untaxed until 2030 and adds 27.2 and 7.7 GtCO$_2$eq, half of the overshoot, almost all of it released before 2030: shale alone carries 44% of the intensive overshoot, the largest share of any segment, and the two segments

together only 4% of the emissions released after 2030. Offshore fields are the opposite case. Their zones are deep and sit high, with median production breakevens of $124–154 in 2030, so a delay-induced offshore field survives the tax and keeps producing for years: offshore fields carry 76% of the 10.3 $GtCO_2eq$ released after 2030 at the timely tax, and deepwater fields alone 45%. Onshore conventional fields contribute mostly through the fields already producing in 2016. The segment pattern translates into a large geographic heterogeneity (Appendix Figure 24). North America, where shale and oil sands are concentrated, accounts for 69% of the intensive overshoot and for very little of the post-2030 band, while South America, with the deepwater developments off the coasts of Brazil and Guyana, carries 38% of the post-2030 band, the largest regional share.

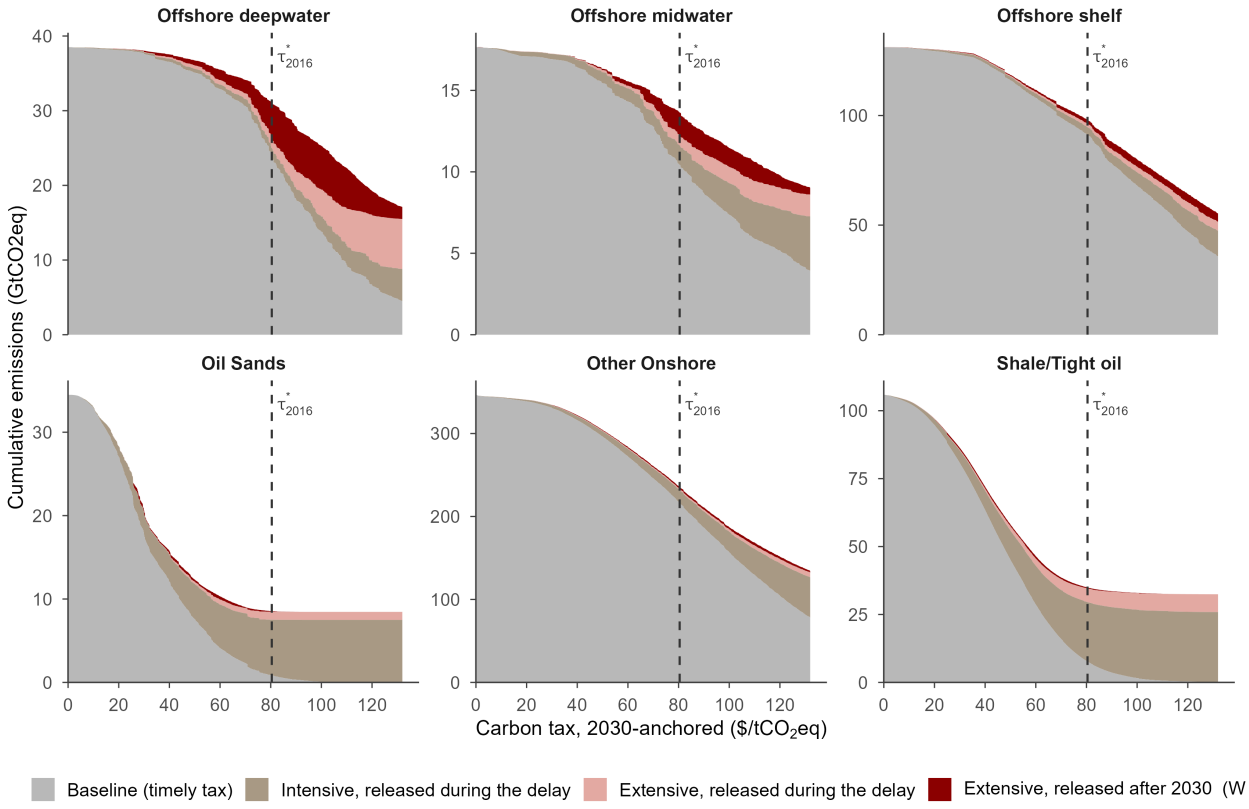


Figure 4: Overshoot caused by the delay, by supply segment.

Notes: Cumulative emissions under the timely policy (grey) and overshoot caused by the delay by margin and date of emission. Panels have different vertical scales. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2030 levels of the tax rising at 2% per year. Appendix Figure 23 shows the overshoot bands alone.

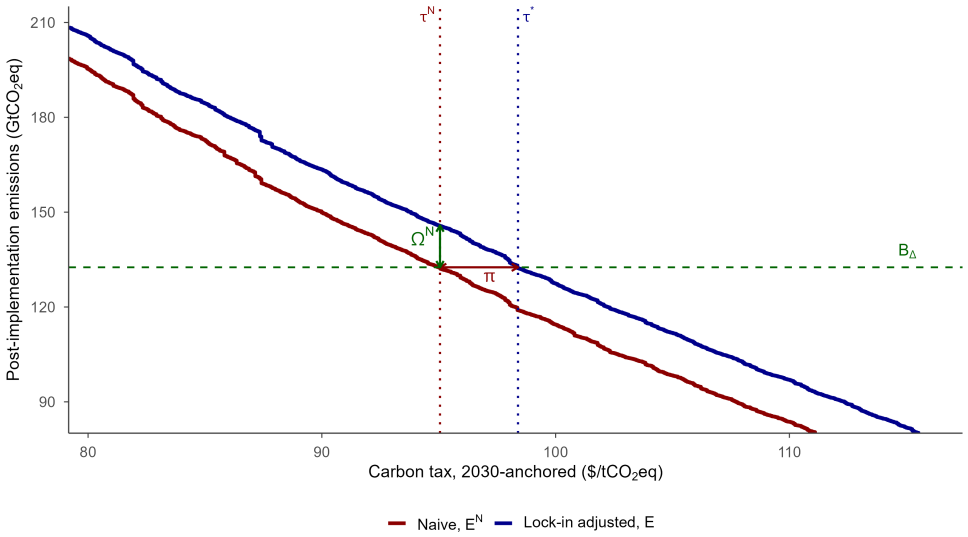
4.2 The residual overshoot and the lock-in premium

Overproduction during the delay raises the level of tax needed to meet the budget from 2030. The 217.7 $GtCO_2eq$ released between 2016 and 2029, against 156.8 under the timely policy, leave 132.6 $GtCO_2eq$ for the rest of the century. Investment during the delay does something more: it locks in reserves that keep producing under high taxes, so that a given tax removes fewer emissions than it would have before the delay. The tax has to rise for both reasons, and a naive planner using a single full-cycle cost per asset only sees the first.

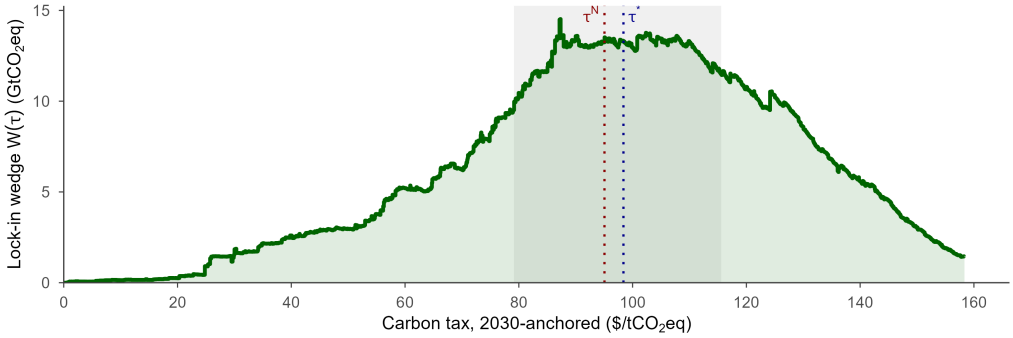
From here on, a field is locked in at a tax τ if it was developed during the delay, would have been deterred by a tax at that level applied from 2016, and still produces after 2030 under τ. At the timely tax, these are the delay-induced fields of Section 4.1. At the higher taxes the delay makes necessary, they also include projects that the timely policy would have built, because the tax reaches their lock-in zone.

Panel (a) of Figure 5 reports post-implementation emissions under the two calibrations: the naive one, E^N, is computed from full-cycle breakevens, as if the fields developed during the delay could still be deterred, and the lock-in-adjusted one, E, from their actual breakevens as of 2030. Both are compared with the remaining budget $B_\Delta = 132.6$ $GtCO_2eq$, drawn as a horizontal line. The line is flat because no tax is levied before 2030, so the choice of the 2030 tax does not change what has already been emitted. The naive planner sets the tax at which E^N meets the budget, \$95.1. At that tax, emissions after 2030, computed with the breakevens the fields actually face, exceed the budget by $\Omega^N = 13.2$ $GtCO_2eq$: this residual overshoot is 10% of the budget remaining in 2030. The two curves are close to parallel over the range shown, about 10 to 14 $GtCO_2eq$ apart.

Panel (b) draws the gap between the two curves, the lock-in wedge $W(\tau)$, over the whole range of taxes, and it has the hump shape described in Section 2.3. It is zero at a zero tax, which deters no field. It rises as the tax draws the capital-intensive fields of the cohort into their lock-in zone, with steps where large fields enter, and forms a plateau of about 13 $GtCO_2eq$ between about \$86 and \$110. Above \$110 it declines: a higher tax now shortens the post-2030 life of locked-in fields, or stops them, faster than it draws new fields into their zone. The naive tax sits on this plateau, so the residual overshoot does not depend on exactly where the budget places it. The plateau may look narrow in tax terms, but it is wide in terms of emissions, because each additional dollar of tax removes 3 to 4 $GtCO_2eq$ of emissions around the naive tax: a budget 10% larger or smaller leaves the residual overshoot within 1 $GtCO_2eq$ of its central value (Section 4.5).



(a) Post-implementation emissions under the full-cycle and lock-in-adjusted calibrations



(b) Lock-in wedge $W(\tau) = E(\tau) - E^N(\tau)$

Figure 5: Full-cycle and lock-in-adjusted calibration of the 2030 tax.

Notes: (a) Post-implementation emissions computed from full-cycle (E^N) and operating (E) breakevens of the fields developed during the delay, against the remaining budget $B_\Delta = 132.6$ GtCO$_2$eq; the dotted lines mark $\tau^N = 95.1$ and $\tau^* = 98.4$. The premium π is the horizontal distance between the two curves at the budget, the residual overshoot Ω^N the vertical distance at τ^N. (b) The lock-in wedge $W(\tau) = E(\tau) - E^N(\tau)$ as a function of the tax; the dotted lines mark the naive tax, at which the wedge gives Ω^N, and the lock-in-adjusted tax, and the shaded band is the range of taxes shown in panel (a). Taxes are 2030 levels of the tax rising at 2% per year.

The lock-in premium. The tax at which E meets the budget, the lock-in-adjusted tax, is \$98.4. The premium is the difference between the naive and the lock-in-adjusted taxes, π = \$3.3/tCO$_2$eq, 3.5% of the naive tax, and it measures the same error as the residual overshoot, in dollars instead of tonnes. Between the two taxes, each dollar removes on average 4.0 GtCO$_2$eq of post-implementation emissions, so that a premium of \$3.3 closes a gap of 13.2 GtCO$_2$eq (equation (11), written with the average slope). The premium looks small next to the tax it adds to for two reasons. Because the oil price does not respond to the tax in the main specification, the tax studied here is the part borne by producers. And because it applies to the lifecycle emissions of all the oil produced worldwide, each dollar

of it moves a large quantity of emissions. The same division makes the premium the less stable of the two measures. It moves with the slope of post-implementation emissions around the naive tax, even when the residual overshoot does not: it falls between 2028 and 2030 while the residual overshoot keeps rising (Section 4.3), and it varies more than the residual overshoot across specifications (Section 4.5).

The cost in dollars. Both the timely and delayed policies meet the same budget, so the cost of the delay can also be measured as what the oil sector loses by meeting it later, with the tax set at the lock-in-adjusted level after the delay. Holding the budget fixed, the delay reduces the surplus of the sector, its revenue net of operating and development costs and before carbon tax payments, by $1.6 trillion in present value, and the sector delivers 11.2 Gbbl less oil for the same emissions.[10] The loss comes from meeting the budget in the wrong order, through two channels. First, the delay lets costly and carbon-intensive fields produce untaxed for as long as it lasts, and the higher tax that follows strands cheaper ones: the Middle East produces 88 Gbbl less over the century, while North America produces 81 Gbbl more. Second, it leads to the development of capital-intensive fields that keep producing after 2030, in place of production that would have been cheaper overall. Put differently, the delay develops more fields than meeting the budget on time requires, and adds $482 billion of investment before 2030, in present value. Section 4.4 bounds the part of the loss that comes from investment.

The error of the naive planner can be valued too. At a social cost of carbon of $200 per tonne, constant in present value as is consistent with a carbon budget (Coulomb et al., 2026) and close to the central estimate of Rennert et al. (2022), the 13.2 $GtCO_2eq$ it leaves above the budget amount to $2.6 trillion of climate damages.[11]

4.3 How lock-in builds up

Figure 6 repeats the calibration of the 2030 tax for every year of implementation between 2016 and 2030, and Appendix Table 14 reports the numbers. Panel (a) splits the increase in the tax relative to the timely path into the part that offsets overproduction, $\tau^N - \tau^*_{2016}$, and the lock-in premium. The overproduction part grows with every year of delay, and faster towards the end, from $3.4 after six years to $14.6 after fourteen. The premium grows until 2028 and then falls slightly, and it accounts for between about a fifth and a third of the

[10] Present values in 2025, at the social discount rate of 2%.

[11] Valued instead at the tax the budget requires, which rises at the social discount rate and is therefore also constant in present value, at $89 per tonne in 2025, they amount to $1.2 trillion.

increase whatever the length of the delay. It does not rise steadily: it is flat between 2022 and 2024, and falls from \$3.5 to \$3.3 between 2028 and 2030 while the residual overshoot keeps rising. Equation (11) explains why: the premium is the residual overshoot divided by the response of emissions to the tax, and at these higher levels, that response grows faster than the residual overshoot.

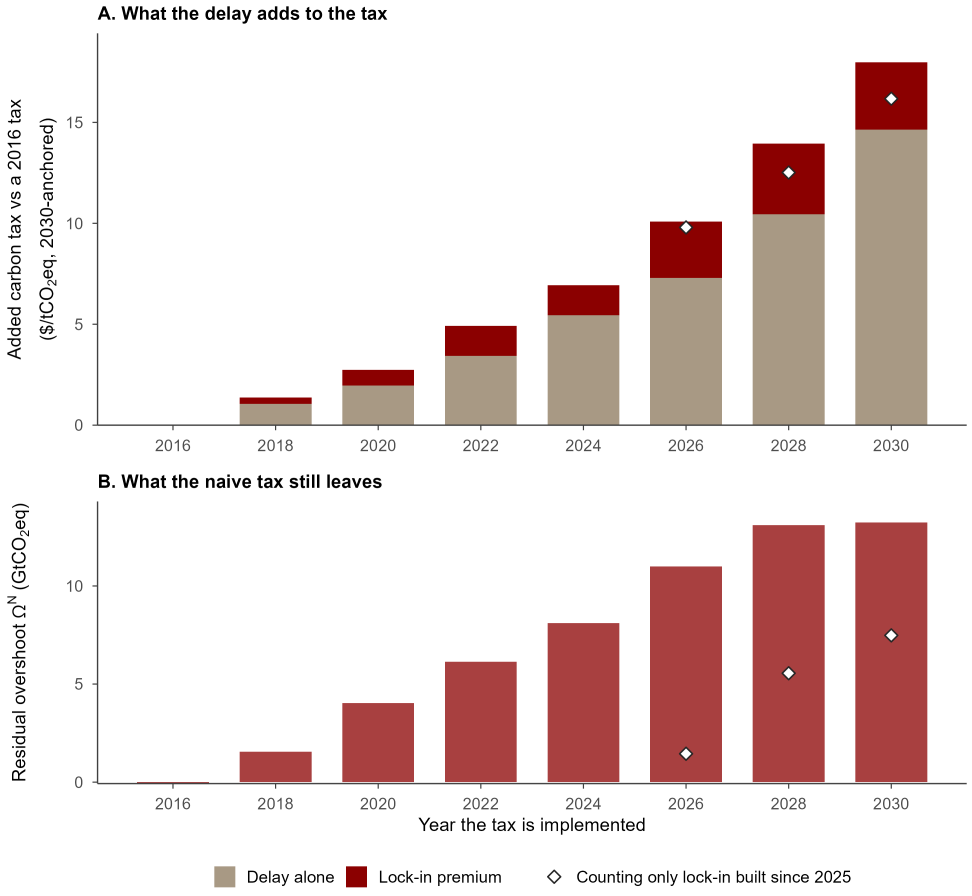


Figure 6: The price of delay, by year of implementation.

Notes: (a) Increase in the tax that meets the remaining budget over the timely tax τ^*_{2016}, split between the part that offsets overproduction during the delay, $\tau^N - \tau^*_{2016}$, and the lock-in premium $\pi = \tau^* - \tau^N$. (b) Residual overshoot Ω^N left by the naive tax. The diamonds measure the delay from 2025, the state of the industry in that year being taken as given, so that only the lock-in of investments made from 2025 counts. Taxes are 2030 levels of the tax rising at 2% per year.

Panel (b) shows that the residual overshoot grows roughly linearly until 2028 and barely afterwards. It reaches 6.1 $GtCO_2eq$ after six years, 13.1 $GtCO_2eq$ after twelve and 13.2 $GtCO_2eq$ after fourteen. Two opposite forces are at work. On one side, each additional year of delay adds new fields that will produce for decades after implementation. On the other side, it also ages every field already built, so that commitments made earlier are partly emitted during the delay rather than after it, and the higher tax needed shortens the lifetime of locked-in fields. As the delay lengthens, the second force slows the build-up. Short delays

therefore matter, because the first force dominates while the delay is short. The diamonds show that a five-year delay starting in 2025 would already leave 7.5 $GtCO_2eq$ above the budget if the lock-in of the investments made in 2025–2029 were ignored, 5.6% of what remains and more than half of the residual overshoot after fourteen years, with a premium of $1.8.

Figure 7 traces the source of lock-in by the timing of investment rather than by the year the tax is implemented. It reports, by supply segment and period of the delay, the post-2030 emissions committed at the naive tax by fields developed during the delay. Each field is attributed to the year of the investment after which its full development can no longer be deterred at that tax (Section 3.4). Most of the residual overshoot is decided late: investments made in 2025–2029 commit 56% of it, against 33% for 2020–2024 and 12% for 2016–2019. Deepwater projects drive this profile. Their contribution to the residual overshoot rises from 0.4 $GtCO_2eq$ in 2016–2019 to 1.6 $GtCO_2eq$ in 2020–2024 and 3.6 $GtCO_2eq$ in 2025–2029, almost half of what is committed in that last period. Overcommitted emissions are also the largest share of commitments in deepwater and midwater projects, as in the model, while onshore conventional projects commit the most in absolute terms, mostly emissions that a timely tax would have let through as well. Shale and oil sands commit almost nothing that survives the naive tax.

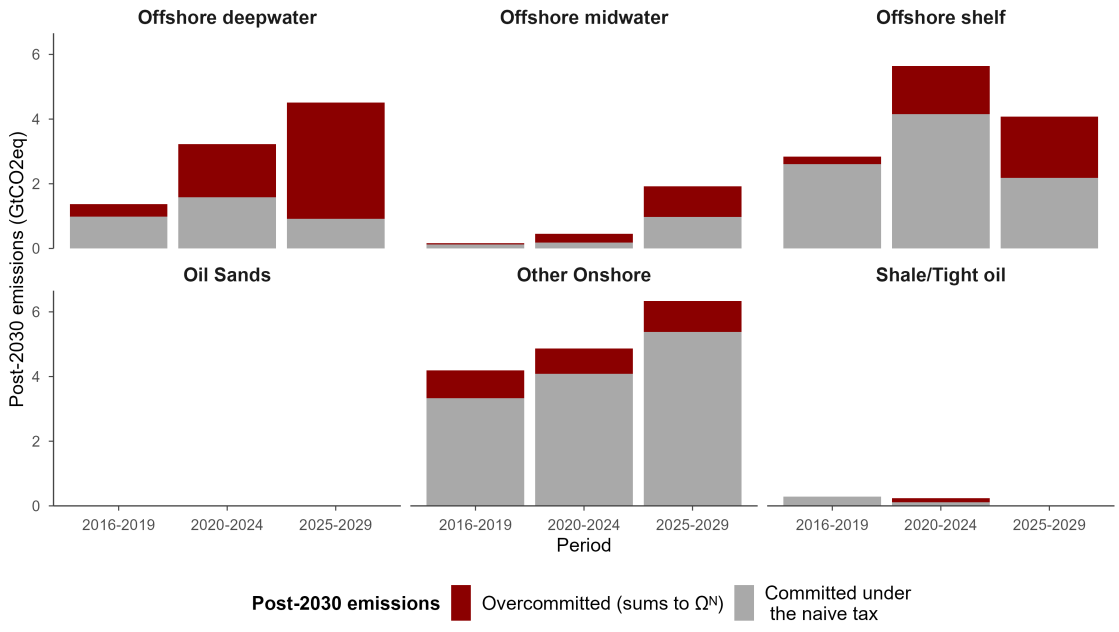


Figure 7: Committed and overcommitted emissions at the naive tax, by supply segment and period of investment.

Notes: Post-2030 emissions committed by investments in new oil assets of each segment during each period, at the naive tax $\tau^N_{2030} = \$95.1$ (2030 level of the tax rising at 2% per year). Emissions are attributed to the year from which the full development of the asset can no longer be deterred at that tax. The grey part would also be committed had a tax at the naive level applied from 2016; the red part would not, and it sums to the residual overshoot Ω^N.

4.4 The role of investors' expectations

The results so far assume that investors expected no tax during the delay. Anticipation changes which fields are developed during the delay, but does not affect production decisions. Investors who expect part of the tax stop developing the projects that would not be worth building even at that anticipated tax, and these are the fields with the lowest development breakevens, typically the most capital-intensive ones, which carry the deepest lock-in (Section 2.5).

Figure 8 reports how the residual overshoot varies with two kinds of expectations. In the first, investors expect the tax to be implemented at any point in time, so they screen their investment assuming implementation the following year, at the level that a planner implementing it then would set, that is the naive tax of a delay ending that year. In the second, investors correctly expect the tax to come in 2030. In both cases, investors may also underestimate the level of the tax, and the horizontal axis reports the share of it they expect. Weak expectations change little. As long as investors expect at most 30% of the tax, the residual overshoot stays within 0.3 $GtCO_2eq$ of its value without anticipation, and it is still close to 12 $GtCO_2eq$ when they expect 60% of it. Weak expectations remove few projects, because at these tax levels most remain worth developing. It falls steeply only when investors expect more than about 70% of the tax, to 10.3 and 10.8 $GtCO_2eq$ at three quarters, and to 6.9 and 7.6 $GtCO_2eq$ at 90%.

Neither kind of expectation removes lock-in, even at its full level. Investors who know that the tax will come in 2030 and expect its full level still leave a residual overshoot of 6.1 $GtCO_2eq$, almost half of its value without anticipation, with a premium of \$1.9: many projects remain worth developing when the tax only starts in 2030. This part of lock-in is the one that no announcement of the 2030 tax could have prevented. Investors who expect the full tax the year after they invest leave 2.8 $GtCO_2eq$, a fifth of the residual overshoot. Early in the delay, the tax they expect is the one a short delay would require, below the tax that the full delay eventually requires, so projects that can bear the first but not the second still lock in. Lock-in disappears only if investors screen their projects with the tax the budget turns out to require, from the date they invest (Section 2.5), which no consistent expectation of the tax implies: investors who expect it in 2030 have no reason to price it before, and investors who expect it the following year expect the lower tax that a shorter delay would require.

This error is also costly for investors themselves, a private cost that the planner's accounting leaves out. Without anticipation, 249 of the 636 locked-in fields that still produce at τ^*_{2030} fall short of the 10% return used to screen them, and they carry more than half of

the residual overshoot.[12] Investors who expect the tax the year after they invest can still fall short, because the tax they expect is lower than the one the full delay requires. Only investors who correctly expect the full tax in 2030 bear almost no such loss: the 6.1 $GtCO_2eq$ they leave in place are a cost for society, not for them.

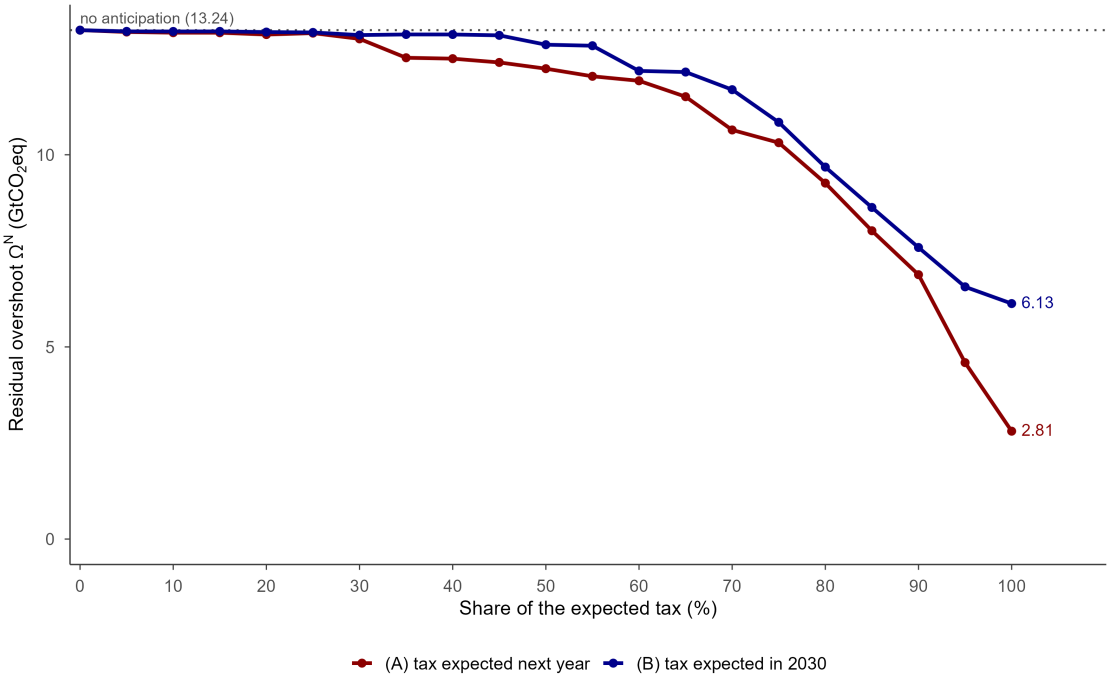


Figure 8: Residual overshoot, by share of the tax anticipated.

Notes: (A) Investors expect the tax to be implemented the year after their investment decision, at the share shown on the horizontal axis of the naive tax of a delay ending that year. (B) Investors expect the tax from 2030, at the same share of the naive tax of 2030. Expected levels come from the runs without anticipation of Figure 6. The dotted line is the residual overshoot without anticipation.

Which of these expectations describes the investors of the delay? Oil companies did report internal carbon prices after Paris. In 2016, Total used $28 per tonne, BP, Shell and Eni $40, and Statoil $50–64 (CDP, 2016). These levels are of the same order as the timely tax, which starts at $61 in 2016, but they apply to a much smaller base: companies charge them on the emissions of their own operations, that is on their share of a project's direct emissions (BP p.l.c., 2022). The tax studied here falls on lifecycle emissions, of which extraction accounts for 13% on average (Section 3.2). Converted to 2025 dollars and expressed per lifecycle tonne, the $40 used by most majors amounts to $6.9, about a tenth of the tax investors would have expected under the first kind of expectations, and the range across companies to 5–17%, depending on the company and the year. At these levels, the residual overshoot is that of investors who expect no tax at all (Figure 8). The targets that BP and

[12] Net present values at 10%, on the realised path: untaxed until 2030 and taxed at τ^*_{2030} afterwards. Of the 651 fields locked in at the naive tax, 636 still produce at τ^*_{2030}.

Equinor adopted later, about $100 per tonne by 2030, amount to about a sixth of the naive tax of 2030 on the same basis, in 2025 dollars, and the internal carbon fees that some firms actually charge their divisions, with a modal value of $10 (Fried et al., 2022), to at most a fifth of the expected tax, even when applied to every tonne. Firms invest according to the carbon price they find credible, and under policy uncertainty this price sits far below the announced one (Dolls et al., 2026). Expectations of this kind would have left almost all of the lock-in in place. Firms may also have anticipated climate policy through lower long-run oil price assumptions rather than through a carbon price, but whatever expectations they held are already embedded in the investments observed up to 2025, which form the path of the delay. A commitment to the 2030 tax, announced in 2016 and believed by investors, would have removed about half of the residual overshoot. The other half could only have been prevented by an earlier tax or by restrictions on the projects themselves, which Section 5.3 examines.

Expectations also have a value in dollars. With the budget held fixed, the surplus the oil sector loses because of the delay is $1,596 billion when investors expect no tax, $1,417 billion when they correctly expect the tax in 2030, and $1,132 billion when they price the full tax from the date they invest. Investment decisions during the delay therefore account for at most $464 billion, 29% of the cost of the delay. This is not an amount invested but a loss of surplus: the part of the $1.6 trillion that the sector would have kept had investors priced the full tax from the date they invest. It covers the production of delay-induced fields before 2030 as well as lock-in. The rest comes from the overproduction of fields already in operation, which no expectation can prevent, since no tax is levied on their production before 2030.

4.5 Robustness

Figure 9 reports the residual overshoot and the lock-in premium under alternative assumptions on costs and the computation of net present values, on carbon intensities and the carbon budget, on the design of the tax and, most importantly, on the response of oil demand, which lets the oil price move with the tax. None of them removes lock-in or reverses the results of the previous sections.

Costs and net present values. The different cost assumptions move the residual overshoot only marginally. As predicted by the model, a higher discount rate deepens every lock-in zone: the residual overshoot is 15.6 $GtCO_2eq$ at 15% and 10.4 $GtCO_2eq$ at 5%. Constant marginal costs remove the rising cost path that phases out locked-in fields over time,

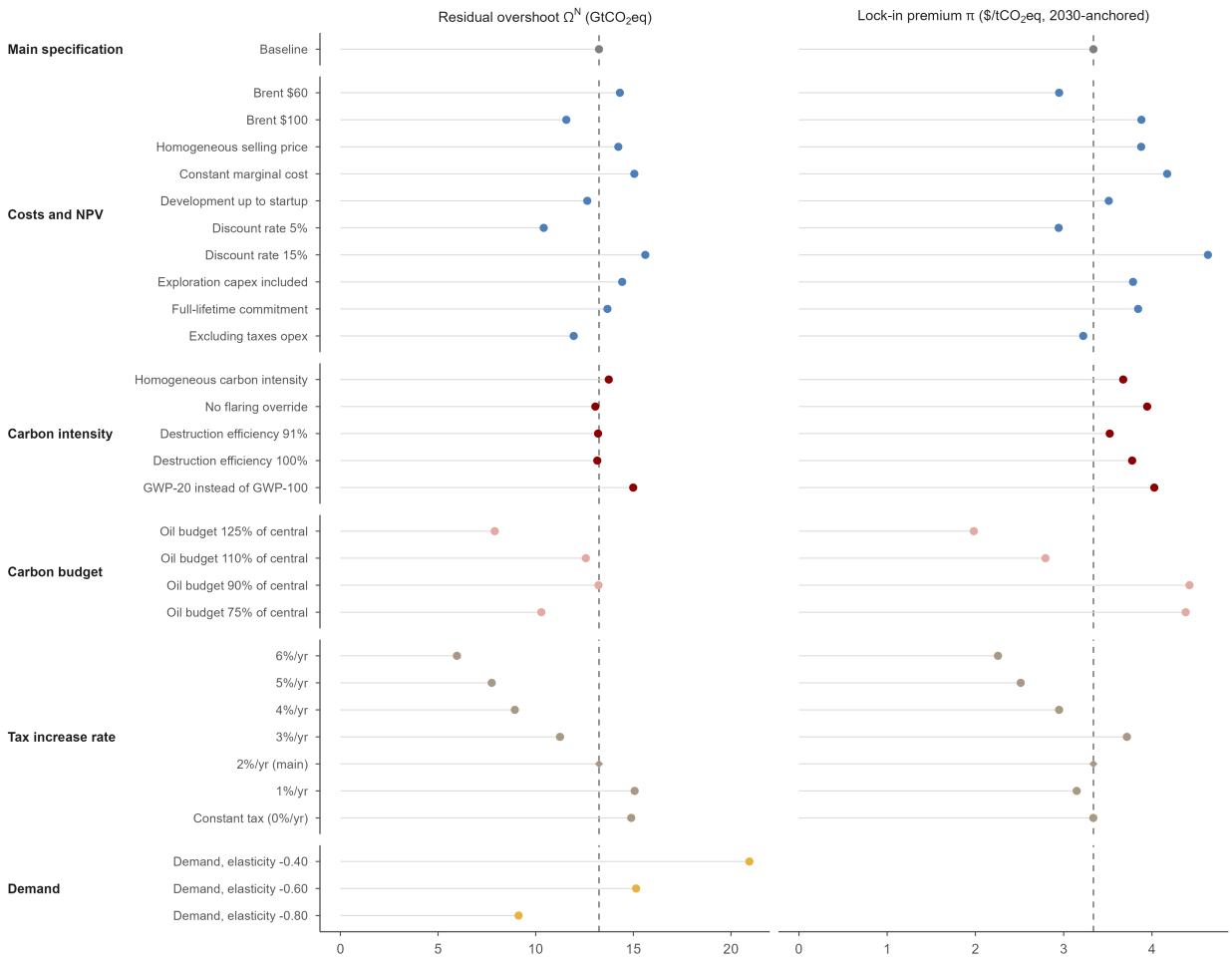


Figure 9: Sensitivity of the residual overshoot and the lock-in premium, by group of assumptions.

Notes: Each row is one specification; the main specification is marked by the dashed line. Premiums are 2030 levels of each tax path. The demand rows report the residual overshoot only (Section 4.5).

and raise it to 15.1 $GtCO_2eq$. Counting as development only the capital spent up to start-up makes every lock-in zone shallower and gives a lower bound on the residual overshoot, 12.6 $GtCO_2eq$, with a premium of $3.5 (Section 3.3). Adding the fields not yet discovered, with their exploration costs treated as development costs, raises it only to 14.4 $GtCO_2eq$, because most of them are not developed before 2030 and are stopped by the tax. A Brent price of $60 raises it to 14.3 $GtCO_2eq$ and a price of $100 lowers it to 11.6 $GtCO_2eq$. A common selling price across assets raises it to 14.2 $GtCO_2eq$, and leaving out of production costs the taxes that companies report as operating expenses lowers it to 11.9 $GtCO_2eq$. Ruling out early phase-out, so that every asset produces until the end of its forecast lifetime once developed and is valued accordingly, as in the accounting of committed emissions (Davis and Socolow, 2014; Tong et al., 2019), gives 13.7 $GtCO_2eq$.

Carbon intensities and budget. Assumptions on carbon intensity move the residual overshoot between 13.1 and 15.0 $GtCO_2eq$, the highest value coming from a 20-years global warming potential conversion, and a common carbon intensity across assets raises it to 13.7 $GtCO_2eq$, which is consistent with locked-in fields being on average cleaner than the rest of the sample. Varying the budget traces the hump of the wedge from another angle. The residual overshoot changes little around the central value, at 12.6 $GtCO_2eq$ with a budget 10% larger and 13.2 $GtCO_2eq$ with a budget 10% smaller. It falls with a much larger budget, 7.9 $GtCO_2eq$ at 125%, because the lower tax deters fewer fields in the timely counterfactual, and with a much smaller one, 10.3 $GtCO_2eq$ at 75%, because the higher tax starts to shut down locked-in fields themselves, as explained in Figure 5b.

Instrument design. The constant tax that meets the same budget under a timely policy is \$105.7/tCO_2eq, and the delay then overshoots the budget by 132.8 $GtCO_2eq$, 38% of the budget, against 71.1 $GtCO_2eq$ and 20% under the increasing tax (Appendix D.4). This comparison is itself a result: the shape of the tax changes the total cost of the delay a lot, and the residual overshoot little. Under the constant tax, the residual overshoot is 14.9 $GtCO_2eq$ and the premium \$3.3, against 13.2 $GtCO_2eq$ and \$3.3 under the increasing tax. Across increase rates, the residual overshoot is roughly flat between 0 and 1%, at 14.9 and 15.1 $GtCO_2eq$, and falls above, to 13.2 $GtCO_2eq$ at 2% and 6.0 $GtCO_2eq$ at 6%. As Section 2.6 explains, the tax increase rate acts on lock-in in two opposite directions: it moves the development boundary and makes more young fields delay-induced, and it shortens the post-implementation life of every locked-in field. The first effect offsets the second up to 1%, and the second dominates above. The shape of the tax matters mostly for the intensive margin, which is released before implementation and which no tax set in 2030 can recover.

Demand response. The main specification holds the oil price at \$80, so that producers bear the whole tax. I relax this assumption with an isoelastic demand for oil, calibrated to reproduce the untaxed production path at \$80, with elasticities of -0.4, -0.6 and -0.8, around the long-run value of -0.6 used by Heal and Schlenker (2019) and covering those of Okullo and Reynès (2016). A clean backstop is added at \$250 a barrel, the price at which substitutes take the whole market. As in Heal and Schlenker (2019), the backstop is the highest cost of the resources that Rystad lists as viable for future extraction. Each tax is solved jointly with the path of the oil price it induces. Demand response acts on the residual overshoot through two channels. Part of the tax now reaches the consumer price, about three quarters of it on average after 2030 for all three elasticities, so producers keep more of each barrel: on the price path of the lock-in-adjusted tax, they receive between \$42 and

$67 a barrel in 2030 net of the carbon charge, against about $35 without demand response. Locked-in fields therefore produce longer, and before the price adjusts the wedge between the two calibrations at the naive tax is 27 to 28 $GtCO_2eq$ for all three elasticities, about twice its value in the main specification. The supply that the naive planner did not expect then lowers the price, which shortens the life of some fields after 2030. Once the price has adjusted, the residual overshoot is 20.9 $GtCO_2eq$ with an elasticity of -0.4, 15.1 $GtCO_2eq$ with -0.6 and 9.1 $GtCO_2eq$ with -0.8, between 6.9% and 15.8% of the budget remaining in 2030, against 10.0% in the main specification. The ordering comes from the backstop. A less elastic demand passes more of the tax into the price in the early years, so that the price reaches the backstop earlier, in 2039 with an elasticity of -0.4 against 2052 with -0.8, and at the backstop an extra barrel of oil displaces a barrel of substitute instead of lowering the price, so that the wedge passes entirely into emissions. In 2030, while the price is still free to move, it falls more when demand is less elastic, by $9.6 a barrel against $4.7. I do not report the premium for these runs, because the share of the tax borne by producers changes over time as the price approaches the backstop, so that no single tax is comparable with the $3.3 of the main specification.

The premium across specifications. The residual overshoot and the premium do not always move in the same direction. Outside the tax increase group, the premium ranges from $2.0 to $4.6, and it varies more across specifications than the residual overshoot, because it is the residual overshoot divided by the average slope of post-implementation emissions between the naive and the adjusted tax, and this slope changes with the assumptions. A Brent price of $100 raises every production breakeven and flattens emissions near the naive tax, so that a smaller residual overshoot requires a larger premium, $3.9 against $2.9 at $60. A budget 10% smaller leaves the residual overshoot unchanged, but places both taxes where an additional dollar removes fewer emissions, which raises the premium to $4.4.

5 Incidence and targeted policy

This section asks where the locked-in fields are, who bears the cost of the delay, and what can be done to alleviate the cost of lock-in. The three questions are linked. Because locked-in fields barely respond to a higher tax (Section 5.1), the premium is paid by other fields, adding a second level of redistribution across fields after the one created by the overproduction during the delay, and this changes who gains and who loses from the delay (Section 5.2). Because the locked-in fields are few and similar, targeted policies can partially achieve what

the delayed tax cannot, provided they target the right projects early enough (Section 5.3).

5.1 Where the lock-in sits

Overcommitted emissions are concentrated in a small number of projects, mostly offshore. At the naive tax, the emissions committed beyond 2030 by locked-in fields come from 651 assets, and 32 of them carry half of the total (Appendix Table 15). Figure 10 shows where these projects are, pooled by province.[13] Offshore projects account for 79% of these emissions, and deepwater projects alone for 43%. South America accounts for 32%, Brazil alone for 21%, and the coasts of Brazil and Guyana and the Gulf of Mexico together for 39%. The Middle East accounts for 18%, mostly through a few large offshore and onshore redevelopments approved during the delay, the largest single asset of the sample being the expansion of the Marjan field in Saudi Arabia. National oil companies may screen projects with other rules than the 10% rate used here, which is one reason the discount rate is varied in Section 4.5. What these projects have in common is high capital intensity combined with operating costs low enough for the tax to fall inside their zone, as in Section 2.2. The timing is also concentrated, since 56% of these emissions are committed by investments made between 2025 and 2029.

The locked-in stock in these assets is also resilient to higher taxes. Figure 11 reports, for each level of the 2030 tax, the quantity of emissions locked in at the naive tax that would still be released, split by supply segment (Appendix Figure 25 gives the aggregate curve). Raising the tax to the lock-in-adjusted level of \$98.4 leaves 91% of them in place, and even a tax of \$105, well above what the budget requires, leaves three quarters in place. For most locked-in projects, the carbon tax arrives too late. An extra dollar of tax removes little of their production, and meeting the budget requires stranding fields elsewhere.

5.2 Who gains from the delay

The delay changes each producer's cumulative emissions, relative to the timely policy, through two channels. The overproduction channel runs through the production the delay allows fields to sell untaxed before 2030, and through the production lost after 2030 when the tax has to rise from τ^{*}_{2016} to τ^{N}_{2030} to compensate for it. The lock-in channel runs through

[13] Province is a level of aggregation available in Rystad smaller than countries, but comprising multiple assets. The median number of assets per dot on the graph is 2, the average 5.1, and the maximum 40 (Western Siberia, Russia).

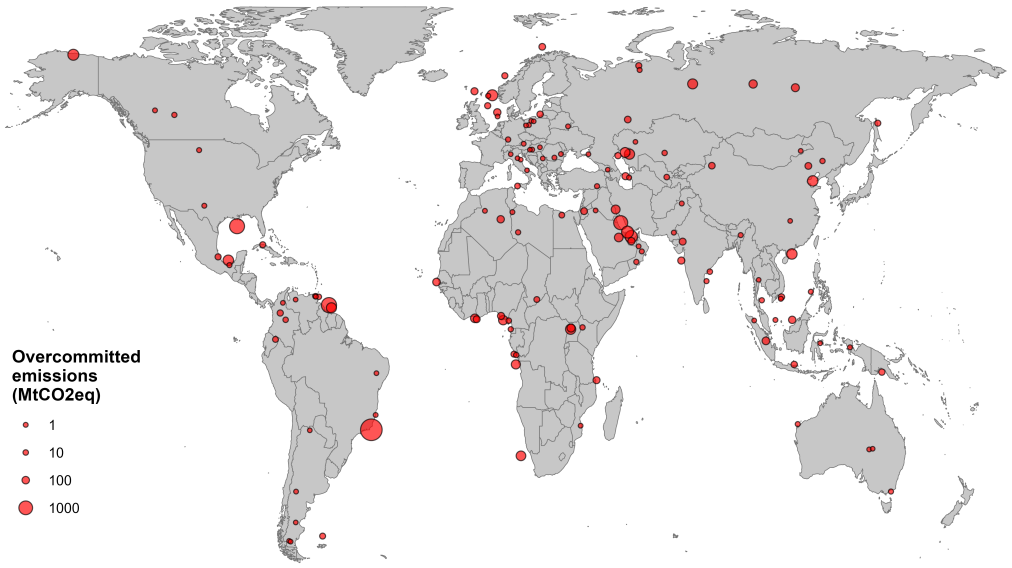


Figure 10: Locked-in projects by province.

Notes: Circle size is the post-2030 emissions of the fields locked in at the naive tax implemented in 2030. A province is a level of aggregation in Rystad's data between the asset and the country. No-anticipation scenario, tax rising at 2% per year.

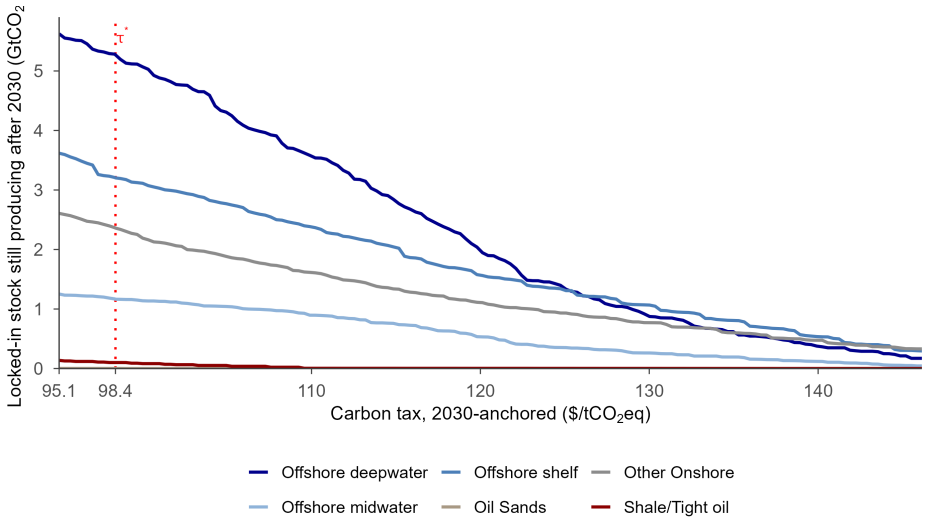


Figure 11: Locked-in emissions surviving a higher tax, by supply segment.

Notes: Post-2030 emissions locked in at the naive tax that are still released when the tax implemented in 2030 is raised to the level on the horizontal axis, by supply segment, in $GtCO_2eq$. The horizontal axis starts at the naive tax, $\tau^N = \$95.1$, and the dotted line marks the lock-in-adjusted tax, $\tau^* = \$98.4$. Taxes are 2030 levels of the tax rising at 2% per year.

the fields locked in at the naive tax. These fields gain their post-2030 locked-in production, and other fields lose the production stranded when the tax has to rise further, from τ^N_{2030} to τ^*_{2030}, to compensate for it. The 60.8 $GtCO_2eq$ excess emissions before 2030 are offset by 60.8 $GtCO_2eq$ from assets stranded after 2030 by the first tax increase, and the 12.1 $GtCO_2eq$ that locked-in fields emit after 2030[14] are offset by 12.3 $GtCO_2eq$ stranded by the second.[15] Table 3 reports the decomposition by continent, in $GtCO_2eq$ and as a share of each continent's own 2016 reserves, and Appendix Table 16 reports it for the largest producing countries.

	Overproduction ($GtCO_2eq$)	Lock-in ($GtCO_2eq$)	Total ($GtCO_2eq$)	Total (% of 2016 reserves)
North America	37.8	1.6	39.4	22.1
Asia	2.9	0.9	3.8	5.8
South America	0.3	3.4	3.7	7.1
Europe	0.7	0.5	1.1	7.9
Oceania	−0.03	0.00	−0.02	−1.5
Africa	−1.4	0.9	−0.5	−1.5
Russia	−6.5	−1.8	−8.3	−12.9
Middle East	−33.6	−5.7	−39.3	−12.4

Table 3: Incidence of the delay by continent.

Notes: Change in cumulative emissions over 2016–2100 relative to a timely tax, no-anticipation scenario, tax rising at 2% per year. The overproduction column nets the production sold untaxed before 2030 against the production stranded when the tax rises from τ^*_{2016} to τ^N_{2030}; the lock-in column nets the post-2030 production of the fields locked in at the naive tax against the production stranded when the tax rises from τ^N_{2030} to τ^*_{2030}. Negative values mean that the production stranded after 2030 to offset a channel exceeds the continent's own gain through it. The two sum to the total effect. The last column is the total change in production (in barrels) as a share of the continent's own 2016 reserves; since carbon intensities differ across fields, it can differ in sign from the change in emissions for small net effects. Values are rounded to one decimal, except for Oceania, whose effects are smaller than 0.05 $GtCO_2eq$, so that columns may not add up exactly.

Three facts stand out. First, the usual view of the delay's incidence, which counts only overproduction, has North America gaining the most and the Middle East paying the most. North American fields, mostly shale and oil sands, sell 41 $GtCO_2eq$ more before 2030 than a timely tax would have allowed and gain 38 $GtCO_2eq$ on net, while the Middle East loses 34 $GtCO_2eq$ on net. Its low-cost fields are not shut down when the tax arrives, but the higher tax needed after the delay slightly shortens the life of every field that still produces, and the low-cost fields of the Middle East, which produce the longest, account for about half of the emissions after 2030 (Table 2). Russia is on the same side as the Middle East, and loses 6.5 $GtCO_2eq$ through this channel. Second, lock-in changes the picture in the middle of

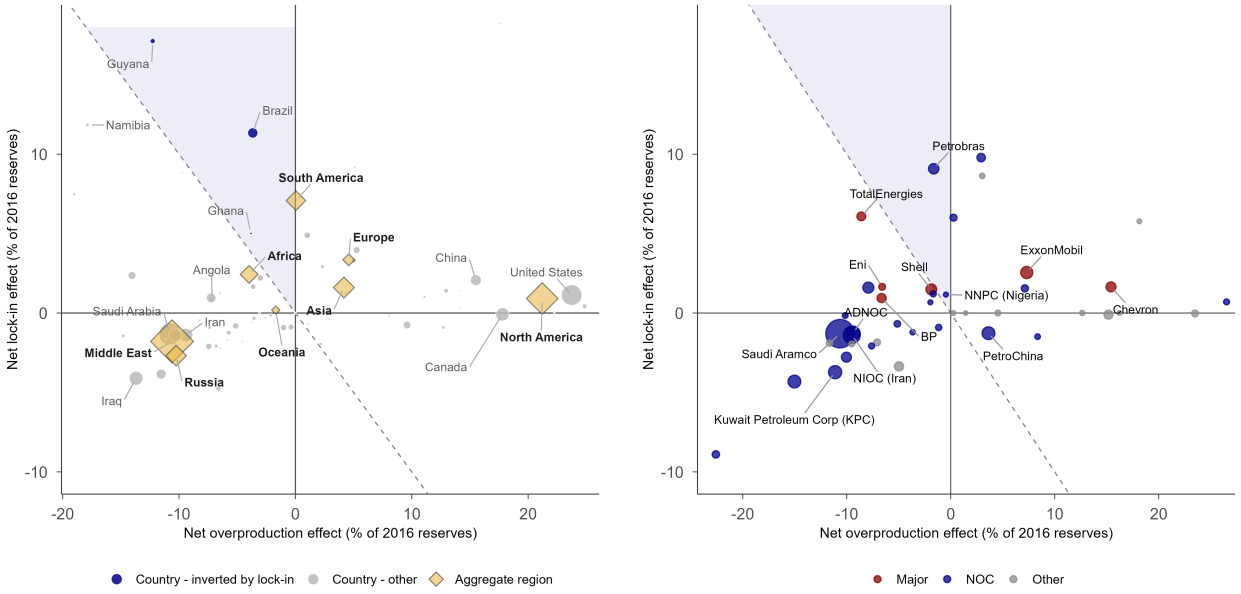
[14] These 12.1 $GtCO_2eq$ are the 13.2 $GtCO_2eq$ of residual overshoot minus the 1.1 $GtCO_2eq$ that the increase in the tax from \$95.1 to \$98.4 removes from the locked-in fields themselves, as shown in Figure 11.

[15] The two columns of Table 3 sum to 0.06 and −0.24 $GtCO_2eq$, against gross flows of 60.8 and about 12 $GtCO_2eq$; the small differences come from steps in the supply curve caused by large assets.

the distribution. It accounts for 4% of the net effect for North America, where few projects would have been deterred by a timely tax, but for more than nine tenths of the net effect for South America. It adds to the losses of Russia and the Middle East rather than offsetting them. The production stranded by the premium falls on the same low-cost, long-lived fields, whose production it ends earlier: 7.9 $GtCO_2eq$ in the Middle East and 2.3 $GtCO_2eq$ in Russia, 83% of the total. These two regions therefore pay for the lock-in of other regions on top of paying for the overproduction of the delay. Measured in dollars, with the surplus of Section 4.2, the incidence has the same signs: the Middle East loses $3.4 trillion, 14% of its surplus under a timely policy, and Russia $0.8 trillion, 17%, while North America gains $2.3 trillion and more than doubles its surplus.

Third, lock-in can reverse the sign of the total. Figure 12 shows the decomposition for continents and for the largest producing countries, normalised by 2016 reserves so that producers of very different size can be compared. The horizontal position is the usual view of the delay's incidence (the overproduction channel), the vertical position what this view misses (the lock-in channel), and producers above the dashed line gain from the delay overall. The United States, China and most of Europe and Asia win on both counts, while Russia and the largest Middle Eastern producers see lock-in add a loss of one to four percent of their reserves to a loss of nine to fourteen percent through overproduction. The shaded area contains the producers that the usual view counts as losers, because a higher tax strands their existing fields, but whose investment during the delay locks-in enough emissions to more than compensate. No continent lies in it, but some countries do. Brazil loses 4% of its reserves through overproduction and gains 11% through its deepwater fields, and ends a net winner. Guyana, whose oil production only started in 2019, loses 12% through overproduction, production that the higher tax strands after 2030, and gains 17% through lock-in. The reversal also holds, in much smaller volumes, for Senegal and Ghana. Namibia, Nigeria and Africa as a whole lie in the upper-left quadrant below the dashed line, where lock-in reduces their loss without reversing it. The reversals are the same whether the effect is measured in barrels, as in the figure, or in emissions, as in Tables 3 and 16.

Firms. The same decomposition can be run by company, attributing each year's production to equity holders according to their share in that year (Appendix Table 17 and panel (b) of Figure 12). National oil companies as a group lose 31 $GtCO_2eq$ through overproduction, Saudi Aramco alone 13.5 $GtCO_2eq$, and lose again through lock-in. Independent producers, mostly North American and focused on shale and oil sands, gain 21 $GtCO_2eq$ through overproduction and little through lock-in. Majors are split in two groups. American majors win on both counts, while for European majors lock-in offsets part of a loss through over-



(a) Regions and countries

(b) Firms

Figure 12: Incidence of the delay as a share of 2016 reserves.

Notes: The horizontal axis is the net effect through overproduction, the vertical axis the net effect through lock-in. (a) Aggregate regions (diamonds) and countries (circles, area proportional to 2016 reserves); countries with less than 500 Mbbl of 2016 reserves are excluded. Senegal, at −6.6% through overproduction and +29.5% through lock-in, lies above the frame. (b) The 50 largest firms by 2016 equity reserves; production is attributed to equity holders at their share in each year, and shares after 2026 are held at their 2026 level. Eight firms lie outside the frame: Midland Oil Company (Iraq) on the left, and seven North American producers on the right, the largest being ConocoPhillips, Canadian Natural Resources, Occidental and EOG Resources. In both panels, the dashed line is where the net effect is zero, and the shaded area contains the producers that lose through overproduction and gain more through lock-in, whose net sign is reversed by lock-in.

production: about all of it for Shell, most of it for TotalEnergies, which loses 0.8 $GtCO_2eq$ and gains 0.7 $GtCO_2eq$, and little of it for BP and Eni. The firm the usual view misclassifies most clearly is Petrobras, whose investment during the delay is in deep offshore projects: it loses 0.2 $GtCO_2eq$ through overproduction and gains 1.4 $GtCO_2eq$ through lock-in, 9% of its 2016 equity reserves. For such firms, the assets that make the delay profitable are those a targeted restriction would have to reach.

5.3 Targeted development restrictions

A worldwide carbon tax on oil producers as envisioned in the paper seems unlikely in the near future, and the international climate regime, which accounts for emissions where fuels are burnt, gives countries little reason to adopt supply-side policies (van Asselt, 2021). Targeted action could nonetheless alleviate a substantial share of the cost arising from infrastructure lock-in. The literature on supply-side policy makes the general case for such instruments, on grounds of abatement certainty (Erickson et al., 2018), public support (Green and

Denniss, 2018; Green et al., 2024) and commitment value (Harstad and Holtsmark, 2026). Measuring lock-in field by field makes it possible to evaluate specific restrictions rather than their principle.

Most instruments studied here act on development rather than production, the exception being targeted early taxes. A ban prevents the approval of new projects in a given perimeter from a given year onwards, including after 2030. Projects already approved are left alone, so a ban never cancels capital that is already partly spent. Table 4 reports, for each instrument, the naive tax, the lock-in-adjusted tax, the premium and the residual overshoot, each instrument being evaluated at its own budget-consistent tax, and Appendix D.9 defines each perimeter and each instrument precisely. The remaining budget also changes, because a project that is not developed does not emit during the delay either, and this second effect is reported in the last column: it is what a restriction buys before implementation, as opposed to what it buys on lock-in. The lock-in-adjusted tax gives a third measure, the stranding a restriction avoids after 2030.

	τ^N ($/tCO$_2$eq)	τ^* ($/tCO$_2$eq)	π ($/tCO$_2$eq)	Ω^N (GtCO$_2$eq)	Remaining budget (GtCO$_2$eq)
No targeted policy (delay only)	95.1	98.4	3.3	13.2	132.6
Ban on new approvals from 2016					
Deepwater	92.5	94.7	2.2	7.5 (−44%)	137.2
Shale	93.0	96.8	3.8	12.9 (−3%)	139.1
The 50 largest locked-in projects (upper bound)	94.3	95.9	1.7	4.8 (−64%)	134.9
The 20 largest locked-in projects (upper bound)	94.7	97.1	2.4	8.3 (−37%)	133.4
Deepwater, OECD perimeter	93.5	96.4	2.9	10.2 (−23%)	135.0
Deepwater, European perimeter	94.6	98.1	3.4	12.2 (−8%)	133.2
Ban on new approvals from 2025					
All new development	85.9	89.3	3.4	8.5 (−36%)	133.0
Deepwater	94.3	97.9	3.5	11.4 (−14%)	132.6
Deepwater, OECD perimeter	94.4	97.9	3.5	11.9 (−10%)	132.6
Shale	95.0	98.4	3.4	13.3 (+0%)	132.6
Other instruments					
Carbon tax on deepwater from 2016	94.2	96.4	2.2	7.4 (−44%)	137.2
Carbon tax on deepwater from 2025	95.6	97.7	2.1	7.7 (−42%)	133.7
Carbon tax on shale from 2016	93.4	97.0	3.6	13.0 (−2%)	163.5
Carbon tax on OECD-perimeter deepwater from 2016	94.6	97.6	3.0	10.2 (−23%)	135.1
Development capital cost +50%, deepwater	94.0	98.1	4.1	15.3 (+16%)	133.0
Development capital cost +50%, all projects	91.4	96.5	5.1	17.7 (+34%)	134.4

Table 4: Targeted development restrictions and alternative instruments.

Notes: Each instrument is evaluated at its own budget-consistent tax. Taxes are 2030 levels of the tax rising at 2% per year, in $/tCO$_2$eq. The OECD and European perimeters cover deepwater projects located in the bloc's territory or operated and majority-owned by firms headquartered there. The change in the residual overshoot relative to the delay without restriction is in parentheses. The remaining budget rises under a restriction because less is emitted before 2030. Premiums are computed before rounding the two taxes, so they can differ by 0.1 from the difference of the rounded values. Appendix Figure 26 reports the full set of instruments.

Three lessons come out of the table, and each of them follows from a property of lock-in established earlier. First, what a restriction achieves depends on the depth of the lock-in zones it removes. A ban on new deepwater approvals from 2016 cuts the residual overshoot

by 44%, from 13.2 to 7.5 $GtCO_2eq$, and the premium by a third. A ban on new shale approvals cuts the residual overshoot by 3% only and raises the premium, to \$3.8, for the reason given in Section 4.5. The shale ban, the closest instrument here to the politically salient ban on fracking, still has an effect. It enlarges the budget remaining in 2030 by 6.6 $GtCO_2eq$, because shale projects that are not developed do not produce during the delay either.[16] But it leaves lock-in untouched, since shale lock-in zones are shallow and sit below the tax. The two ban targets pursue different objectives. Policies aimed at shale buy budget while policies aimed at capital-intensive offshore projects lower the cost of lock-in.

The second lesson is that the concentration documented in Section 5.1 makes highly targeted restrictions effective. Banning the approval of the 50 assets that carry the most locked-in emissions cuts the residual overshoot by 64%, more than a ban on all new deepwater development, and brings the premium down to \$1.7. Banning the 20 largest cuts it by 37%. The two bans rank differently on the stranding they avoid: the deepwater ban lowers the tax the budget requires to \$94.7, against \$95.9 for the ban on the 50 largest projects, because it also removes projects that are not locked in, including deepwater projects approved after 2030, and enlarges the budget remaining in 2030 by 4.6 $GtCO_2eq$. Correctly identifying the projects most likely to lock in large emissions can therefore achieve a lot, but doing so in advance could prove complicated, since it depends on the tax that eventually comes and on projects not yet discovered. Targeting deepwater projects as a whole is the safer route.

The third lesson is that a restriction does not have to be worldwide, but it does have to reach the firms that build capital-intensive projects. Consider a ban on new deepwater approvals on which OECD countries could act: projects located in OECD countries and those that are both operated and majority-owned by firms headquartered there. It reaches 293 of the 434 deepwater projects not yet approved in 2016, including those approved after 2030, and 54% of their reserves, of which only 17% lie on OECD territory and 37% abroad, through the firms. The same rule applied to Europe, defined here as the European Union, Norway and the United Kingdom, reaches 19% of these reserves, of which 1% on European territory. The two bans cut the residual overshoot by 23% and 8%, roughly in proportion to the deepwater reserves they cover. A coalition can therefore reach much more deepwater development through its firms than on its territory, and the firms matter more as time passes: of the deepwater reserves still to be approved in 2025, the OECD perimeter covers 60%, mainly through its firms. Ahlvik et al. (2026) estimate that 47–73% of the emission reductions from an OECD-only tax on oil production are offset by non-OECD production in

[16] Since a large share of shale assets are already producing by 2016, the reduction in delay-era overproduction remains limited. However, shale production within an asset is maintained or increased through continuous drilling of new wells. A policy banning the drilling of any new well would hence affect shale overproduction much more.

the medium term, through the oil price. The model excludes this channel, so the perimeter-based rows of Table 4 are upper bounds. Yet the offset should be smaller than in their setting, where non-OECD production is untaxed, because the restricted projects produce mostly after 2030, when every producer faces the tax.

Alternative instruments. Two instruments are worth comparing with the bans. Early action through a tax targeting deepwater projects alone from 2016, at the level the budget requires, achieves the same reduction in the residual overshoot as the deepwater ban, 44%. The same instrument applied to shale from 2016 leaves lock-in where it is but enlarges the remaining budget by 30.9 $GtCO_2eq$, by far the largest budget effect of any instrument in the table: it confirms that segment-specific early taxation is a policy for the intensive margin when it targets shale, and for lock-in when it targets offshore projects. This contrast also points to a limit of the model. An early tax on shale, or on OECD production more generally, removes a large volume of barrels while the rest of the world is still untaxed, and the price response it would trigger, which the model excludes, would bring part of that production back elsewhere. A restriction on deepwater development is less exposed to this objection, because what it removes would have been produced after 2030, when every producer faces the tax.

The second instrument, raising development costs through a tax on investment, deters some projects but deepens the lock-in of those that are still built. Applied to all new projects, a 50% increase in development costs enlarges the budget remaining in 2030 by 1.8 $GtCO_2eq$ and lowers the tax the budget requires from \$98.4 to \$96.5, but it raises the residual overshoot by 34% and the premium to \$5.1, and applied to deepwater projects only it raises the residual overshoot by 16% and the premium to \$4.1. Raising development costs lowers a project's development breakeven without changing its operating cost, which deepens the lock-in zone of every asset. The naive planner reads the shock as deterring more assets that are in reality already built by 2030, which is why the naive tax falls, from \$95.1 to \$91.4 when the shock applies to all projects, while the emissions these projects still deliver after 2030 rise through the lower tax. Higher development costs therefore change little in what is actually emitted: they deter a few projects, while those still built produce after 2030 as they would have anyway. What they change is how wrong a calibration on full-cycle costs becomes, since more of these projects now look deterrable to the naive planner.

Timing. Restrictions lose most of their effect when they come late. Applied from 2025, the deepwater ban cuts the residual overshoot by 14% instead of 44%, and even a ban on all new approvals cuts it by only 36%, because most of the lock-in of the delay is approved

by then. Approval precedes commitment, however: a project is approved before most of its capital is spent, and its emissions are committed only once enough of that capital is sunk for it to be completed under the tax (Section 3.4). A carbon tax on deepwater assets from 2025 still cuts the residual overshoot by 42%, and all the post-2030 emissions it removes come from projects that would have been committed in 2025–2029 (Figure 7). The deadline for acting on lock-in is therefore set by commitment rather than approval, although a tax that stops projects under construction also falls on fields already producing and strands capital already spent, which raises compensation questions (Paletta et al., 2025).

Incidence. Restrictions also change the incidence described in Section 5.2, and they change both sides of it at once. By removing part of the lock-in, they remove the gain that this lock-in gives to the producers who hold it. Because the premium falls, they also reduce the production stranded elsewhere to make up for it. The deepwater ban from 2016 cuts the post-2030 emissions of locked-in fields from 12.1 to 7.0 $GtCO_2eq$ worldwide, and the ban on the 50 largest projects cuts them to 4.6 $GtCO_2eq$. By the accounting of Section 5.2, the stranding that compensates for them falls by the same amounts, benefiting mainly the Middle East and Russia. South America and Africa, which gain 3.9 and 1.5 $GtCO_2eq$ gross through lock-in without any restriction, keep only 0.7 and 0.7 $GtCO_2eq$ of it under the deepwater ban. The producers that gain from lock-in are therefore those that would oppose a restriction on it, and the producers that pay for it through a higher tax are those that would gain from it. An early tax on shale works on the other channel and therefore has a different incidence. It removes most of the overproduction gain of North America during the delay, which enlarges the remaining budget by 30.9 $GtCO_2eq$ and lowers the tax the budget then requires, so the stranding borne after 2030 by the Middle East and Russia falls while lock-in is left untouched.

Scope for smaller initiatives. Convincing large producing countries to adopt such bans may prove difficult, since governments that depend on oil revenue are the least likely to adopt any constraint on production (Lujala et al., 2022), but the concentration documented above changes what smaller initiatives can achieve. Most bans adopted so far have come from public campaigns against specific projects (Piggot, 2018; Carter and McKenzie, 2020). With 32 assets carrying half of the post-2030 locked-in emissions, campaigns against specific projects can matter, and Table 4 puts a number on what they can achieve: stopping the approval of the 50 largest of them would have cut the residual overshoot by almost two thirds. The Willow project in Alaska is an example. The project was widely debated in the United States, and was finally approved by the Biden administration in 2023. It embeds

228 $MtCO_2eq$ of locked-in emissions that survive the naive tax, 1.7% of the total. Large offshore developments can weigh even more: the six phases of the Buzios deepwater project in Brazil that are locked in at the naive tax[17] carry alone 1.1 $GtCO_2eq$, 8% of the total. Many such projects are owned and operated by firms headquartered in OECD countries, as shown above, and their development depends on technologies these firms master. OECD countries could therefore restrict these specific projects and encourage the redirection of the corresponding capital, which is large: the delay adds $482 billion of investment before 2030, discounted to 2025, spread across regions, with 24% in North America, 22% in South America, 21% in Asia and 14% in Africa.

6 Conclusion

This paper measures what delaying carbon pricing costs in the oil sector once the lock-in effects from fields developed during the delay are taken into account. During the delay, firms develop fields that a timely tax would have deterred, and once their development cost is sunk, a tax high enough to have prevented them is no longer high enough to stop them. Models of oil supply that give each field a single cost per barrel can only account for excess emissions during the delay, but cannot account for future emissions locked-in by investments. Separating the development and production decisions of 25,336 oil fields brings this lock-in back into the cost of the delay. Delaying a 1.5°C-aligned carbon tax from 2016 to 2030 overshoots the oil carbon budget by 20%, mostly through overproduction during the delay. After the delay, a tax set on full-cycle costs leaves 13 $GtCO_2eq$ above the remaining budget, a tenth of it, and closing this gap requires a lock-in premium of $\$3.3/tCO_2eq$.

Lock-in concentrates in fields that are expensive to develop and cheap to produce from, which in the data means mostly offshore projects, deepwater ones above all, while the overproduction during the delay comes mostly from fields that are cheap to develop, or already developed in 2016, and expensive to run, shale and oil sands above all. Because locked-in fields are resilient to a higher tax, the premium is paid by low-cost fields elsewhere, mostly in the Middle East and Russia. For producers whose recent investment is in deep offshore projects, such as Brazil and Guyana, lock-in turns what the usual view counts as a loss from the delay into a gain.

Most of this lock-in is decided in the last years before implementation, and the invest-

[17] Rystad splits Buzios into 17 assets. Six are locked in, all approved between 2019 and 2022. None of the others is, including the four phases approved in 2012 and the phase approved in 2028, much of whose development runs past 2030.

ments made from 2025 alone still leave 7.5 $GtCO_2eq$ above the budget if their lock-in is ignored. Even investors who had correctly expected the tax in 2030 would have left almost half of the residual overshoot in place, and the internal carbon prices that oil companies reported after Paris imply expectations close to no tax at all. Because lock-in is concentrated in a few hundred projects, and half of it in 32, restrictions on specific projects can do part of what the delayed tax cannot. A ban on new deepwater approvals from 2016 cuts the residual overshoot by 44% and the premium by a third, while a ban on new shale projects barely changes it, and a tax on deepwater production still cuts it by 42% from 2025, when most of these projects are approved but not yet committed. When a delay in carbon pricing is expected, early restrictions on capital-intensive projects can therefore complement the tax that eventually comes.

These estimates have three limitations, the first two of which lead to understating the cost of the delay. First, the analysis is confined to oil, so the results should be read as if comparable policies applied to gas and coal at the same time; to the extent that a tax on oil alone shifts demand towards other fossil fuels, total emissions are understated. Second, the model excludes the green paradox, the response of producers who anticipate the tax by bringing production and investment forward, so the total cost of the delay computed here is likely a lower bound. Third, the paths of development and production after 2025 are those forecast by Rystad, which embed Rystad's own assumptions on prices, costs and the timing of projects, and each field follows its forecast profile once developed. The results are therefore conditional on that forecast, in the same way as the main specification is conditional on a constant oil price, which Section 4.5 relaxes.[18] The same separation between development and operating costs applies wherever capital is sunk before climate policy arrives, from LNG terminals to gas power plants, and the lock-in premium could be measured there in the same way.

[18]The model also leaves aside the optimal fiscal response of resource-owning governments to an external carbon price (Section 3.6), and the political economy that produces delays in the first place.

References

Ahlvik, L., Andersen, J. J., Hamang, J. H., and Harding, T. (2026). Quantifying supply-side climate policies. *Review of Economic Studies*. Forthcoming.

Ahlvik, L. and Harding, T. (2025). Resource investments and the timing of tax deductions. *Journal of the Association of Environmental and Resource Economists*, 12(1):1–32.

Anderson, S. T., Kellogg, R., and Salant, S. W. (2018). Hotelling under pressure. *Journal of Political Economy*, 126(3):984–1026.

Asker, J., Collard-Wexler, A., De Cannière, C., De Loecker, J., and Knittel, C. R. (2024). Two wrongs can sometimes make a right: The environmental benefits of market power in oil. NBER Working Paper 33115, National Bureau of Economic Research.

Asker, J., Collard-Wexler, A., and De Loecker, J. (2019). (Mis)Allocation, Market Power, and Global Oil Extraction. *American Economic Review*, 109(4):1568–1615.

Bailey, W., Couët, B., Lamb, F., Simpson, G., and Rose, P. (2000). Taking a calculated risk. *Oilfield Review*, 12(3):20–35.

Berntsen, M., Bøe, K. S., Jordal, T., and Molnár, P. (2018). Determinants of oil and gas investments on the Norwegian Continental Shelf. *Energy*, 148:904–914.

Bertram, C., Johnson, N., Luderer, G., Riahi, K., Isaac, M., and Eom, J. (2015). Carbon lock-in through capital stock inertia associated with weak near-term climate policies. *Technological Forecasting and Social Change*, 90:62–72.

BP p.l.c. (2022). Annual report and Form 20-F 2021. BP p.l.c., London. Note 1, Significant accounting policies, judgements, estimates and assumptions.

Carter, A. V. and McKenzie, J. (2020). Amplifying "Keep It in the Ground" First-Movers: Toward a Comparative Framework. *Society & Natural Resources*, 33(11):1339–1358.

CDP (2016). Embedding a carbon price into business strategy. Technical report, CDP, London.

Chermak, J. M. and Patrick, R. H. (1995). A well-based cost function and the economics of exhaustible resources: The case of natural gas. *Journal of Environmental Economics and Management*, 28(2):174–189.

Coulomb, R. and Henriet, F. (2018). The grey paradox: How fossil-fuel owners can benefit from carbon taxation. *Journal of Environmental Economics and Management*, 87:206–223.

Coulomb, R., Henriet, F., and Reitzmann, L. (2026). "Bad" Oil, "Worse" Oil, and Carbon Misallocation. *Review of Economic Studies*, 93(1):404–437.

Daubanes, J. X. and Lasserre, P. (2023). How should the use of nonrenewables be taxed under a public budget constraint? *Resource and Energy Economics*, 73:101375.

Davis, S. J. and Socolow, R. H. (2014). Commitment accounting of CO_2 emissions. *Environmental Research Letters*, 9(8):084018.

De Cannière, C. (2025). Pump it? Market power and the energy transition in the global oil market. Working paper, Toulouse School of Economics.

Dietz, S. and Venmans, F. (2019). Cumulative carbon emissions and economic policy: in search of general principles. *Journal of Environmental Economics and Management*, 96:108–129.

Dolls, M., Link, S., Liski, M., and Wolf, G. (2026). When do firms invest green? Carbon prices, uncertainty, and policy credibility. EconPol Policy Report 58, CESifo and ifo Institute, Munich.

Elvidge, C. D., Zhizhin, M., Baugh, K., Hsu, F.-C., and Ghosh, T. (2016). Methods for global survey of natural gas flaring from visible infrared imaging radiometer suite data. *Energies*, 9(1):14.

Erickson, P., Kartha, S., Lazarus, M., and Tempest, K. (2015). Assessing carbon lock-in. *Environmental Research Letters*, 10(8):084023.

Erickson, P., Lazarus, M., and Piggot, G. (2018). Limiting fossil fuel production as the next big step in climate policy. *Nature Climate Change*, 8(12):1037–1043.

Fattouh, B. (2007). OPEC pricing power: The need for a new perspective. Working Paper WPM 31, Oxford Institute for Energy Studies.

Financial Accounting Standards Board (2010). Accounting Standards Update 2010-03, Extractive Activities—Oil and Gas (Topic 932): Oil and Gas Reserve Estimation and Disclosures. Financial Accounting Standards Board, Norwalk, CT.

Fried, S., Novan, K., and Peterman, W. B. (2022). Climate policy transition risk and the macroeconomy. *European Economic Review*, 147. Earlier version: "The Macro Effects of Climate Policy Uncertainty", Finance and Economics Discussion Series 2021-018, Federal Reserve Board.

Ghosh, S., Jean, L., and Saha, P. (2025). Voice until the last exit: Ownership structure and the limits of voluntary environmental commitments. Working paper.

Gollier, C. (2012). *Pricing the Planet's Future: The Economics of Discounting in an Uncertain World*. Princeton University Press.

Green, F., Bois von Kursk, O., Muttitt, G., and Pye, S. (2024). No new fossil fuel projects: The norm we need. *Science*, 384(6699):954–957.

Green, F. and Denniss, R. (2018). Cutting with both arms of the scissors: the economic and political case for restrictive supply-side climate policies. *Climatic Change*, 150(1-2):73–87.

Grubb, M., Wieners, C., and Yang, P. (2021). Modeling myths: On DICE and dynamic

realism in integrated assessment models of climate change mitigation. *WIREs Climate Change*, 12(3):e698.

Harstad, B. and Holtsmark, K. (2026). The gas trap: Outcompeting coal versus renewables. *Journal of Political Economy*, 134(7):2166–2214.

Heal, G. and Schlenker, W. (2019). Coase, Hotelling and Pigou: The incidence of a carbon tax and CO_2 emissions. NBER Working Paper 26086, National Bureau of Economic Research.

International Monetary Fund (2024). Regional Economic Outlook: Middle East and Central Asia, April 2024: An Uneven Recovery amid High Uncertainty. International Monetary Fund, Washington, DC.

IPCC (2006). *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2: Energy*. Institute for Global Environmental Strategies (IGES), Hayama, Japan.

Jahn, F., Cook, M., and Graham, M. (2008). The field life cycle. *Developments in Petroleum Science*, 55:1–7.

Jakob, M., Luderer, G., Steckel, J., Tavoni, M., and Monjon, S. (2012). Time to act now? Assessing the costs of delaying climate measures and benefits of early action. *Climatic Change*, 114(1):79–99.

Kellogg, R. (2026). The end of oil. *Review of Economic Studies*. Conditionally accepted. Earlier version: NBER Working Paper 33207, 2024.

Kotlikoff, L., Kubler, F., Polbin, A., Sachs, J., and Scheidegger, S. (2021). Making carbon taxation a generational win win. *International Economic Review*, 62(1):3–46.

Leighty, W. and Lin, C.-Y. C. (2012). Tax policy can change the production path: A model of optimal oil extraction in Alaska. *Energy Policy*, 41:759–774.

Lemoine, D. and Rudik, I. (2017). Steering the climate system: using inertia to lower the cost of policy. *American Economic Review*, 107(10):2947–2957.

Luderer, G., Pietzcker, R. C., Bertram, C., Kriegler, E., Meinshausen, M., and Edenhofer, O. (2013). Economic mitigation challenges: how further delay closes the door for achieving climate targets. *Environmental Research Letters*, 8(3):034033.

Lujala, P., Le Billon, P., and Gaulin, N. (2022). Phasing out fossil fuels: Determinants of production cuts and implications for an international agreement. *Global Environmental Politics*, 22(4):95–128.

Masnadi, M. S., El-Houjeiri, H. M., Schunack, D., Li, Y., Englander, J. G., Badahdah, A., Monfort, J.-C., Anderson, J. E., Wallington, T. J., Bergerson, J. A., et al. (2018). Global carbon intensity of crude oil production. *Science*, 361(6405):851–853.

McCollum, D., Bauer, N., Calvin, K., Kitous, A., and Riahi, K. (2014). Fossil resource

and energy security dynamics in conventional and carbon-constrained worlds. *Climatic Change*, 123(3-4):413–426.

McGlade, C. and Ekins, P. (2015). The geographical distribution of fossil fuels unused when limiting global warming to 2 °C. *Nature*, 517(7533):187–190.

Meinshausen, M., Meinshausen, N., Hare, W., Raper, S. C., Frieler, K., Knutti, R., Frame, D. J., and Allen, M. R. (2009). Greenhouse-gas emission targets for limiting global warming to 2 °C. *Nature*, 458(7242):1158–1162.

Mercure, J.-F., Pollitt, H., Viñuales, J. E., Edwards, N., Holden, P., Chewpreecha, U., Salas, P., Sognnaes, I., Lam, A., and Knobloch, F. (2018). Macroeconomic impact of stranded fossil fuel assets. *Nature Climate Change*, 8(7):588–593.

Metcalf, G. E. (2019). On the economics of a carbon tax for the United States. *Brookings Papers on Economic Activity*, 2019(1):405–484.

Nordhaus, W. D. (2007). A review of the Stern Review on the economics of climate change. *Journal of Economic Literature*, 45(3):686–702.

Okullo, S. J. and Reynès, F. (2016). Imperfect cartelization in OPEC. *Energy Economics*, 60:333–344.

Paletta, J., Cunha, B. S., Draeger, R., Schaeffer, R., and Szklo, A. (2025). Stranded assets and compensation in oil and gas upstream projects: Conceptual and practical issues. *Energy and Climate Change*, 6:100178.

Pierru, A., Smith, J. L., and Zamrik, T. (2018). OPEC's impact on oil price volatility: The role of spare capacity. *The Energy Journal*, 39(2):173–196.

Piggot, G. (2018). The influence of social movements on policies that constrain fossil fuel supply. *Climate Policy*, 18(7):942–954.

Rennert, K., Errickson, F., Prest, B. C., Rennels, L., Newell, R. G., Pizer, W., Kingdon, C., Wingenroth, J., Cooke, R., Parthum, B., Smith, D., Cromar, K., Diaz, D., Moore, F. C., Müller, U. K., Plevin, R. J., Raftery, A. E., Ševčíková, H., Sheets, H., Stock, J. H., Tan, T., Watson, M., Wong, T. E., and Anthoff, D. (2022). Comprehensive evidence implies a higher social cost of CO_2. *Nature*, 610(7933):687–692.

Riahi, K., Kriegler, E., Johnson, N., Bertram, C., den Elzen, M., Eom, J., Schaeffer, M., Edmonds, J., Isaac, M., Krey, V., Longden, T., Luderer, G., Méjean, A., McCollum, D. L., Mima, S., Turton, H., van Vuuren, D. P., Wada, K., Bosetti, V., Capros, P., Criqui, P., Hamdi-Cherif, M., Kainuma, M., and Edenhofer, O. (2015). Locked into Copenhagen pledges — Implications of short-term emission targets for the cost and feasibility of long-term climate goals. *Technological Forecasting and Social Change*, 90:8–23.

Semieniuk, G., Holden, P. B., Mercure, J.-F., Salas, P., Pollitt, H., Jobson, K., Vercoulen, P.,

Chewpreecha, U., Edwards, N. R., and Viñuales, J. E. (2022). Stranded fossil-fuel assets translate to major losses for investors in advanced economies. *Nature Climate Change*, 12(6):532–538.

Seto, K. C., Davis, S. J., Mitchell, R. B., Stokes, E. C., Unruh, G., and Ürge-Vorsatz, D. (2016). Carbon Lock-In: Types, Causes, and Policy Implications. *Annual Review of Environment and Resources*, 41(1):425–452.

Stefanski, R., Ahlvik, L., Andersen, J. J., Harding, T., and Trew, A. (2025). Extracting wedges: Misallocation and taxation in the oil industry. CESifo Working Paper 11915, CESifo.

Stern, N. (2007). *The Economics of Climate Change: The Stern Review*. Cambridge University Press.

Tang, B.-J., Zhou, H.-L., Chen, H., Wang, K., and Cao, H. (2017). Investment opportunity in China's overseas oil project: An empirical analysis based on real option approach. *Energy Policy*, 105:17–26.

Tong, D., Zhang, Q., Zheng, Y., Caldeira, K., Shearer, C., Hong, C., Qin, Y., and Davis, S. J. (2019). Committed emissions from existing energy infrastructure jeopardize 1.5 °C climate target. *Nature*, 572(7769):373–377.

Trout, K., Muttitt, G., Lafleur, D., Van De Graaf, T., Mendelevitch, R., Mei, L., and Meinshausen, M. (2022). Existing fossil fuel extraction would warm the world beyond 1.5 °C. *Environmental Research Letters*, 17(6):064010.

Unruh, G. C. (2000). Understanding carbon lock-in. *Energy Policy*, 28(12):817–830.

van Asselt, H. (2021). Governing fossil fuel production in the age of climate disruption: Towards an international law of 'leaving it in the ground'. *Earth System Governance*, 9:100118.

van der Ploeg, F. and Rezai, A. (2020). The risk of policy tipping and stranded carbon assets. *Journal of Environmental Economics and Management*, 100:102258.

Venables, A. J. (2014). Depletion and Development: Natural Resource Supply with Endogenous Field Opening. *Journal of the Association of Environmental and Resource Economists*, 1(3):313–336.

Wachtmeister, H. and Höök, M. (2020). Investment and production dynamics of conventional oil and unconventional tight oil: Implications for oil markets and climate strategies. *Energy and Climate Change*, 1:100010.

Weitzman, M. L. (1998). Why the far-distant future should be discounted at its lowest possible rate. *Journal of Environmental Economics and Management*, 36:201–208.

Welsby, D., Price, J., Pye, S., and Ekins, P. (2021). Unextractable fossil fuels in a 1.5 °C

world. *Nature*, 597(7875):230–234.

Wood Mackenzie (2013). Impacts of Delaying IDC Deductibility (2014–2025). Prepared for the American Petroleum Institute, Washington, DC.

A Model extensions

A.1 Closed forms for a parametric cohort

This appendix collects the closed-form expressions behind the statements of Section 2. They rely on two simplifications that the body does not use: no discounting where indicated, and a parametric cohort of fields. None of the definitions in the body depend on them.

The discounted profit per barrel. Evaluating the integral in (4) with $a^* = a^*(\tau)$,

$$\bar{v}_\delta(\tau) = \frac{p - c_0 - \theta\tau}{\delta}\left(1 - e^{-\delta a^*}\right) - \frac{\gamma}{\delta^2}\left[1 - e^{-\delta a^*}\left(1 + \delta a^*\right)\right], \tag{13}$$

using $\int_0^{a^*} e^{-\delta a} da = (1 - e^{-\delta a^*})/\delta$ and $\int_0^{a^*} a\, e^{-\delta a} da = [1 - e^{-\delta a^*}(1 + \delta a^*)]/\delta^2$. By Leibniz, $\partial\bar{v}_\delta/\partial\tau = -\theta \int_0^{a^*} e^{-\delta a} da = -\theta(1 - e^{-\delta a^*})/\delta < 0$, the boundary term vanishing because the margin is zero at $a = a^*$. Since a^* does not depend on δ, $\partial\bar{v}_\delta/\partial\delta = -\int_0^{a^*} a\, e^{-\delta a}(p - c_0 - \theta\tau - \gamma a)\, da < 0$, the integrand being positive on $(0, a^*)$. As $\delta \to 0$, $\bar{v}_\delta(\tau) \to (p - c_0 - \theta\tau)^2/2\gamma$.

Depth without discounting. With $\bar{v}_0(\tau) = (p - c_0 - \theta\tau)^2/2\gamma$, the development breakeven, the depth and the width of the zone are explicit:

$$\theta\, \bar{\tau}_D = p - c_0 - \sqrt{2\gamma\kappa}, \qquad \theta\, D = \sqrt{2\gamma\kappa}, \qquad a^*(\bar{\tau}_D) = \sqrt{2\kappa/\gamma}, \qquad (\delta = 0). \tag{14}$$

Because $\bar{v}_\delta$ decreases with δ, a higher discount rate lowers $\bar{\tau}_D$ and deepens the zone, so the square root is a lower bound on the depth of lock-in at any positive discount rate. The log-log slope of depth per barrel on capital intensity is $1/2$ at $\delta = 0$ and above it at any positive rate.

The cohort. Let a mass N of fields be developed during the delay, with capital intensity distributed with c.d.f. F_κ on $[0, \kappa_1]$, independent of the age at implementation a_Δ, which is uniform on $(0, \Delta]$. All other characteristics, θ, $\bar{q}$, γ and $p - c_0$, are common across fields. The most capital-intensive field of the cohort is pinned down by the model. A field is developed during the delay only if it is worth developing at the tax investors expect to face, which in the baseline is no tax at all, so the highest capital intensity in the cohort is the one at which

development just breaks even without a tax:

$$\kappa_1 = \bar{v}_\delta(0), \qquad \text{equivalently} \qquad a^*\Big(\bar{\tau}_D(\kappa_1)\Big) = \frac{p - c_0}{\gamma} = a^*(0), \tag{15}$$

with $\kappa_1 = (p - c_0)^2/2\gamma$ at $\delta = 0$. The lock-in zone of this field stays open over its whole untaxed life. Under anticipation, the zero tax is replaced by τ^A. Only one feature of F_κ appears in what follows, the share of delay-induced fields

$$G(\tau) \equiv \Pr\Big[\kappa > \bar{v}_\delta(\tau)\Big] = 1 - F_\kappa\Big(\bar{v}_\delta(\tau)\Big), \tag{16}$$

the fraction of the cohort a timely tax τ would have deterred. Since fields only differ in κ and a_Δ, the development question depends on κ alone and the survival question on a_Δ alone. Every row of Table 1 therefore aggregates into $G(\tau)$ or $1 - G(\tau)$ times an integral over ages, and the expressions below hold for any F_κ. With κ uniform and $\delta = 0$, $G(\tau) = 1 - (1 - \tau/\bar{\tau}_P(0))^2$.

The wedge. Take a tax at which $a^*(\tau) \leq \Delta$, so that the oldest fields of the cohort have already been retired when the tax arrives. A delay-induced field of age a_Δ then produces for $(a^* - a_\Delta)^+$ years after implementation, and integrating over the uniform age gives

$$W(\tau) = \frac{N\theta\bar{q}}{2\Delta}\, G(\tau)\, a^*(\tau)^2. \tag{17}$$

At lower taxes, when all fields of the cohort are still producing at implementation, $a^{*2}/2$ is replaced by $\Delta(a^* - \Delta/2)$. The wedge is zero at $\tau = 0$, where $G = 0$, and at $\tau = \bar{\tau}_P(0)$, where $a^* = 0$, and positive in between. With κ uniform, W is proportional to $u(1-u)$ with $u \equiv (1 - \tau/\bar{\tau}_P(0))^2$, so it is single-peaked.

Actual and naive emissions. In the same range of taxes, the emissions of the cohort after implementation and the emissions computed by the naive planner are

$$E(\tau) = \frac{N\theta\bar{q}}{2\Delta}\, a^*(\tau)^2, \qquad E^N(\tau) = \frac{N\theta\bar{q}}{2\Delta}\, a^*(\tau)^2 \Big(1 - G(\tau)\Big), \qquad E(\tau) - E^N(\tau) = W(\tau), \tag{18}$$

which is (8) for the parametric cohort.

The premium in closed form. At $\delta = 0$, with κ uniform and $a^*(\tau^N) \leq \Delta$, solving $E^N(\tau^N) = E(\tau^*) = B_\Delta$ and eliminating B_Δ gives

$$\begin{aligned} \bar{\tau}_P(0) - \tau^N &= \sqrt{\bar{\tau}_P(0)\left(\bar{\tau}_P(0) - \tau^*\right)}, \qquad \text{equivalently} \\ \frac{\pi}{\bar{\tau}_P(0)} &= \sqrt{x} - x, \qquad x \equiv 1 - \frac{\tau^*}{\bar{\tau}_P(0)} \qquad (\delta = 0). \end{aligned} \tag{19}$$

Three properties follow. The premium depends on the primitives only through $\bar{\tau}_P(0)$ and τ^*: two cohorts with different cost slopes, carbon intensities and capital-intensity distributions but the same production breakeven and the same budget-consistent tax have the same premium. It is strictly positive whenever $\tau^* > 0$, since $\bar{\tau}_P(0) - \tau^N$ is the geometric mean of $\bar{\tau}_P(0)$ and $\bar{\tau}_P(0) - \tau^*$ and lies strictly between them. Relative to the production breakeven, it is largest at $x = 1/4$, when the budget-consistent tax is three quarters of the production breakeven.

Anticipation. When investors screen projects with an anticipated tax τ^A, the cohort ends at $\kappa_1 = \bar{v}_\delta(\tau^A)$. With κ uniform and $\delta = 0$, G and a^* keep the same form once $\bar{\tau}_P(0)$ is replaced by $\bar{\tau}_P(0) - \tau^A$ and taxes are measured from τ^A, so (19) holds with these substitutions as long as $\tau^N \geq \tau^A$. The premium is zero when investors screen with τ^* or with a higher tax, since no field developed during the delay is then delay-induced at τ^*.

A.2 An increasing tax

This appendix gives the algebra behind Section 2.6. The tax increases at rate r from its 2016-anchored level τ. A field that is a_Δ years old at implementation earns, x years after implementation, a margin per barrel of

$$p - c_0 - \gamma\,(a_\Delta + x) - \theta\tau e^{r(\Delta + x)}. \tag{20}$$

Three objects of the constant-tax model are affected.

The survival boundary. The field is phased out when (20) reaches zero. The corresponding age has no closed form but is unique, since the margin decreases with x, and it is lower than under a constant tax because the tax keeps rising. To first order in r, the margin falls by $\gamma + r\theta\tau e^{r\Delta}$ per year after implementation: the operating cost rises by γ and the tax in force at implementation, $\tau e^{r\Delta}$, rises by $r\tau e^{r\Delta}$. This is the statement used in Section 2.6.

Whether a field survives implementation depends on its margin at $x = 0$ only, so the survival boundary is still $\bar{\tau}_P(a_\Delta)$ once the tax is measured at its implementation level $\tau e^{r\Delta}$.

The development boundary. Under a timely policy, a field developed at date s faces the tax $\tau e^{r(s+a)} = (\tau e^{rs})\, e^{ra}$ at age a. Its discounted profit per barrel of capacity therefore depends on τ and s only through τe^{rs}, the tax in force at its start-up, and the development condition is a threshold on that product. Writing $\bar{\tau}_D^0$ for the development breakeven of the field if it were developed at $s = \Delta$, the development breakeven of a field that is $a_\Delta = \Delta - s$ years old at implementation is exactly $\bar{\tau}_D(a_\Delta) = \bar{\tau}_D^0\, e^{ra_\Delta}$. Fields developed early in the delay would have faced a lower tax under a timely policy and are more likely to have been developed anyway. Because the tax keeps rising after start-up, $\bar{\tau}_D^0$ depends on the whole tax path and not only on the margin at start-up, so the depth of the zone now depends on the tax. Figure 2 shows the resulting partition.

Aggregation. The four types of Section 2.3 and the identity $W = E - E^N$ are unchanged, and intensive fields still carry no overcommitted emissions: a field that the timely tax would also have developed produces for the same period under both policies, whatever the tax path. What is lost is the factorisation of the overshoot into $G(\tau)$ times an integral over ages, because whether a field is delay-induced now depends on its age at implementation as well as on its capital intensity. The wedge and the premium are computed numerically over (κ, a_Δ):

$$W(\tau) = \frac{N\theta\bar{q}}{\Delta} \int_0^\Delta \int_{\bar{v}_\delta(\tau, a_\Delta)}^{\kappa_1} \left(a^*(\tau, a_\Delta) - a_\Delta\right)^+ dF_\kappa(\kappa)\, da_\Delta, \tag{21}$$

where $\bar{v}_\delta(\tau, a_\Delta)$ is the discounted profit per barrel of capacity of a field of age a_Δ at implementation under the timely tax path, and $a^*(\tau, a_\Delta)$ is its phase-out age. E, E^N, τ^N, τ^* and π follow as in Section 2.4.

A.3 A development period

In Section 2 development is instantaneous. In the data it takes several years, and a field may be partly built when the tax arrives. Let development take B years, with development spending spread evenly over them, so that a field whose development began at date $s > \Delta - B$ has paid a share $(\Delta - s)/B$ of its development cost at implementation, and a field whose development began earlier has paid all of it. Production starts at $s + B$, so a completed field is $a_\Delta = \Delta - s - B$ years old at implementation.

Nothing changes for a field completed before implementation: its lock-in zone at implementation is $(\bar{\tau}_D, \bar{\tau}_P(a_\Delta)]$ and Section 2.2 applies with the shifted age. A field under construction at implementation is completed if its discounted profit per barrel of capacity covers the development cost that remains to be paid, rather than the full development cost:

$$\bar{v}_\delta(\tau) \;\geq\; \kappa\left(1 - \frac{\Delta - s}{B}\right), \qquad \text{equivalently} \qquad \tau \;\leq\; \tau_L(s), \quad \bar{v}_\delta\Big(\tau_L(s)\Big) = \kappa\left(1 - \frac{\Delta - s}{B}\right). \tag{22}$$

The completion boundary $\tau_L(s)$ falls from $\bar{\tau}_P(0)$ at $s = \Delta - B$, when nothing remains to be paid and the field is completed at any tax that leaves it a positive margin, to $\bar{\tau}_D$ at $s = \Delta$, when nothing has been paid and the full development condition applies. Without discounting, $\theta\,\tau_L(s) = p - c_0 - \sqrt{2\gamma\kappa\,(s - \Delta + B)/B}$, so the boundary falls steeply as soon as part of the development cost remains to be paid, and flattens as s approaches Δ. The threshold on remaining development cost used in Section 3.4 is its empirical counterpart.

Figure 13 shows the partition in (s, τ) at a given capital intensity. Two types are added to the four of Section 2.3. Fields under construction at implementation whose completion satisfies (22) are completed and locked in (Type IV-a). They produce over their whole life after implementation, since they only start producing once the tax is in place. Fields under construction whose completion does not satisfy (22) are abandoned, with their capital stranded and no emissions (Type IV-b). A development period can only add fields to the locked-in set, since a partly built field is kept at any tax below $\tau_L(s) \geq \bar{\tau}_D$, so the case of instantaneous development in the body is a lower bound. Lock-in is largest for fields whose development began just before $\Delta - B$, which have just started producing when the tax arrives.

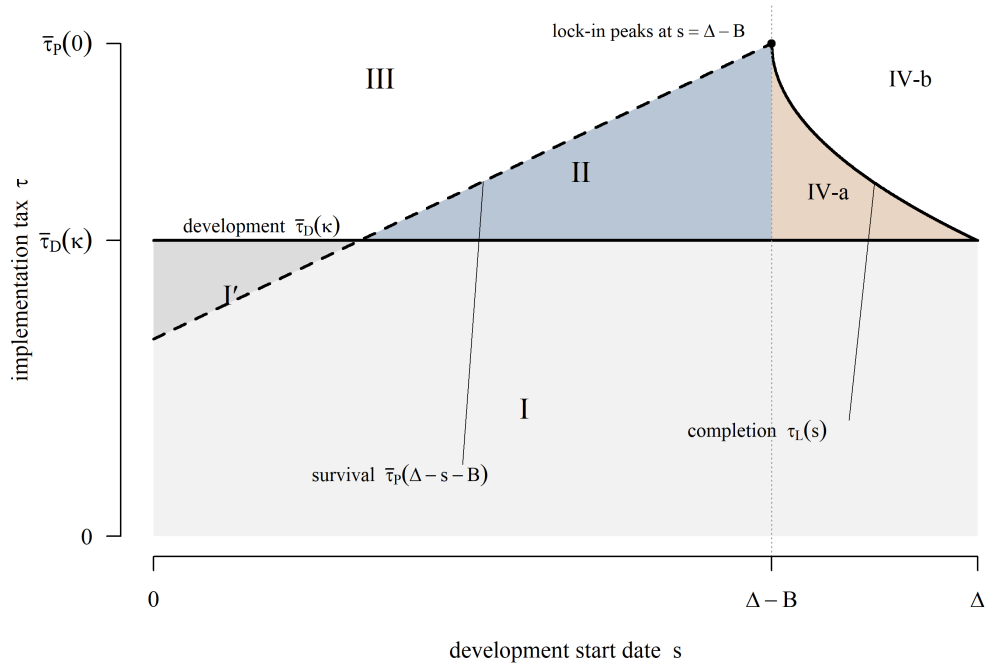


Figure 13: The taxonomy with a development period, at a given capital intensity.

Notes: The horizontal axis is the date s at which development starts; fields starting after $\Delta - B$ are under construction at implementation. The development boundary $\bar{\tau}_D$ is horizontal; the survival boundary $\bar{\tau}_P(\Delta - s - B)$ rises with s because later fields are younger at implementation; the completion boundary $\tau_L(s)$ falls from $\bar{\tau}_P(0)$ at $s = \Delta - B$ to $\bar{\tau}_D$ at $s = \Delta$. Region II is the locked-in set among completed fields, IV-a among partly built fields that are completed, IV-b the partly built fields abandoned at implementation. Lock-in is largest at $s = \Delta - B$.

B Data appendix

B.1 Costs

Figure 14 reproduces Rystad's definition of each cost item in the UCube database. Rystad splits the costs of an asset into capital expenses, for facilities and for wells; operating expenses, for production, transportation and selling, general and administrative costs (SG&A); exploration expenses; and government take. Only the expenses attributed to oil operations are kept. Exploration expenses are excluded from the main specification, since exploration precedes the decisions studied here, and the sensitivity analysis adds them back for the fields not yet discovered (Section 4.5). Government take is excluded as well, except for the taxes that companies report as operating expenses, for the reasons given in Section 3.6.

Development expenses. The development cost of an asset, the empirical counterpart of C, is the facility and well capital spent from the start of development to two years after the

Economics Category	The Economics Category splits the Economics variable (=Revenue) into higher granularity than Economics Group, e.g. capital expenses is subdivided into facility and well expenses. The sum of all items equals the revenue. Use Economics Category to analyze breakdown of cost models or specific cost elements.
Free Cash Flow	FCF is the incurred cash flow available to investors and creditors, thus largely used in DCF valuation analysis. FCF = revenue-capex-opex-gvt take. FCF can be compared with cash from operations and investments activity in the cash flow statement from companies (only upstream). The present value of (all discounted) FCF equals NPV (Net Present Value also called Gross Asset Value) that can be compared with Enterprise Value (=Market cap+Debt for upstream activities), or compared with asset transaction costs. The discount factor in UCube is 10% flat for all assets.
Government Profit Oil	Gvt PO is the PSA equivalent to petroleum taxes, but paid in kind. Gvt PO reduces the company's entitelement production and is thus treated as a royalty effect in company reports.
Royalty effects	Royalty effects is the sum of all gross taxes, including royalty and oil and export duties.
Income Tax	Income tax is a sum of all profit based taxes, ie corporate taxes, where the tax rate is equal to the country's corporate tax rate, and special petroleum taxes. The tax base is the net sales revenue minus opex and depreciation. For petroleum taxes there may be applied an additional deduction called uplift. In many cases, some profit based taxes can be deducted in the tax base for other profit taxes. The income taxes in the UCube assumes there is no interest expenses.
Bonuses	Signature bonus for the license at award, as officially reported.
Well Capex	Well capex is capitalized costs related to well construction, including drilling costs, rig lease, well completion, well stimulation, steel costs and materials.
Facility Capex	Development capex except well construction costs, includes costs to develop, install, maintain and modify surface installations and infrastructure. As reported by operators, field partners or officials, or modelled. Distribution of costs over time is largely modelled, one exception is Norway, where all historic costs are as reported in national budgets.
Exploration Capex	Costs incurred to find and prove hydrocarbons: seismic, wildcat and appraisal wells, general engineering costs, based on reports and budgets or modelled.
Abandonment cost	Abandonment cost represents the costs associated with shutting down and dismantling the surface and subsea facilities such as surface equipment, platforms, subsea structures and pipelines. Activities included is topside and jacket removal as well as pipeline cleaning and decommissioning, onshore disposal, surveys and monitoring, and project management costs. Note that the Abandonment cost only includes the remaining P&A work for wells, and not full cycle P&A work. Abandonment cost is mostly modelled and assumed to be incurred and expensed over couple of years after the production has ceased. The estimate is proportional to initial greenfield capex.
Production Opex	Represents operational expenses directly related to the production activity. The category includes materials, tools, maintenance, equipment lease costs and operation related salaries. Depreciation and other non-cash items are not included.
Transportation Opex	Represents the costs of bring the oil and gas from the production site/processing plant to the pricing point (only upstream transportation). The category includes transport fees and blending costs.
SG&A Opex	Represent operation expenses not directly associated with field operations. The category includes administrative staff costs, office leases, related benefits (stocks and stock option plans) and professional expenses (legal, consulting, insurance). Only E&P related SG&A are included.
Taxes Opex	Represent gross tax reported by companies under operational costs and applies for US and Brazil. For US the category represent ad valorem taxes (county based) and severance taxes (state based).

Figure 14: Rystad economic data, split items.

Notes: Descriptions are provided by Rystad alongside the UCube database.

start of production, following Rystad's own definition. On most projects, a large part of this capital is spent in the first years of production, on the wells drilled to reach plateau, and it is committed by the time a tax can reach the field. Shale and tight oil assets follow a different rule: only facility expenses incurred up to two years after start-up count as development, and all well expenses are production costs. A shale asset keeps drilling new wells for as long as it produces, and each well produces most of its oil within a year or two, so drilling behaves like a recurring cost of production rather than a long-lived investment.

Development expenses are discounted to the first year of development in the investment

decision (12), together with the revenues and production costs of the asset, so the time between the first expenses and the first barrel enters the value of a project. It matters most for capital-intensive projects: a deepwater project typically spends for three to five years before it produces, and at a 10% discount rate these years weigh on the value of developing it, while they play no role once it is built. Two adjustments keep the timing of expenses consistent with the timing of decisions. First, the development window is capped at $\min(T_1 - T_A, 6)$ years before start-up, where T_A is the approval year and T_1 the first year of production; six years is the 90th percentile of $T_1 - T_A$ in the delay cohort, weighted by 2016 reserves. Expenses incurred before this window, typically on projects whose development stopped and resumed years later, are gathered in its first year without discounting. Discounted from their actual dates at 10%, they would weigh far more than the revenues of the project, which arrive much later. Projects whose start-up was postponed after the 2020 price collapse are few, 43 assets, so the cap is not driven by them. Second, the development expenses of the first three years of production, from T_1 to $T_1 + 2$, are moved to $T_1 - 1$ and discounted accordingly, which leaves the net present value of the project unchanged. They are treated as committed when production starts, so that the decision to develop and the decision to produce never apply to the same year.

An asset is developed if its net present value at the anticipated tax is non-negative. Rearranging (12) gives the highest anticipated tax at which this holds, the empirical counterpart of the development breakeven $\bar{\tau}_D$. Because development takes several years and can stop and resume, this threshold is computed at each development stage on the expenses that remain to be spent, and the asset is developed only if the anticipated tax is below the threshold at every stage. The threshold rises as the remaining cost falls, so an asset that clears it when development starts keeps clearing it as long as expectations do not change. When the tax is implemented, an asset under development whose remaining-cost threshold lies below the tax is no longer developed: the capital already spent is lost and the infrastructure is abandoned. This remaining-cost threshold is the empirical counterpart of the completion boundary of Appendix A.3. For a given tax, the first year in which the threshold exceeds the tax is the year of investment responsible for the asset's lock-in: from that year on, the asset would be completed even if the tax were implemented immediately (Section 4.3).

Production costs. Production costs include all operating expenses (production, transportation and SG&A), the taxes that companies report as operating expenses, and all capital expenses incurred more than two years after start-up, together with all well expenses for shale and tight oil. Rystad forecasts long lives for most assets, because it ends production when

the breakeven price of the full project, development expenses included, is minimised. Once the development cost is sunk, the relevant breakeven excludes it, and it is often minimised by stopping earlier, since operating costs rise over time and capital expenses per barrel rise as production declines. The production of each asset is therefore split into stages with strictly increasing breakeven prices. From the first year of production, I compute the last year of production that minimises the breakeven price of the remaining production, which closes the first stage, and repeat the operation from the following year until the end of the asset's life. The breakeven of each stage, converted into a tax with the asset's price discount and carbon intensity, is the empirical counterpart of the production breakeven $\bar{\tau}_P(a)$ of equation (2). An asset produces as long as the price net of the tax covers the breakeven of its current stage, and is phased out for good in the first year in which it does not.

Excluding shale, the 2,682 fields of the delay cohort have 9.3 production stages on average (8 at the median, and 14.7 when weighted by reserves, since larger fields produce for longer), and their first stage lasts 6.2 years on average and 3 years at the median. The 2,756 shale and tight oil assets mostly have a single stage, 1.7 on average and 1 at the median. All their well capital counts as a production cost and drilling is heaviest in the first years, so the breakeven of the remaining production is lowest when production runs to the end of the asset's life, and the production breakeven of a shale asset is in most cases constant. The model thus treats shale close to the single-cost convention of existing models, as its economics imply: with little capital sunk before production, there is little to lock in.[19]

B.2 Asset-specific price discounts

Table 5 reports the reserve-weighted average discount to Brent by oil type, which Section 3.3 discusses.

B.3 Anticipation scenarios

The main specification assumes investors do not anticipate the tax during the delay. Figure 8 considers two kinds of expectations, indexed by the share s of the expected tax that investors anticipate. In the first, investors deciding on a project in year t expect the tax to be implemented in year $t+1$, at the level a naive planner would set for a delay ending that

[19] This leaves aside the production of wells drilled just before the tax arrives, which continues for a year or two after it. Treating these wells as small sunk investments would add a short tail of locked-in shale production, so the main specification is conservative on this point.

Oil type	Average discount	2016 reserves (Gbbl)
Regular	−5.4%	760
NGL	−50.6%	222
Light	−2.7%	203
Condensate	−5.9%	113
Sour (> 3%)	−4.6%	52
Bitumen	−29.0%	47
Heavy oil 15–19	−14.0%	34
Heavy oil 20–22	−10.4%	22
Extra heavy oil	−12.4%	21
Synthetic crude	−7.4%	19
Total	−12.9%	1,493
Excluding NGL	−6.3%	1,271

Table 5: Discounts relative to the Brent price by oil type.

Notes: Reserve-weighted averages of the discounts observed and forecast by Rystad over 2016–2100, estimation sample. Discount data are missing for a small part of the sample's production, so the reserves in the table (1,493 Gbbl) fall slightly short of the 1,555 Gbbl of the estimation sample, the total of Table 2 plus the never-developed fields.

year, τ^N_{t+1}, taken from the runs without anticipation of Figure 6. They screen the project with $s\,\tau^N_{t+1}$, on a path rising at 2% per year from $t+1$, over its whole lifetime production. In the second, investors expect the tax from 2030 only, and screen the project with $s\,\tau^N_{2030}$ from 2030 onwards. Expected levels are taken as given, and are not updated for the change in the naive tax that anticipation itself causes. In every scenario the anticipated tax enters only the investment decision (12), through T_τ and τ; production decisions after implementation are taken under the implemented tax. The limit case in which investors screen with the lock-in-adjusted tax τ^*_{2030} from the date they invest reproduces $\pi = 0$ exactly.

C Field-level carbon intensities

While Rystad's databases provide detailed asset-level data on oil production, they lack comprehensive carbon intensity (CI) estimates. This appendix fills this gap by combining engineering-based models (OPGEE, PRELIM, OPEM), satellite-derived flaring data (VIIRS), and predictive regressions to estimate lifecycle CI for all assets, including those not yet in production. Carbon emissions related to oil can be split into 3 stages (Gordon et al., 2015). The upstream stage includes all emissions arising from extraction processes to the transport of crude oil to the refinery inlet. The midstream stage corresponds to refining processes and is heavily dependent on the refinery configuration. Finally, the downstream

stage comprises emissions from the transport of oil products to their end users and from their final use. While downstream emissions usually represent around 80% of total emissions, the upstream stage features important variability from one field to the other.

For each of these stages, I rely on a state-of-the-art engineering model used by the Oil-Climate Index (Gordon et al., 2015), run on a sample of currently produced oils, to estimate their carbon intensity. I then fit a regression model for each stage to assign a carbon intensity to all other Rystad assets.

Figure 15 reports cumulative 2016 reserves ordered by lifecycle carbon intensity. Most reserves lie between 400 and 500 $kgCO_2eq/bbl$, with a short tail of cleaner assets and a longer tail of more carbon-intensive ones. Fields developed during the delay and fields already producing in 2016 have similar distributions.

C.1 Upstream CI

C.1.1 Flaring data

Upstream emissions exhibit substantial heterogeneity across assets, and the major contributor to this heterogeneity is the flaring-to-oil ratio (FOR). Flaring corresponds to the combustion on-site of co-produced gas[20], and represents more than 150 billion cubic meters worldwide in 2024 (World Bank, 2025). To compute precise estimates of upstream CI, I use satellite data on flares to derive field-specific flaring-to-oil ratios.

Satellite flare observations and imputed emissions. Rystad's UCube database does not contain information on gas flared volumes at the asset level. This is mainly due to the fact that most oil and gas operators do not publicly report flared volumes. Moreover, under-reporting by field operators remains an important issue making the use of self-reported flaring less reliable.

To fill this gap, I use geocoded flaring volumes calculated by the Earth Observation Group (EOG) at the Payne Institute for Public Policy, within the Colorado School of Mines. The EOG analyzes satellite data from the Visible Infrared Imaging Radiometer Suite (VIIRS), on board satellites launched in 2012 and 2017, which detects heat from gas flares at night.

[20]Co-produced gas is often used to power equipment on site, or sold, if the asset is connected to the local gas market or has the equipment to liquefy and ship it. If not, excess gas is then either reinjected, to increase pressure in the reservoir, flared, or directly vented. Given the higher global warming potential of methane, flaring is preferable to venting, and venting is banned in most countries except for rare emergencies. Gas reinjection is often costly, making flaring an easier decision in many oil fields.



Figure 15: Cumulative 2016 oil reserves ordered by lifecycle carbon intensity, by development timing.

Flaring volumes are then estimated using the VIIRS Nightfire algorithm (VNF), as detailed by Elvidge et al. (2016) and further refined by Zhizhin et al. (2021). I use the 141,841 observations with positive flared volumes between 2012 and 2024 to assign them to specific assets from Rystad's database.

Rystad assets geographic information. In order to assign flaring observations to Rystad assets, I need geographic information on these assets. I complement data from the UCube with two sets of geocoded data obtained from the same company, one in 2022 and the other in 2025.

First, geocoded boundaries were shared by Rystad in 2022 for 41,762 assets, among which 16,801 were generic (i.e. a 500m radius circle). Since the universe of Rystad assets has evolved between 2022 and 2025, 1,916 generic shapes were added for assets missing in the shapefile, based on their geocoded centroid.

Second, data were obtained on the location of wells completed between 2000 and 2025. This dataset contains the latitude and longitude of 1,733,698 producer, injector or wildcat wells, from 39,494 assets. In some regions, the location of wells corresponds to the asset centroid, meaning that Rystad probably does not have its precise location. In total, 857,900 unique locations exist in the dataset. Crucially, well locations are precisely defined for onshore US and Canada assets, where all asset boundaries are generic, making this dataset a useful complement to the first one.

Matching satellite observations to Rystad assets. The assignment of satellite observations to Rystad assets relies on both assets' eligibility (the existence or not of surface wells, excluding for example subsea tiebacks and not yet completed wells) and geographic distances to existing wells and assets' boundaries. For each observation, distances to all completed wells and assets in a 20km radius were computed. A score reflecting the probability of a detected flare to originate from each nearby asset is then computed. For observations with existing wells in a 2km radius, only information on wells was used to compute this score. For other observations, distances to both assets and wells were incorporated in the score. When scores were too close to decide between assets, flared volumes were assigned based on each eligible asset's score and production capacity. Ghosh et al. (2025) provide more detailed information on the 5-stage matching procedure used to assign flaring observations to Rystad assets.

Overall, all observations with at least one eligible well or asset in a 20km radius are matched, leaving 7,261 observations unmatched (corresponding to 4.7% of total flared volumes between 2012 and 2024). Among matched observations, around two thirds were assigned to a single asset (70.4% of matched flared volumes) and the rest split between multiple assets.

Computing flaring-to-oil ratios (FOR). Once the matching procedure is done, total flared volumes over the 2012-2024 period are computed and divided by total oil production to obtain asset-specific FOR. For not-yet-active assets over this period, an average FOR is assigned. Two categories of assets producing over the period were also attributed averages: assets with at least 75% of years in the sample with positive production outages, mainly due to wars in Libya and Syria (109 assets), and assets whose production started less than

2 years before the end of the period (911 assets). The average FOR is computed at the lowest scale possible, for assets of similar size if possible[21], and always by oil type. Averages are capped at 12,000 scf/bbl, corresponding to twice as much gas flared as oil produced, in barrels of oil equivalent. Table 6 reports the number and type of average FORs assigned to Rystad assets, and the corresponding oil reserves.

FOR origin	Number of assets	Share of oil reserves in 2016
From satellite matching	20,571	64.3%
Area × Oil Type × Field Size	5,133	4.2%
Province × Oil Type × Field Size	7,350	7.3%
Province × Oil Type	10,497	8.9%
Country × Oil Type × Field Size	6,662	3.3%
Country × Oil Type	4,250	4.8%
Region × Oil Type × Field Size	2,948	3.4%
Region × Oil Type	3,136	3.1%
World × Oil Type × Field Size	946	0.9%
World × Oil Type	2	0.0%

Table 6: Origin of FOR values.

Table 7 shows the binned distribution of computed FOR. Most assets display relatively low levels of flaring with around a third of them having a FOR equal to 0 (representing around 22% of oil reserves). In these assets, the 0 value can be explained by either complete use of co-produced gas or particularly low gas-to-oil ratio, but it can also be due to flared volumes so low (or so intermittent) that it falls below the satellite detection threshold. Among assets producing mainly oil (assets with gas-to-oil ratios below 10,200 scf/bbl), the reserve-weighted average FOR is 104.4 scf/bbl, and the median is 13.4 scf/bbl.

FOR (kscf/bbl)	Assets count	Reserves (Gbbl)
0	11,651	345.6
0–0.1	10,820	748.7
0.1–1	10,974	435.4
1–10	1,069	21.0
>10	291	2.5

Table 7: Distribution of flaring-to-oil ratios.

[21] Assets are split into 10 categories depending on their reserve size, with thresholds at 1, 3, 10, 30, 100, 300, 1000, 3000 and 10000 MMboe.

C.1.2 OPGEE model

The Oil Production Greenhouse Gas Emissions Estimator (OPGEE) is an engineering-based model developed at Stanford University and used by the Oil-Climate Index (Gordon et al., 2015)[22]. It measures GHG emissions from extraction, processing and transport of crude oil in grams of CO_2 equivalent per megajoule of oil produced.

Data inputs and matching with Rystad assets. Running the model requires a wide range of inputs for each field, from extraction methods to gas composition. When information is missing, the model computes smart default values (deduced from other variables when possible). I recover publicly available input data from Masnadi et al. (2018) on 958 oil fields worldwide, and match it to Rystad assets manually. Among the 958 original fields, 737 were matched to 1,136 assets in the Rystad database. Most unmatched fields correspond to marginal heavily-depleted oil fields in the US. The difference between the number of OPGEE fields and matched Rystad assets comes from differing levels of aggregation. In many instances, a unique OPGEE field is split into multiple assets in Rystad data. On rare occasions, multiple OPGEE fields were matched to a unique Rystad asset.

After completing the matching procedure, I complement the 737-field dataset with information on 6 variables from Rystad data in order to improve the accuracy of my estimates. These variables are: production level, field age, gas-to-oil ratio, flaring-to-oil ratio, number of producer wells and number of injection wells. Since Rystad databases do not contain information on the quantity of gas reinjected, used on site, or vented, I assume the amount of gas produced to be equal to the sum of flared and sold gas in order to compute gas-to-oil ratios. Finally, extraction methods reported in Masnadi et al. (2018) input data were adjusted if necessary to match information retrieved from Rystad[23].

Computing CI. The resulting input dataset was fed into the OPGEE model to export results. Two distinct carbon emissions accounting methods are available within the model: energy-based allocation (EBA) and carbon product displacement (CPD). In the EBA approach CI are defined such that $CI_{EBA} = \frac{\text{TotalEmissions}_{\text{field}}}{\text{Prod}_{\text{oil}}+\text{Prod}_{\text{gas}}}$, with Prod_{oil} and Prod_{gas} in MJ. This means that total emissions from the oil field are computed, and allocated to oil and gas sold on the market according to their energy content. In the CPD approach, the definition becomes $CI_{CPD} = \frac{\text{TotalEmissions}_{\text{field}}-\text{Prod}_{\text{gas}}\cdot\overline{CI}_{\text{gas}}}{\text{Prod}_{\text{oil}}}$. Co-produced gas gives rise to carbon credits

[22]Latest versions of the model are available at: https://eao.stanford.edu/research-project/opgee-oil-production-greenhouse-gas-emissions-estimator

[23]This is for example the case for water reinjection. If Rystad data reports existing injection wells, at least water or gas injection must be indicated in OPGEE input data. When this was not the case, water reinjection was assumed to happen, since it is more common in oil fields than gas reinjection.

corresponding to the energy content of gas produced multiplied by $\overline{CI}_{gas}$, a worldwide average carbon intensity of gas production computed from the CA-GREET model. The latter approach can lead to negative oil CI, which is why the retained approach was the EBA one.

Figure 16 displays CI computed using the model. The production-weighted average carbon intensity is equal to 59 $kgCO_2eq/bbl$ (indicated by the dashed red line on the figure). The variance among fields is considerable, with 10% and 90% weighted deciles ranging from 35 $kgCO_2eq/bbl$ to 113 $kgCO_2eq/bbl$. Three key features influencing CI are indicated on the graph:

- **FOR**: as expected, large CI values often originate in large quantities of co-produced gas flared. The influence of FOR on CI largely depends on the assumptions made on destruction removal efficiency (DRE). This value measures the efficiency of flares, i.e. the share of gas effectively combusted and thus transformed into CO_2. Since the global warming potential of methane is 28 times larger than the one of carbon dioxide, assumptions on DRE greatly affect computed CI. I keep the default value of 96.1% retained by the model to compute baseline CI, and provide in sensitivity analysis results using values of 91.1% and 100% to bracket the estimates found in the literature (Plant et al., 2022; Gvakharia et al., 2017; Zavala-Araiza et al., 2021).[24]

- **Steam injection**: Heavier oils (API below 20) tend to be thicker than lighter ones, and often require the injection of steam in order to lower their viscosity and allow their extraction. This is captured by the steam-to-oil ratio (SOR), corresponding to the number of barrels of vaporized water injected to extract one barrel of oil. High SOR values imply larger quantities of liquids extracted, and therefore more energy consumed to extract oil.

- **Upgrading into synthetic oil**: some particularly viscous oils (bitumen or extra heavy) are extracted without first reducing their viscosity (for instance using steam injection). This is mainly the case for Canadian oil sands, which are often directly extracted in open air mines. However, they are often too viscous to flow through pipelines and be handled by existing refineries. These oils are therefore "upgraded" into synthetic crude on site. This requires using chemical treatments and/or distillation in order to break long hydrocarbon chains into shorter ones, thus reducing viscosity. These processes are often very energy intensive, translating into large CI for fields using

[24]While playing a much smaller role than FOR, the quantity of co-produced gas (GOR) can still influence extraction emissions through fugitive emissions of methane related to its processing and transport on site.

them (Brandt and Farrell, 2007). Produced synthetic crude is however easier to refine, translating into relatively lower midstream CI[25].

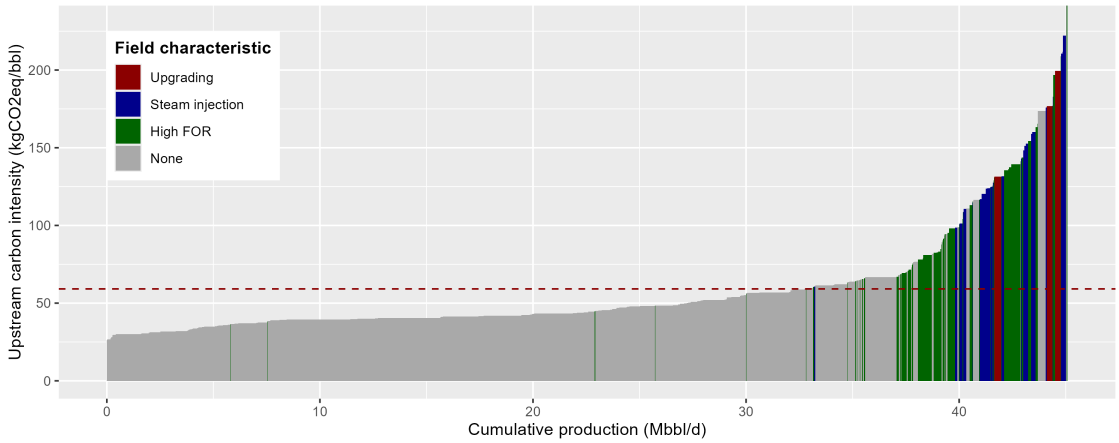


Figure 16: OPGEE-computed CI.

Notes: The figure depicts CI computed in the OPGEE model, with cumulative production in 2015 on the x-axis. The horizontal dashed red line indicates the production-weighted average.

C.1.3 Estimation model

Model and sample restrictions. Once the 737 OPGEE-computed CI are attributed to the corresponding 1,136 Rystad assets, I fit a regression model to explain across-field variations. Following the logic of the EBA approach, the dependent variable in the model thus consists of the field carbon intensity multiplied by its "energy denominator" $ED_f = \frac{Prod_{oil,f}+Prod_{gas,f}}{Prod_{oil,f}}$, with production in energy units, so that $CI_f \times ED_f$ measures upstream emissions per barrel of oil. The retained model is:

$$\begin{aligned} CI_f^{OPGEE} * ED_f \quad &= \sum_{i=1}^{10} \beta_i^C ShareOilType_{i,f} + \beta_{11} Offshore_f + \beta_{12} SOR_f \\ &+ \beta_{13} GOR_f + \beta_{14} FOR_f + \epsilon_f, \end{aligned} \tag{23}$$

where f denotes a Rystad asset, $ShareOilType_f$ the share of each oil type in its production[26], $Offshore_f$ a dummy indicating if the field is offshore, SOR_f its steam-to-oil ratio (in

[25]Midstream CI for synthetic crudes are on average 30% lower when compared to other extra heavy or bitumen oils, and still 8% lower when compared to all types of oil.

[26]An asset produces a unique type of crude oil, among 8 possible (synthetic, bitumen, extra heavy, heavy (15-19), heavy (20-22), regular, light and sour). It can however produce simultaneously condensate and/or natural gas liquids (NGL), which are sold as oil, similar to light crude.

bbl of water/bbl of oil), GOR_f its gas-to-oil ratio (in kscf/bbl) and FOR_f its flaring-to-oil ratio (in kscf/bbl).

Oil-type variables, together with the offshore dummy, capture various aspects of the extraction methods and facility characteristics on site. This is especially relevant for synthetic crude, where upgrading processes such as distillation or chemical treatments significantly increase CI. SOR, GOR and FOR have a more straightforward and linear impact on carbon emissions, and explain most of the remaining variance. SOR data used for the estimation is the one used as input in the OPGEE model since precise values are not available in Rystad data.

The sample used in the regressions was restricted to fields with water-to-oil ratio below 40 bbl of water/bbl of oil, and gas-to-oil ratio below 40,000 scf/bbl. The first restriction excludes fields almost fully depleted, characterized by exponentially increasing WOR[27]. The second one is motivated by the fact that my focus is on oil producing fields. Oil fields are defined by Rystad as reservoirs with GOR below 10,200 scf/bbl. I exclude fields with GOR higher than about four times this threshold in order to keep all fields producing non-negligible oil quantities. Altogether, these two restrictions exclude 20 Rystad assets, making the final sample 1,116 assets (associated with 719 OPGEE fields). The regressions are weighted so that each unique OPGEE-computed value carries the same weight. If a unique OPGEE value was matched to multiple Rystad assets, weights were computed based on oil reserves size and sum to 1.

Regression results. Table 8 shows coefficient estimates from the regression for two sets of CI: the first corresponds to my baseline estimates, using a 100-year horizon for global warming potential, and the second uses a 20-year horizon (associated with higher global warming potential for methane as compared to carbon dioxide, due to its shorter lifetime). R-squared values are close to 1, and predicted R-squared are also quite close to 1, indicating that the model's predictive power is relatively high.

As expected, synthetic crude production is associated with large CI, and coefficients for heavier crudes tend to be larger than the ones of lighter crudes. While coefficients associated with bitumen and extra heavy crudes are not significantly higher than others, production of these two types often implies the use of steam injection, which means that most of the variation is captured by the SOR variable. The negative coefficient associated with the offshore dummy reflects the absence of land-use transformation, source of GHG

[27] I exclude these fields because Rystad data does not contain information on this ratio. The threshold of 40bbl/bbl is particularly high and only excludes 7 fields from the sample.

Variables	EBA - AR6 100y	EBA - AR6 20y
Oil type: Synthetic	157.096***	192.358***
	(6.514)	(9.943)
Oil type: Bitumen	52.854***	78.442***
	(4.555)	(6.952)
Oil type: Extra Heavy	48.122***	78.368***
	(4.169)	(6.363)
Oil type: Heavy (15-19)	54.778***	96.841***
	(3.73)	(5.693)
Oil type: Heavy (20-22)	58.861***	97.629***
	(3.561)	(5.436)
Oil type: Regular	46.522***	79.708***
	(1.105)	(1.687)
Oil type: Light	46.939***	77.204***
	(1.862)	(2.842)
Oil type: Sour (> 3%)	43.308***	71.654***
	(5.14)	(7.846)
Oil type: Condensate	43.535***	72.446***
	(4.235)	(6.465)
Oil type: NGL	48.05***	83.603***
	(6.641)	(10.137)
Offshore	-6.048***	-8.435***
	(1.273)	(1.943)
SOR (bbl water/bbl oil)	26.317***	27.369***
	(0.983)	(1.501)
GOR (kscf/bbl)	4.601***	8.7***
	(0.154)	(0.236)
FOR (kscf/bbl)	83.908***	146.98***
	(0.533)	(0.813)
Corrected R^2	0.970	0.977
Predicted R^2	0.936	0.943

Table 8: Regression results from upstream emissions estimation.

Notes: The table displays coefficients from the regressions using equation (23), with standard errors in parentheses. R^2 are corrected in order to account for the absence of intercept, as well as predicted R^2.

emissions. The three remaining variables explain most of the variance. Steam injection greatly increases emissions (the average steam-to-oil ratio is 3 bbl of water/bbl of oil). While having a relatively low magnitude, GOR still plays an important role in explaining the variance in carbon intensity across fields. Fields producing gassy oils, even when selling their co-produced gas, are subject to higher quantities of fugitive emissions. Finally, FOR plays the most crucial role in explaining the variance across fields, as noted by Masnadi et al. (2018).

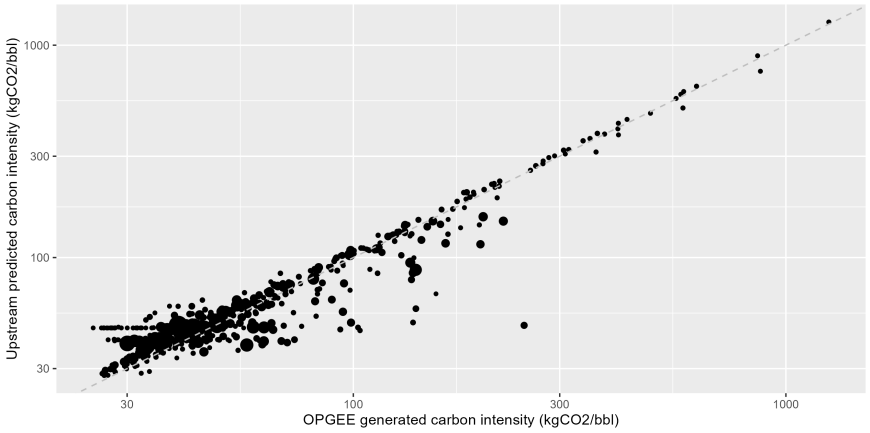


Figure 17: Predicted vs OPGEE-computed upstream CI.

Notes: The figure displays upstream CI computed using the OPGEE model on the x-axis and predicted using the model on the y-axis. Only assets matched with OPGEE fields are included. Both axes are transformed using the decimal logarithm. The size of each point reflects the log volume of oil reserves for each asset.

Regressions using 20-year horizon CI as the dependent variable display similar results. By construction, reducing the time horizon can only increase CI: while carbon dioxide emissions are not affected, methane emissions lead to higher global warming potential. Consequently, variables associated at least partially with methane emissions see their associated coefficients increase in large proportions. This is particularly visible when comparing the SOR, GOR and FOR variables. While steam injection does not entail methane emissions, gas production leads to fugitive methane emissions and flaring does not fully convert methane into carbon dioxide (depending on the retained DRE value as explained above). The large increase in the coefficients associated with GOR and FOR, while the SOR coefficient remains nearly unchanged, is thus consistent with this mechanism.

Sensitivity analysis. Figure 17 highlights the strong correlation between OPGEE-computed CI and CI predicted using the model for the same assets sample. To verify the influence of outliers, I re-estimate the coefficients of equation (23) excluding each observation one by one. Figure 18 displays estimated coefficients, with the baseline estimate indicated in red. While the coefficients of heavier oil types vary because few assets produce them, results indicate that my estimates are overall robust to changes in the sample.

Finally, I test the sensitivity of estimated coefficients to the exclusion of the most influential observations. I compute the Cook's distance of each observation, defined by $CD_i = \sum_{j=1}^{N} \left(\widehat{y}_{\,j} - \widehat{y}_{\,j(i)}\right)^2 / ps^2$, where $\widehat{y}_{\,j}$ and $\widehat{y}_{\,j(i)}$ are the fitted response values from using the full sample and after excluding field i, respectively; s^2 is the mean squared error of the regression model, p the number of coefficients, and N the number of observations. Table 9

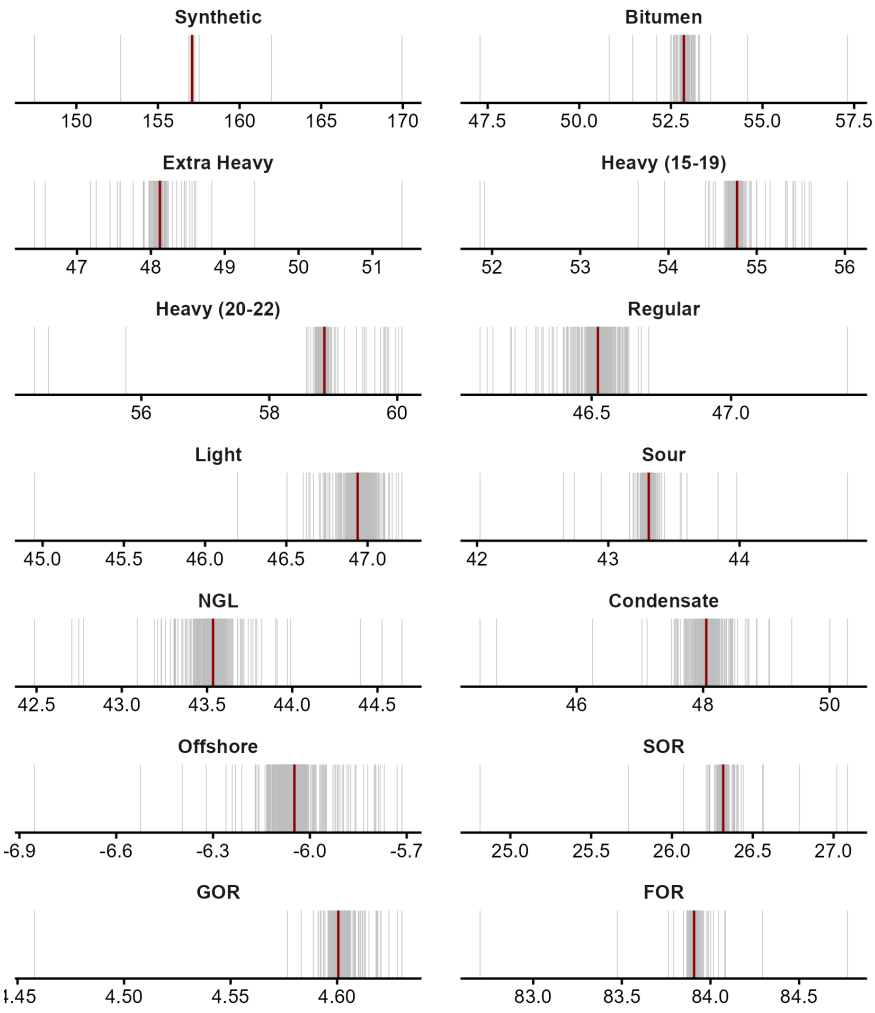


Figure 18: Upstream CI estimation: Estimates of the coefficients on the explanatory variables removing the OPGEE fields one-by-one.

Notes: The panels show the estimated coefficients on each of the explanatory variables in equation (23) removing observations one-by-one. Each grey line corresponds to a different estimation with the same number of observations: $N(OPGEE_{sample})-1$. The thick red lines show the original estimates obtained with the full sample of OPGEE publicly-available fields (OPGEE sample).

displays re-estimated coefficients excluding the 1, 10 and 50 most influential observations. Heavier oil types see their estimates vary slightly, due to the small number of assets producing them in my OPGEE sample, while SOR, GOR and FOR coefficients are particularly stable. Predicted R-squared stay high, even when removing the 50 most influential observations. Overall, re-estimated coefficients remain consistent with my original estimates using the full sample.

	Synthetic	Bitumen	Extra Heavy	Heavy 15–19	Heavy 20–22
0	157.1*** (6.51)	52.85*** (4.55)	48.12*** (4.17)	54.78*** (3.73)	58.86*** (3.56)
1	157.12*** (6.29)	53.14*** (4.40)	48.48*** (4.02)	55.42*** (3.60)	59.49*** (3.44)
10	133.06*** (6.51)	56.86*** (3.70)	51.35*** (3.37)	53.23*** (3.07)	50.40*** (2.99)
50	154.73*** (14.33)	51.56*** (2.88)	47.81*** (2.50)	49.14*** (2.40)	49.60*** (2.64)
	Regular	Light	Sour (>3%)	Condensate	NGL
0	46.52*** (1.11)	46.94*** (1.86)	43.31*** (5.14)	43.54*** (4.24)	48.05*** (6.64)
1	47.42*** (1.07)	46.87*** (1.80)	43.98*** (4.96)	44.40*** (4.09)	49.04*** (6.41)
10	47.62*** (0.90)	45.37*** (1.50)	44.17*** (4.12)	45.48*** (3.41)	53.37*** (5.37)
50	44.59*** (0.65)	41.86*** (1.07)	44.73*** (3.05)	43.83*** (2.51)	46.00*** (3.87)
	Offshore	SOR	GOR	FOR	Pred. R^2
0	-6.05*** (1.27)	26.32*** (0.98)	4.60*** (0.15)	83.91*** (0.53)	0.934
1	-6.85*** (1.23)	26.22*** (0.95)	4.62*** (0.15)	82.70*** (0.53)	0.931
10	-7.39*** (1.03)	24.97*** (0.81)	4.51*** (0.13)	82.67*** (0.54)	0.918
50	-5.21*** (0.74)	26.02*** (0.67)	4.54*** (0.09)	83.93*** (0.46)	0.910

Table 9: Upstream CI estimation: coefficient stability to influential observations (Cook's distance).

Notes: Each panel shows re-estimated coefficients using equation (23) after excluding the 0, 1, 10, and 50 most influential observations. Cook's distances were computed to determine the influence of each observation. The last panel indicates predicted R-squared for each of the re-estimated models.

C.1.4 Predicting upstream carbon intensity for all Rystad assets

Using estimated coefficients, predicted upstream CI were computed for each Rystad asset. Data on reserves by oil type, offshore status, and GOR was directly available in Rystad databases. Asset-specific FOR were either derived from satellite observations or predicted for not-yet-producing assets, as described in Appendix C.1.1. Values for other needed variables were assigned as described below:

- SOR: on top of assets matched to OPGEE fields, assets using steam injection were identified and assigned a SOR value of 3, in line with both the default value in the OPGEE model and the average SOR value in OPGEE fields data from Masnadi et al. (2018). All assets containing "Steam", "Thermal" or "SAGD" in their name were identified as using steam injection. In addition, the list of Enhanced Oil Recovery projects made available by the IEA was used to manually identify remaining steam injecting assets (McGlade et al., 2018).
- ED_f: energy denominators for each asset were approximated by $ED_f = 1 + \frac{GOR_f - FOR_f}{6000}$, using a 6,000 MJ average energy content per barrel. This approximation is remarkably close to energy denominators derived from the OPGEE model, with a correlation coefficient of 0.975 within the OPGEE matched assets sample.

When a unique Rystad asset was matched with a unique OPGEE field, the carbon intensity computed using the OPGEE model was kept. For all remaining assets, predicted values prevail. Figure 19 shows the 2012-2024 production-weighted distribution of CI for both samples, with weighted means depicted by the vertical lines. Although averages are remarkably similar, the predictive model does not seem to fully replicate the variance across OPGEE-computed values. The general shape of the distribution remains coherent with OPGEE outputs.

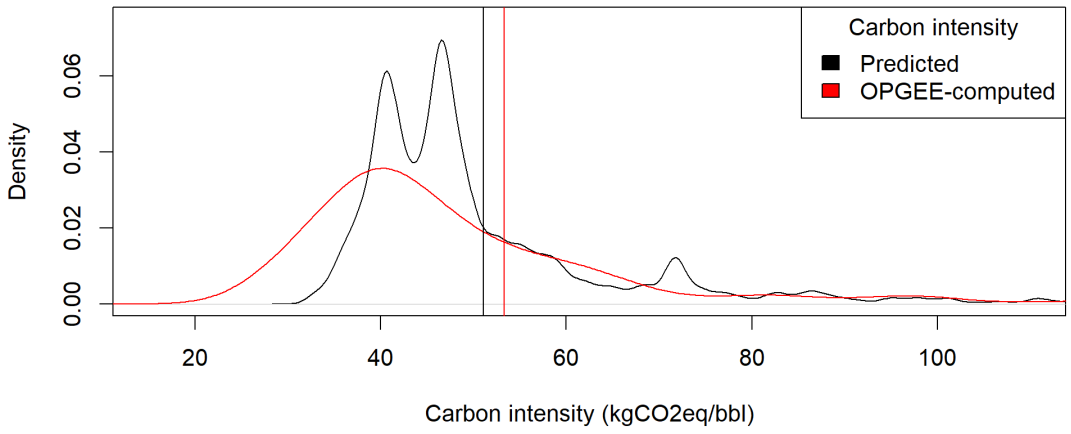


Figure 19: Distribution of CI: Predicted vs OPGEE-computed.

Notes: The figure shows in red the distribution of baseline CI (100-year horizon, EBA) for the OPGEE sample, i.e. assets matched with OPGEE fields and for which the OPGEE value was kept, and in black predicted CI for all remaining assets. Densities are weighted by oil production volumes over the 2012-2024 period. Vertical lines indicate weighted averages for both samples.

C.2 Midstream-downstream CI

Midstream emissions correspond to emissions from refining crude oil, while downstream emissions include emissions from both transport to end-users and final use (often combustion) of the different oil products obtained from refining. Refining crude oil yields multiple oil products depending on its characteristics and refinery configurations. Midstream and downstream emissions are typically computed at the crude stream level, which corresponds to the actual crude stream arriving at the refinery inlet. In some instances, a crude stream originates in a single oil field, while the majority of crude streams consist of the output of multiple fields blended together. For instance, the Brent crude used as reference in oil pricing is a blend of crude oils produced in 13 separate assets in the North Sea.

C.2.1 PRELIM and OPEM models

Midstream and downstream CI can be computed using the Petroleum Refinery Life Cycle Inventory Model (PRELIM) and Oil Products Emissions Module (OPEM) respectively[28]. The versions used in this study to compute CI are PRELIM 1.6 and OPEM 1.1. The PRELIM model requires detailed information on crude assays, such as API gravity, sulphur content, and various boiling points. This data is typically not made available by producers, and publicly available data usually does not follow the same format and is therefore unusable in the models. The models' authors therefore included data on 657 unique crude assays, which I match to the Rystad assets they originate from.

Running the model. The PRELIM model allows for a large array of refinery configurations, designed to handle different types of oils and aimed at maximizing the production of specific oil products. Since I do not have sufficient information to match Rystad assets to the refineries treating their oil production, I follow the assumptions of the model on refinery configurations. Among the three main configurations used worldwide, the model selects the adapted configuration based on crude oil characteristics in terms of API gravity and sulphur content. All heavy crudes (API gravity below 22) are assigned the "deep conversion" configuration, entailing additional treatments aimed at reducing the length of carbon chains yielded. Light crudes with low sulphur levels (API gravity above 32 and sulphur content below 0.5%) are assigned the "hydroskimming" configuration, associated with fewer treatment stages and in turn lower emissions. All other crudes are refined using the "medium conversion" configuration. The importance of refinery configuration is illustrated in Figure 20, depicting PRELIM-computed midstream CI. Intensive treatment of crude in the case of deep conversion is associated with higher emissions, while hydroskimming units yield much smaller emissions.

Once midstream emissions are computed, PRELIM output sheets are imported to the OPEM model in order to estimate downstream CI of each crude stream. Emissions from transport and final combustion (for oil products used as fuel) are computed, giving me for a given barrel of crude oil its final products yield and associated emissions. Final combustion represents the largest share of emissions over the oil lifecycle, and displays smaller variability across crudes.

[28]Information and most recent versions of the models can be found at: https://ociplus.rmi.org/methodology.

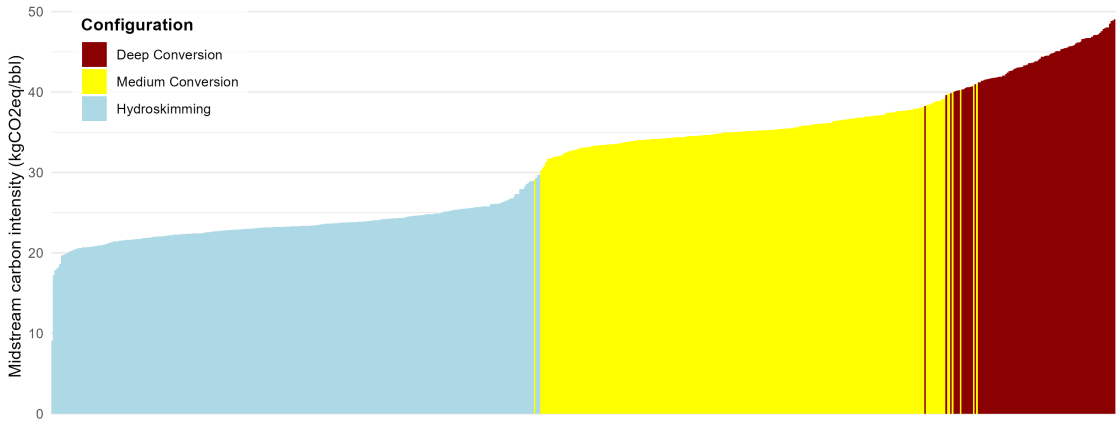


Figure 20: PRELIM-computed midstream CI.

C.2.2 Matching PRELIM crude assays to Rystad assets

Crude streams available in the PRELIM model are matched to Rystad assets using data made available by Rystad on crude streams and associated assets. The matching is performed using fuzzy matching methods on crude names. After cleaning crude names from both sources, Jaro-Winkler distances were calculated and capped at 0.25. Matches were then manually verified, leaving me with 493 crude assays from PRELIM (from the original 657) matched to 392 crude streams from Rystad[29]. Additionally, 25 unmatched PRELIM assays were manually matched to 41 Rystad assets with no specified crude streams. Overall, 518 PRELIM crude assays were linked to 8,386 Rystad assets, representing 64.4% of world production in 2024.

A unique crude stream often consists of a blend of crude oils from multiple assets, but in certain cases (mainly in Canada), production in a unique asset is split between different crudes. For these assets, midstream and downstream CI from PRELIM/OPEM were averaged weighting by the share of production being allocated to each crude stream.

C.2.3 Estimation models

Model and sample restrictions. Using PRELIM output, I fit a regression model to explain the variance in midstream and downstream CI separately. In both cases, I use the

[29]The difference in number comes from unique crude streams being split into different crude assays in PRELIM differing by the data source. For instance, the "Hibernia" crude in Canada is split into 3 assays, corresponding to information on this specific crude obtained from ExxonMobil, Statoil and Chevron.

two most influential variables, API gravity and sulphur content, which are also available at the asset level in Rystad data. Refinery configuration plays the most important role in explaining variations. I follow the assumptions made in the PRELIM model and assign "deep conversion", "hydroskimming" and "medium conversion" based on the same rules on API gravity and sulphur content. Both variables (in log form for sulphur content) are subsequently introduced in polynomial form to account for as much of the residual variation as possible. Higher-order terms are iteratively dropped to avoid overfitting based on predicted R-squared. The final models used to explain midstream and downstream carbon emissions are as follows:

$$\begin{aligned} CI_f^{PRELIM} &= \sum_1^3 \beta_i^C RefineryConfig_{i,f} + \beta_4 API_f + \beta_5 API_f^2 + \beta_6 API_f^3 \\ &+ \beta_7 \log(Sulphur) + \beta_8 \log(Sulphur)^2 + \epsilon_f \end{aligned} \tag{24}$$

$$\begin{aligned} CI_f^{OPEM} &= \sum_1^3 \beta_i^C RefineryConfig_{i,f} + \beta_4 API_f + \beta_5 API_f^2 + \beta_6 API_f^3 + \beta_7 API_f^4 \\ &+ \beta_8 \log(Sulphur) + \beta_9 \log(Sulphur)^2 + \beta_{10} \log(Sulphur)^3 + \epsilon_f \end{aligned} \tag{25}$$

Regression results. Table 10 shows results from both estimations. All coefficients are significant and predicted R-squared are 0.95 and 0.88 respectively. As expected, refinery configuration explains a large share of variations across crudes, especially in the case of midstream emissions. Adding polynomial terms in API gravity and log sulphur content improves the variance explained marginally, as the relatively low within R-squared indicates. By contrast, API gravity and sulphur content play a much more important role in explaining the variance in downstream emissions.

Sensitivity analysis. Figure 21 compares midstream and downstream CI to the values predicted by the model. Midstream values appear grouped according to each refinery configuration, illustrating the crucial role played by the dummies in the regression results. While the inclusion of API gravity and sulphur content in the model explains a large share of within-group variation for medium and deep conversion, it does not appear to be the case for crudes using the hydroskimming configuration (bottom-left of the midstream panel), explaining the low within R-squared value. Downstream CI are more evenly distributed across the graph, and predicted values remain close to OPEM-computed ones.

Contrary to upstream CI, midstream and downstream CI computed using PRELIM and OPEM models do not contain visible outliers. The risk of overfitting particular values is thus low. Tables 11 and 12 report re-estimated coefficients excluding most-influential observations for midstream and downstream CI respectively. In both cases, estimates remain significant

Variables	Midstream	Downstream
Configuration: Deep conversion	56.552***	530.269***
	(1.374)	(17.341)
Configuration: Hydroskimming	40.093***	587.056***
	(1.608)	(17.764)
Configuration: Medium conversion	51.361***	515.626***
	(1.604)	(18.037)
API	-0.933***	-16.122***
	(0.124)	(2.159)
API^2	0.016***	0.781***
	(0.003)	(0.096)
API^3	-9.62e-05***	-0.017***
	(2.74e-05)	(0.002)
API^4		1.11e-04***
		(1.16e-05)
log(Sulphur)	-0.988***	-7.784***
	(0.103)	(0.832)
$log(Sulphur)^2$	-0.135***	-2.404***
	(0.025)	(0.474)
$log(Sulphur)^3$		-0.19***
		(0.072)
Corrected R^2	0.948	0.887
Predicted R^2	0.947	0.883
Within R^2	0.300	0.875

Table 10: Regression results from midstream and downstream CI estimations.

Notes: The table displays coefficients from the regressions using equations (24) and (25), with standard errors in parentheses. R^2 are corrected in order to account for the absence of intercept, as well as predicted R^2. Within R^2 are computed using fixed-effects regressions to capture the effect of configuration dummies.

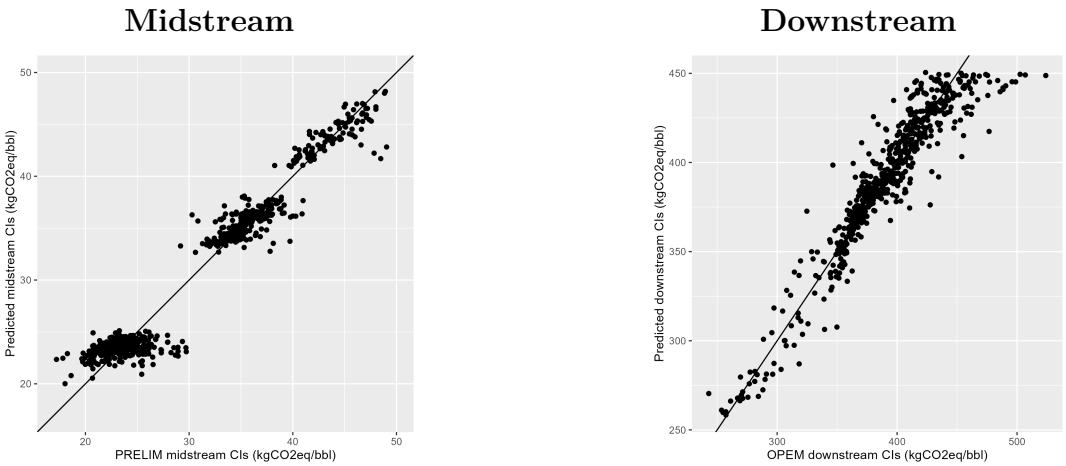


Figure 21: Comparison between PRELIM/OPEM CI and predicted ones.

and stable enough to exclude the risk of overfitting.

	Deep Conversion	Hydroskimming	Medium Conversion	API
0	56.55*** (1.37)	40.09*** (1.61)	51.36*** (1.60)	-0.93*** (0.12)
1	56.38*** (1.30)	40.05*** (1.52)	51.22*** (1.51)	-0.91*** (0.12)
10	56.69*** (1.23)	40.58*** (1.44)	51.73*** (1.44)	-0.93*** (0.11)
50	56.56*** (1.05)	40.80*** (1.22)	51.91*** (1.21)	-0.89*** (0.10)
	API^2	API^3	log(Sulphur)	$log(Sulphur)^2$
0	0.0159***	-9.62e−05***	-0.99*** (0.10)	-0.13*** (0.03)
1	0.0152***	-9.03e−05***	-1.00*** (0.10)	-0.13*** (0.02)
10	0.0153***	-8.77e−05***	-1.10*** (0.09)	-0.18*** (0.02)
50	0.0126***	-5.57e−05***	-1.18*** (0.08)	-0.22*** (0.02)

Table 11: Midstream CI estimation: coefficient stability to influential observations (Cook's distance).

Notes: Each panel reports coefficients re-estimated after excluding the 0, 1, 10, and 50 most influential observations according to Cook's distance. Standard errors are reported in parentheses, except for the higher-order API terms.

	Deep Conversion	Hydroskimming	Medium Conversion	API	API^2
0	530.27*** (17.34)	587.06*** (17.76)	515.63*** (18.04)	-16.12*** (2.16)	0.78*** (0.10)
1	543.82*** (18.09)	599.01*** (18.32)	527.93*** (18.62)	-18.39*** (2.33)	0.90*** (0.11)
10	544.71*** (16.65)	600.12*** (16.83)	528.63*** (17.12)	-18.46*** (2.15)	0.91*** (0.10)
50	555.93*** (14.05)	609.59*** (14.18)	540.02*** (14.41)	-19.59*** (1.83)	0.94*** (0.09)
	API^3	API^4	log(Sulphur)	$log(Sulphur)^2$	$log(Sulphur)^3$
0	-1.66e−02***	1.11e−04***	-7.78*** (0.83)	-2.40*** (0.47)	-0.19*** (0.07)
1	-1.92e−02***	1.30e−04***	-7.81*** (0.83)	-2.24*** (0.48)	-0.16** (0.07)
10	-1.93e−02***	1.31e−04***	-7.60*** (0.76)	-2.40*** (0.45)	-0.15** (0.07)
50	-1.95e−02***	1.30e−04***	-7.31*** (0.60)	-2.74*** (0.36)	-0.21*** (0.06)

Table 12: Downstream CI estimation: coefficient stability to influential observations (Cook's distance).

Notes: Each panel reports coefficients re-estimated after excluding the 0, 1, 10, and 50 most influential observations according to Cook's distance. Standard errors are reported in parentheses, except for the higher-order API terms.

C.2.4 Predicting midstream/downstream CI

Rystad assets matched to PRELIM crudes are assigned midstream and downstream CIs computed using the PRELIM and OPEM models. These assets represent 59% of oil reserves in 2016. For remaining assets, I use the coefficients estimated in both regressions to predict their CI, at the crude level in the case of crude streams unmatched to PRELIM crude assays (17.3% of reserves) and at the asset level if the crude stream is unspecified (11.7%).

Several assets, representing 12% of world reserves, lack information on either API gravity or sulphur content. For these assets, reserve-weighted averages from similar oils were

computed, according to the following order: reservoir, sub-basin, similar API within the same country, same main oil type within the same country, and finally same main oil type. The vast majority of assets are attributed values computed within the same reservoir or sub-basin. Since geographically close assets are more likely to be blended together before refining, averages computed from nearby assets are preferable. Table 13 sums up the methods and granularity levels of midstream and downstream CI predictions.

Method	Granularity level	Share of 2016 reserves
PRELIM/OPEM	Crude stream	58.9%
Predicted	Crude stream	17.3%
	Asset-specific	11.7%
Average	Reservoir	5.4%
	Sub-basin	5.7%
	API category × country	0.7%
	Oil type × country	0.2%
	Oil type	$< 0.1\%$

Table 13: Data sources and aggregation levels for midstream and downstream carbon intensity values.

D Additional results

D.1 Overshoot against emissions under the timely policy

Figure 22 draws the bands of Figure 3 on top of cumulative emissions under the timely policy. Emissions under the timely policy fall from 674 $GtCO_2eq$ at a zero tax to the 350.3 $GtCO_2eq$ budget at \$80.4, so that at the timely tax the overshoot adds 20% to what the budget allows. The top band, the part a tax set in 2030 can still act on, is a thin layer on top of a much larger stock, which is why a small error in the calibration of the 2030 tax matters.

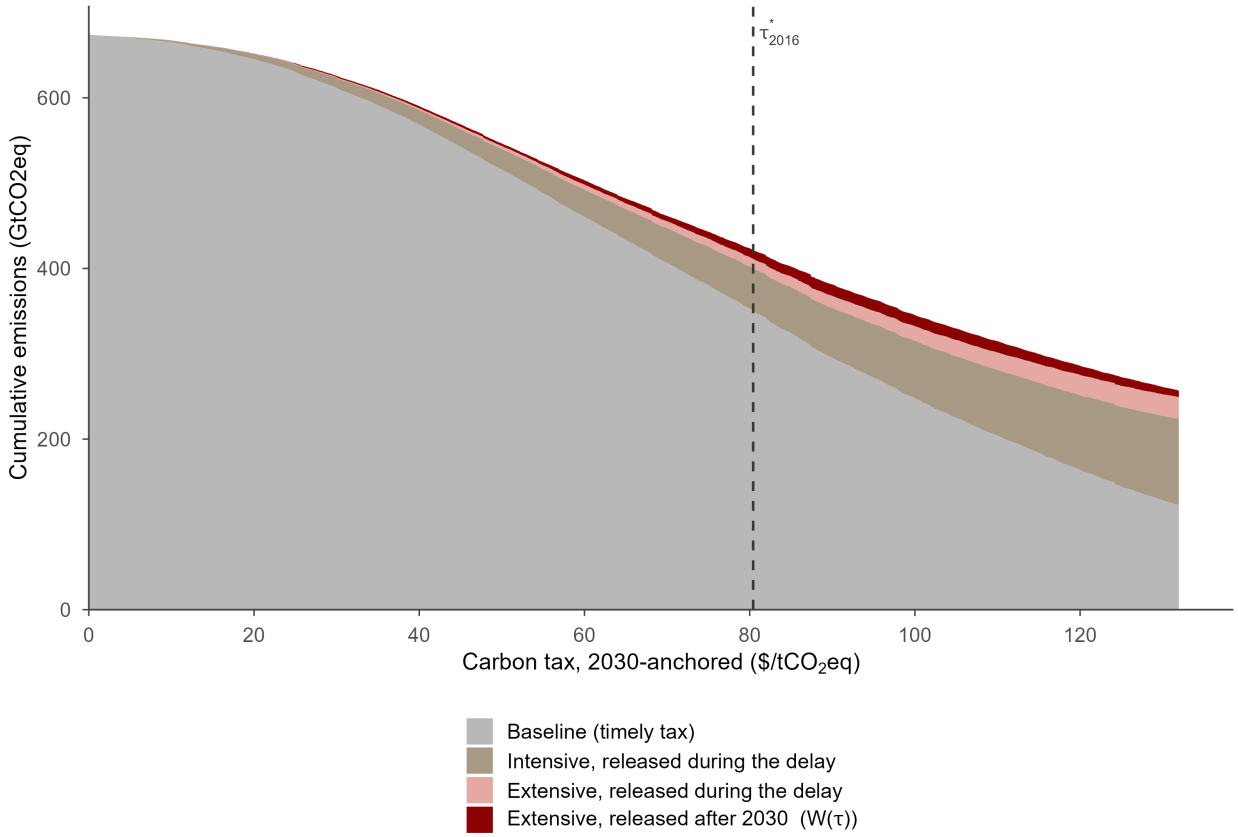


Figure 22: Overshoot caused by the delay and emissions under a timely policy.

Notes: Cumulative emissions under the timely policy (grey) and overshoot caused by the delay, by margin and date of emission. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2030 levels of the tax rising at 2% per year.

D.2 Overshoot by supply segment and region

Figure 23 shows the overshoot bands of Figure 4 without the emissions under the timely policy, which makes the post-2030 band of the onshore and shelf segments visible. Figure 24 shows the same decomposition by region, which Section 4.1 discusses.

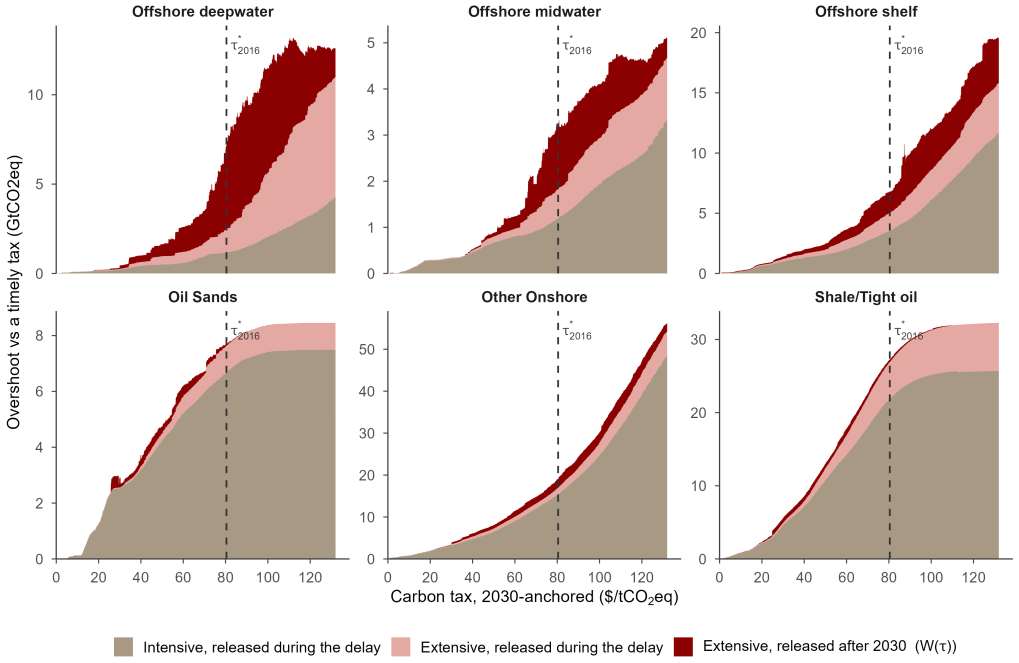


Figure 23: Overshoot caused by the delay by supply segment, margin and date of emission.

Notes: Panels have different vertical scales. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2030 levels of the tax rising at 2% per year.

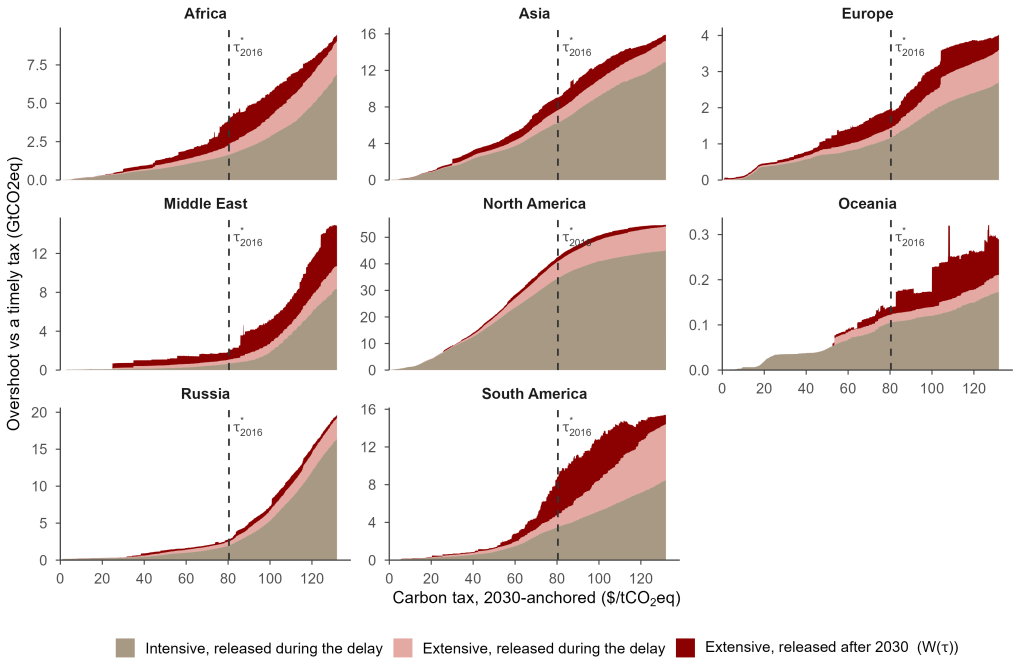


Figure 24: Overshoot caused by the delay by region, margin and date of emission.

Notes: Panels have different vertical scales. The dashed line is the tax that meets the budget under a timely policy. Taxes are 2030 levels of the tax rising at 2% per year.

D.3 The premium and the residual overshoot by year of implementation

Table 14 gives the numbers behind Figure 6: for each year of implementation, the remaining budget, the lock-in-adjusted tax, the premium and the residual overshoot left by the naive tax, for delays measured from 2016 and from 2025.

			From 2016		From 2025	
Year	Budget ($GtCO_2eq$)	Adjusted tax ($/$tCO_2eq$)	Premium ($/$tCO_2eq$)	Residual overshoot ($GtCO_2eq$)	Premium ($/$tCO_2eq$)	Residual overshoot ($GtCO_2eq$)
2016	350.3	80.4	–	–	–	–
2018	320.1	81.8	0.3	1.6 (0.5%)	–	–
2020	289.3	83.2	0.8	4.0 (1.4%)	–	–
2022	260.3	85.3	1.5	6.1 (2.4%)	–	–
2024	229.5	87.4	1.5	8.1 (3.5%)	–	–
2026	198.1	90.5	2.8	11.0 (5.6%)	0.3	1.4 (0.7%)
2028	165.6	94.4	3.5	13.1 (7.9%)	1.4	5.6 (3.4%)
2030	132.6	98.4	3.3	13.2 (10.0%)	1.8	7.5 (5.6%)

Table 14: Lock-in premium and residual overshoot by year of implementation.

Notes: Tax rising at 2% per year; taxes are 2030 levels. The remaining budget is the initial budget net of emissions released before implementation on the business-as-usual path. Overshoot percentages are relative to the remaining budget. "From 2016" and "From 2025" indicate the year from which the delay is measured, the state of the industry in that year being taken as given.

D.4 Constant carbon tax

This appendix reports the results for a constant tax that meets the same budget under a timely policy, $\$105.7/tCO_2eq$. The delay then overshoots the budget by 132.8 $GtCO_2eq$, 38% of the budget: 91.4 $GtCO_2eq$ on the intensive margin, 22.3 $GtCO_2eq$ from delay-induced fields before 2030 and 19.2 $GtCO_2eq$ from the same fields after 2030. The intensive margin is larger than under the increasing tax because a constant tax meeting the same budget is higher in the early years, so the timely policy removes more production during the delay. In 2030, the naive tax is \$139.1 and the lock-in-adjusted tax \$142.4, so that the premium is \$3.3, as under the increasing tax, and the residual overshoot is 14.9 $GtCO_2eq$, against 13.2 $GtCO_2eq$. The lock-in component of the cost of delay is therefore of the same size under both designs, while the total cost of the delay almost doubles (Section 4.5).

D.5 Survival of locked-in emissions under a higher tax

Figure 25 reports, for all locked-in fields together, the emissions locked in at the naive tax that survive each level of the 2030 tax. The stock falls to 91% of its initial level at the lock-in-adjusted tax, and still stands at 76% at \$105, 64% at \$110 and 39% at \$120. Section 5.1 reads the same object by supply segment. The curve falls slowly over the range the budget makes relevant and steeply above it, which is the aggregate counterpart of the fact that a locked-in field stops producing only when the tax exceeds what its operating costs can bear.

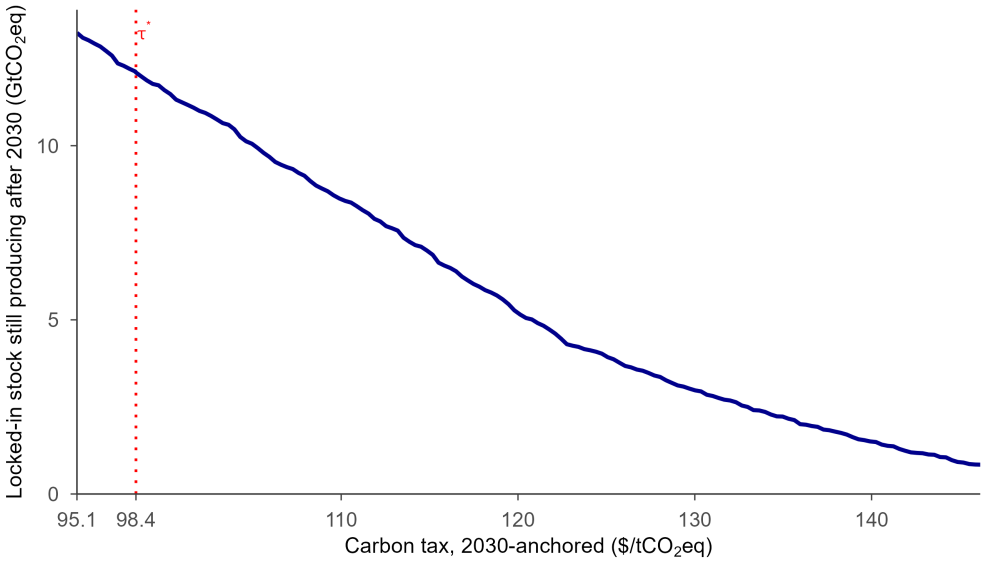


Figure 25: Locked-in emissions surviving a higher tax, all locked-in fields.

Notes: Post-2030 emissions locked in at the naive tax that are still released when the tax implemented in 2030 is raised to the level on the horizontal axis, all delay-induced fields, in $GtCO_2eq$. The horizontal axis starts at the naive tax, $\tau^N = \$95.1$, and the dotted line marks the lock-in-adjusted tax, $\tau^* = \$98.4$. Taxes are 2030 levels of the tax rising at 2% per year.

D.6 The largest locked-in assets

Table 15 lists the twenty assets that carry the most post-2030 emissions at the naive tax, out of the 651 that carry any. They account for 37.6% of the total, and fifty assets account for 63.4%, which is what the size-based bans of Section 5.3 remove. The year of responsibility is the year from which the asset would be completed even if the tax were implemented immediately (Section 3.4).

Asset	Country	Segment	Year of responsibility	Emissions ($GtCO_2eq$)	Share	Cumulative
Marjan (Expansion)	Saudi Arabia	Offshore shelf	2021	0.86	6.5%	6.5%
Tengiz FGP	Kazakhstan	Onshore conventional	2019	0.31	2.3%	8.8%
Atapu (P-84)	Brazil	Offshore deepwater	2027	0.30	2.3%	11.1%
Buzios XI (P-83)	Brazil	Offshore deepwater	2025	0.25	1.9%	13.0%
Venus Phase 1	Namibia	Offshore deepwater	2029	0.25	1.9%	14.9%
Sepia (P-85)	Brazil	Offshore deepwater	2028	0.24	1.8%	16.7%
Willow (GMT)	United States	Onshore conventional	2026	0.23	1.7%	18.5%
Argos (Mad Dog Phase 2) (GC825)	United States	Offshore deepwater	2019	0.22	1.7%	20.2%
Buzios IX (P-80)	Brazil	Offshore deepwater	2025	0.22	1.7%	21.8%
Longtail	Guyana	Offshore deepwater	2027	0.22	1.7%	23.5%
Buzios X (P-82)	Brazil	Offshore deepwater	2025	0.22	1.7%	25.2%
Al Shaheen Ruya 3 (Batch 1)	Qatar	Offshore shelf	2025	0.21	1.6%	26.8%
Hail (Sour Gas)	UAE	Offshore shelf	2025	0.19	1.5%	28.3%
Bacalhau (x-Carcara) & Bacalhau Norte Phase 2	Brazil	Offshore deepwater	2029	0.19	1.4%	29.7%
Hammerhead	Guyana	Offshore midwater	2028	0.19	1.4%	31.1%
Trion	Mexico	Offshore deepwater	2025	0.19	1.4%	32.5%
Tilapia (Jaguar)	Guyana	Offshore deepwater	2026	0.17	1.3%	33.8%
Azadegan North Phase 2	Iran	Onshore conventional	2027	0.17	1.3%	35.1%
Bacalhau (x-Carcara) Phase 1	Brazil	Offshore deepwater	2022	0.17	1.3%	36.4%
Gran Morgu (x-Krabdagu)	Suriname	Offshore midwater	2026	0.17	1.2%	37.6%

Table 15: The twenty largest locked-in assets.

Notes: Post-2030 emissions of delay-induced fields surviving the naive tax implemented in 2030, no-anticipation scenario, tax rising at 2% per year. Asset names are those of the Rystad database.

D.7 Incidence by country

Table 16 reports the decomposition of Table 3 for the largest producers of each region and for the producers discussed in Section 5.2.

	Overproduction ($GtCO_2eq$)	Lock-in ($GtCO_2eq$)	Net ($GtCO_2eq$)	Net (% of 2016 reserves)
Net winners through overproduction mainly				
United States	27.18	1.31	28.49	24.9
Canada	10.18	−0.04	10.14	17.7
China	4.27	0.57	4.84	17.6
Net winners through lock-in mainly				
Norway	0.13	0.37	0.50	5.9
Sign reversed by lock-in				
Brazil	−0.64	2.26	1.61	7.7
Guyana	−0.68	0.97	0.29	4.9
Senegal	−0.02	0.09	0.07	22.9
Ghana	−0.03	0.04	0.01	1.2
Net losers despite lock-in gains				
Nigeria	−0.21	0.17	−0.04	−0.8
Angola	−0.39	0.05	−0.34	−6.3
Kazakhstan	−1.55	0.29	−1.26	−11.7
Qatar	−1.44	0.23	−1.21	−6.3
Net losers on both counts				
Saudi Arabia	−13.55	−1.63	−15.18	−12.1
Iraq	−7.23	−2.17	−9.41	−17.8
Russia	−6.51	−1.79	−8.29	−12.9
Iran	−4.44	−0.63	−5.07	−10.8
UAE	−3.97	−0.53	−4.50	−11.7
Kuwait	−2.63	−0.90	−3.54	−15.4
Azerbaijan	−0.24	−0.17	−0.40	−11.4
Algeria	−0.30	−0.04	−0.34	−5.9

Table 16: Incidence of the delay by country.

Notes: Change in cumulative emissions over 2016–2100 relative to a timely tax, no-anticipation scenario, tax rising at 2% per year. The last column is the net change in production (in barrels) as a share of the country's 2016 reserves.

D.8 Incidence by firm

Table 17 reports the decomposition of Table 3 for the twenty largest companies by 2016 equity reserves. Production is attributed to equity holders at their share in each year, so a firm keeps what it lifted while it held the asset, and ownership beyond 2026 is held at its 2026 level. Figure 12 plots the same decomposition for the fifty largest firms, normalised by their own reserves.

D.9 Definition of the targeted instruments

Each instrument of Table 4 restricts or taxes a set of assets from a starting year, and is then evaluated at the tax the remaining budget requires under that restriction.

Company	Type	Headquarters	Reserves (Gbbl)	Overproduction ($GtCO_2eq$)	Lock-in ($GtCO_2eq$)	Net ($GtCO_2eq$)	Net (% res.)
Saudi Aramco	NOC	Saudi Arabia	289.7	−13.45	−1.62	−15.08	−11.9
NIOC	NOC	Iran	99.1	−4.41	−0.61	−5.02	−10.9
ADNOC	NOC	UAE	64.8	−2.58	−0.35	−2.93	−10.7
Kuwait Petroleum Corp	NOC	Kuwait	56.1	−2.71	−0.93	−3.64	−14.8
PetroChina	NOC	China	52.7	0.96	−0.31	0.65	2.4
Rosneft	NOC	Russia	52.3	−3.63	−1.07	−4.70	−19.3
ExxonMobil	Major	United States	46.4	1.77	0.56	2.33	9.9
QatarEnergy	NOC	Qatar	37.5	−1.23	0.29	−0.93	−6.3
Shell	Major	United Kingdom	36.2	−0.20	0.26	0.06	−0.4
Petrobras	NOC	Brazil	33.1	−0.19	1.40	1.20	7.5
Chevron	Major	United States	29.9	2.29	0.21	2.51	17.1
Gazprom	NOC	Russia	29.8	−1.27	−0.35	−1.62	−12.8
Lukoil	Integrated	Russia	24.7	−0.55	−0.40	−0.95	−8.3
BP	Major	United Kingdom	24.4	−0.79	0.12	−0.67	−5.7
TotalEnergies	Major	France	22.5	−0.79	0.67	−0.12	−2.5
CNOOC	NOC	China	20.5	0.36	0.96	1.32	12.7
ConocoPhillips	Independent	United States	16.8	2.44	0.29	2.73	35.2
Basra Oil Company	NOC	Iraq	14.8	−1.60	−0.63	−2.23	−31.5
Pemex	NOC	Mexico	14.2	0.49	0.10	0.59	8.7
Canadian Natural Resources	Independent	Canada	13.0	2.21	−0.00	2.21	29.0

Table 17: Incidence of the delay by firm.

Notes: Change in cumulative emissions over 2016–2100 relative to a timely tax, no-anticipation scenario, tax rising at 2% per year. The twenty largest companies by 2016 equity reserves, excluding one residual category of unattributed United States partners. The last column is the net change in production (in barrels) as a share of the firm's own 2016 equity reserves.

Perimeters. A segment restriction applies to all assets of the segment, using the classification of Table 2. A regional perimeter covers the assets located in the bloc's territory and, outside it, the assets that firms headquartered in the bloc operate and own at least half of; Europe is the European Union together with Norway and the United Kingdom. A size restriction applies to the assets that carry the most post-2030 emissions at the naive tax in the delay without restriction, ranked as in Table 15; the twenty largest carry 37.6% of the residual overshoot and the fifty largest 63.4%.

Instruments. A ban on new approvals removes from the analysis every asset of the perimeter approved from the starting year onwards, so assets already approved are left alone and no capital already committed is cancelled. An early carbon tax applies to the assets of the perimeter from the starting year, at the level the budget requires, while the rest of the industry faces the tax only from 2030; it therefore acts on the development of new assets and on the production of assets already in operation. A shock to development capital raises the development cost of every asset of the perimeter by the stated percentage from the starting year, leaving operating costs unchanged. In every case, the naive planner observes the restriction and its effect on development costs when calibrating the 2030 tax.

D.10 Other targeted instruments

Figure 26 reports the naive tax, the lock-in-adjusted tax, the premium between them and the residual overshoot for every instrument evaluated in Section 5.3. It covers the development bans, by segment, by regional perimeter and by size of project, the early carbon taxes applied to the same perimeters, and the two shocks to development capital, each of them starting in 2016 or in 2025 and each evaluated at its own budget-consistent tax.

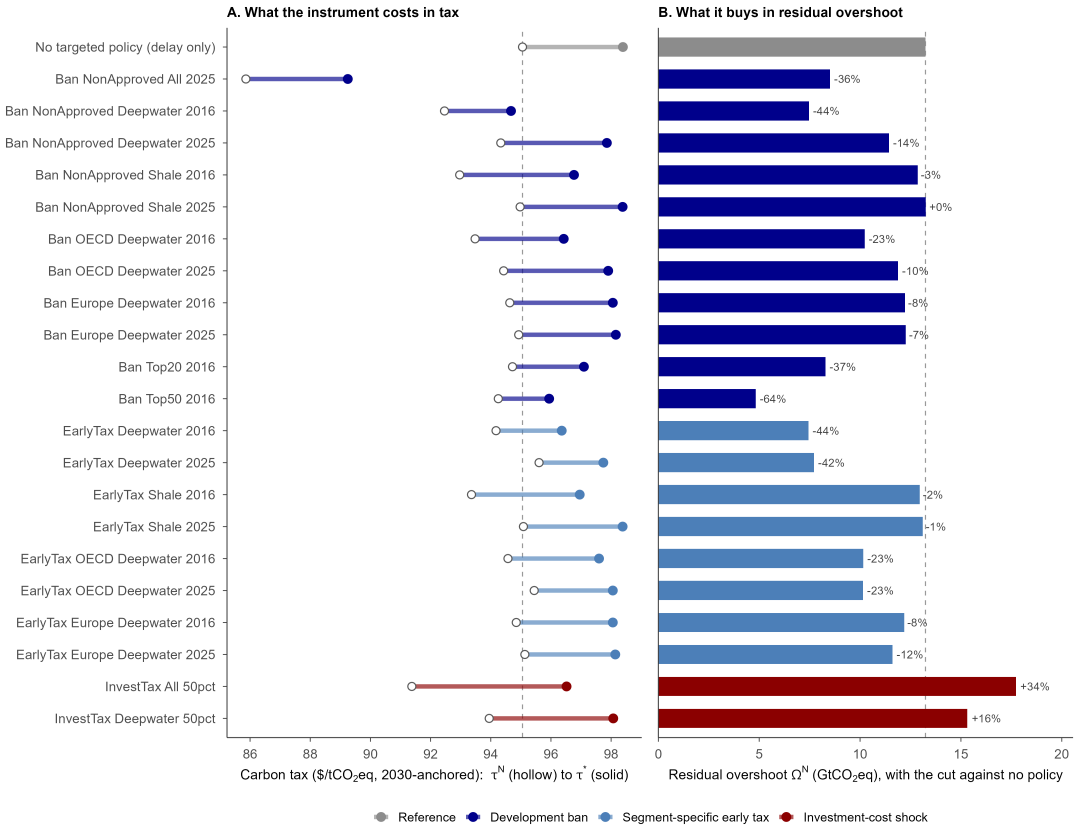


Figure 26: Targeted instruments.

Notes: Panel (a) runs from the naive tax (hollow) to the lock-in-adjusted tax (solid), so that the length of each segment is the premium. Panel (b) gives the residual overshoot and its change relative to the delay without restriction. Each instrument is evaluated at its own budget-consistent tax. Taxes are 2030 levels of the tax rising at 2% per year.

References

Brandt, A. R. and Farrell, A. E. (2007). Scraping the bottom of the barrel: greenhouse gas emission consequences of a transition to low-quality and synthetic petroleum resources. *Climatic Change*, 84(3):241–263.

Elvidge, C. D., Zhizhin, M., Baugh, K., Hsu, F.-C., and Ghosh, T. (2016). Methods for global survey of natural gas flaring from visible infrared imaging radiometer suite data. *Energies*, 9(1):14.

Ghosh, S., Jean, L., and Saha, P. (2025). Voice until the last exit: Ownership structure and the limits of voluntary environmental commitments. Working paper.

Gordon, D., Brandt, A. R., Bergerson, J., and Koomey, J. (2015). *Know your oil: creating a global oil-climate index*. Carnegie Endowment for International Peace Washington, DC.

Gvakharia, A., Kort, E. A., Brandt, A., Peischl, J., Ryerson, T. B., Schwarz, J. P., Smith, M. L., and Sweeney, C. (2017). Methane, black carbon, and ethane emissions from natural gas flares in the Bakken Shale, North Dakota. *Environmental Science & Technology*, 51(9):5317–5325.

Masnadi, M. S., El-Houjeiri, H. M., Schunack, D., Li, Y., Englander, J. G., Badahdah, A., Monfort, J.-C., Anderson, J. E., Wallington, T. J., Bergerson, J. A., et al. (2018). Global carbon intensity of crude oil production. *Science*, 361(6405):851–853.

McGlade, C., Sondak, G., and Han, M. (2018). Whatever happened to enhanced oil recovery? IEA Commentary, International Energy Agency, Paris.

Plant, G., Kort, E. A., Brandt, A. R., Chen, Y., Fordice, G., Gorchov Negron, A. M., Schwietzke, S., Smith, M., and Zavala-Araiza, D. (2022). Inefficient and unlit natural gas flares both emit large quantities of methane. *Science*, 377(6614):1566–1571.

World Bank (2025). Global gas flaring tracker report, July 2025. Technical report, World Bank.

Zavala-Araiza, D., Omara, M., Gautam, R., Smith, M. L., Pandey, S., Aben, I., Almanza-Veloz, V., Conley, S., Houweling, S., Kort, E. A., et al. (2021). A tale of two regions: methane emissions from oil and gas production in offshore/onshore Mexico. *Environmental Research Letters*, 16(2):024019.

Zhizhin, M., Matveev, A., Ghosh, T., Hsu, F.-C., Howells, M., and Elvidge, C. (2021). Measuring gas flaring in Russia with multispectral VIIRS Nightfire. *Remote Sensing*, 13(16):3078.